

2

Building Blocks of Managerial Accounting

Figure 2.1 Building Blocks of Managerial Accounting. How costs behave and how managers can estimate future costs are the building blocks of managerial accounting. (credit: modification of "Blocks" by "Hey Paul"/Flickr, CC BY 2.0)

Chapter Outline

LO **2.1** Distinguish between Merchandising, Manufacturing, and Service Organizations
LO **2.2** Identify and Apply Basic Cost Behavior Patterns
LO **2.3** Estimate a Variable and Fixed Cost Equation and Predict Future Costs

Why It Matters

Many 16-year-olds in the United States eagerly anticipate having a car of their own and the freedom that comes from having their own means of transportation. For many, this means not having to bum a ride from a friend, take a bus, hire Uber or Lyft, or worse, borrow the parents' car. However, as appealing as having one's own set of wheels sounds, it comes with an array of costs that many young drivers do not anticipate. Some of the costs associated with buying and owning a car are fixed, and some vary with the level of activity. For example, a driver pays car payments and insurance premiums every month whether or not the car is driven, but the cost of maintenance and gas can be controlled by driving less. A driver cannot control the price of gasoline or the mechanic's hourly wage but can control how much of each is used each month.

Just as car owners incur a variety of costs—fixed, variable, controllable, and uncontrollable—businesses incur these types of costs as well. The goal of managerial accountants is to use this cost information to assist management in both long- and short-term decision-making. Managerial accounting follows standards and best practices for reporting cost data that are less formal than those used for financial accounting. This means management often has the discretion to determine how costs are used internally.

Since businesses collect and analyze cost data for internal use, there may be distinct differences among businesses in how they estimate and treat certain costs. What does not change, regardless of how cost data is used, are generally agreed upon cost classifications managers use for decision-making. In short, most businesses incur the same *type* of costs, but how each firm classifies and manages these costs can vary widely.

2.1 Distinguish between Merchandising, Manufacturing, and Service Organizations

Most businesses can be classified into one or more of these three categories: manufacturing, merchandising, or service. Stated in broad terms, manufacturing firms typically produce a product that is then sold to a merchandising entity (a retailer) For example, Proctor and Gamble produces a variety of shampoos that it sells to retailers, such as Walmart, Target, or Walgreens. A service entity provides a service such as accounting or legal services or cable television and internet connections.

Some companies combine aspects of two or all three of these categories within a single business. If it chooses, the same company can both produce and market its products directly to consumers. For example, Nike produces products that it directly sells to consumers and products that it sells to retailers. An example of a company that fits all three categories is Apple, which produces phones, sells them directly to consumers, and also provides services, such as extended warranties.

Regardless of whether a business is a manufacturer of products, a retailer selling to the customer, a service provider, or some combination , all businesses set goals and have strategic plans that guide their operations. Strategic plans look very different from one company to another. For example, a retailer such as Walmart may have a strategic plan that focuses on increasing same store sales. Facebook's strategic plan may focus on increasing subscribers and attracting new advertisers. An accounting firm may have long-term goals to open offices in neighboring cities in order to serve more clients. Although the goals differ, the process all companies use to achieve their goals is the same. First, they must develop a plan for how they will achieve the goal, and then management will gather, analyze, and use information regarding costs to make decisions, implement plans, and achieve goals.

Table 2.1 lists examples of these costs. Some of these are similar across different types of businesses; others are unique to a particular business.

Costs

Type of Business	Costs Incurred
Manufacturing Business	• Direct labor • Plant and equipment • Manufacturing overhead • Raw materials
Merchandising Business	• Lease on retail space • Merchandise inventory • Retail sales staff
Service Business	• Billing and collections • Computer network equipment • Professional staff

Table 2.1 Some costs, such as raw materials, are unique to a particular type of business. Other costs, such as billing and collections, are common to most businesses, regardless of the type.

Knowing the basic characteristics of each cost category is important to understanding how businesses measure, classify, and control costs.

Merchandising Organizations

A merchandising firm is one of the most common types of businesses. A **merchandising firm** is a business that purchases finished products and resells them to consumers. Consider your local grocery store or retail clothing store. Both of these are merchandising firms. Often, merchandising firms are referred to as *resellers* or *retailers* since they are in the business of reselling a product to the consumer at a profit.

Think about purchasing toothpaste from your local drug store. The drug store purchases tens of thousands of tubes of toothpaste from a wholesale distributor or manufacturer in order to get a better per-tube cost. Then, they add their mark-up (or profit margin) to the toothpaste and offer it for sale to you. The drug store did not manufacture the toothpaste; instead, they are reselling a toothpaste that they purchased. Virtually all of your daily purchases are made from merchandising firms such as Walmart, Target, Macy's, Walgreens, and AutoZone.

Merchandising firms account for their costs in a different way from other types of business organizations. To understand merchandising costs, Figure 2.2 shows a simplified income statement for a merchandising firm:

Sales Revenue
– Cost of Goods Sold
= Gross Profit
– Operating Expenses
= Operating Profit

Figure 2.2 Simplified Income Statement for a Merchandising Firm. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

This simplified income statement demonstrates how merchandising firms account for their sales cycle or process. *Sales revenue* is the income generated from the sale of finished goods to consumers rather than from the manufacture of goods or provision of services. Since a merchandising firm has to purchase goods for resale, they account for this cost as *cost of goods sold*—what it cost them to acquire the goods that are then sold to the customer. The difference between what the drug store paid for the toothpaste and the revenue generated by selling the toothpaste to consumers is their *gross profit*. However, in order to generate sales revenue, merchandising firms incur expenses related to the process of operating their business and selling the merchandise. These costs are called *operating expenses*, and the business must deduct them from the gross profit to determine the *operating profit*. (Note that while the terms "operating profit" and "operating income" are often used interchangeably, in real-world interactions you should confirm exactly what the user means in using those terms.) Operating expenses incurred by a merchandising firm include insurance, marketing, administrative salaries, and rent.



Figure 2.3 Shopping Mall. Merchandising firms must identify and manage their costs to remain competitive and attract customers to their business. (credit: "stairs shopping mall" by "jarmoluk"/Pixabay, CC0)

CONCEPTS IN PRACTICE

Balancing Revenue and Expenses

Plum Crazy is a small boutique selling the latest in fashion trends. They purchase clothing and fashion accessories from several distributors and manufacturers for resale. In 2017, they reported these revenue and expenses:

Rent	$12,000	Sales revenue	$150,000
Advertising	4,000	Cost of goods sold	60,000
Utilities	1,500	Supplies	3,000
Salaries and wages	35,000	Miscellaneous	1,200

Before examining the income statement, let's look at Cost of Goods Sold in more detail. Merchandising companies have to account for inventory, a topic covered in Inventory (http://openstax.org/books/principles-financial-accounting/pages/10-why-it-matters). As you recall, merchandising companies carry inventory from one period to another. When they prepare their income statement, a crucial step is identifying the actual cost of goods that were sold for the period. For Plum Crazy, their Cost of Goods Sold was calculated as shown in Figure 2.4.

PLUM CRAZY
Cost of Goods Sold
For the Year Ended December 31, 2017

Beginning Merchandise Inventory	$ 23,500
+ Purchases	115,000
Goods Available for Sale	138,500
– Ending Merchandise Inventory	(78,500)
Cost of Goods Sold	$ 60,000

Figure 2.4 Plum Crazy's Cost of Goods Sold Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Once the calculation of the Cost of Goods Sold has been completed, Plum Crazy can now construct their income statement, which would appear as shown in Figure 2.5.

PLUM CRAZY
Income Statement
For the Year Ended December 31, 2017

Sales Revenue		$150,000
Cost of Goods Sold		60,000
Gross Profit		90,000
Advertising	$ 4,000	
Rent	12,000	
Salaries and Wages	35,000	
Supplies	3,000	
Utilities	1,500	
Miscellaneous	1,200	
Operating Expenses	56,700	56,700
Net Income		$ 33,300

Figure 2.5 Plum Crazy's Income Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Since merchandising firms must pass the cost of goods on to the consumer to earn a profit, they are extremely cost sensitive. Large merchandising businesses like Walmart, Target, and Best Buy manage costs by buying in bulk and negotiating with manufacturers and suppliers to drive the per-unit cost.

CONTINUING APPLICATION

Introduction to the Gearhead Outfitters Story

Gearhead Outfitters, founded by Ted Herget in 1997 in Jonesboro, AR, is a retail chain which sells outdoor gear for men, women, and children. The company's inventory includes clothing, footwear for hiking and running, camping gear, backpacks, and accessories, by brands such as The North Face, Birkenstock, Wolverine, Yeti, Altra, Mizuno, and Patagonia. Ted fell in love with the outdoor lifestyle while working as a ski instructor in Colorado and wanted to bring that feeling back home to Arkansas. And so, Gearhead was born in a small downtown location in Jonesboro. The company has had great success over the years, expanding to numerous locations in Ted's home state, as well as Louisiana, Oklahoma, and Missouri.

While Ted knew his industry when starting Gearhead, like many entrepreneurs he faced regulatory and financial issues which were new to him. Several of these issues were related to accounting and the wealth of decision-making information which accounting systems provide.

For example, measuring revenue and expenses, providing information about cash flow to potential lenders, analyzing whether profit and positive cash flow is sustainable to allow for expansion, and managing inventory levels. Accounting, or the preparation of financial statements (balance sheet, income statement, and statement of cash flows), provides the mechanism for business owners such as Ted to make fundamentally sound business decisions.

LINK TO LEARNING

Walmart is inarguably a retail giant, but how did the company become so successful? Read the article about how low costs have allowed Walmart to keep prices low while still making a large profit (https://openstax.org/l/50Walmart) to learn more.

Manufacturing Organizations

A **manufacturing organization** is a business that uses parts, components, or raw materials to produce finished goods (Figure 2.6). These finished goods are sold either directly to the consumer or to other manufacturing firms that use them as a component part to produce a finished product. For example, Diehard manufactures automobile batteries that are sold directly to consumers by retail outlets such as AutoZone, Costco, and Advance Auto. However, these batteries are also sold to automobile manufacturers such as Ford, Chevrolet, or Toyota to be installed in cars during the manufacturing process. Regardless of who the final consumer of the final product is, Diehard must control its costs so that the sale of batteries generates revenue sufficient to keep the organization profitable.

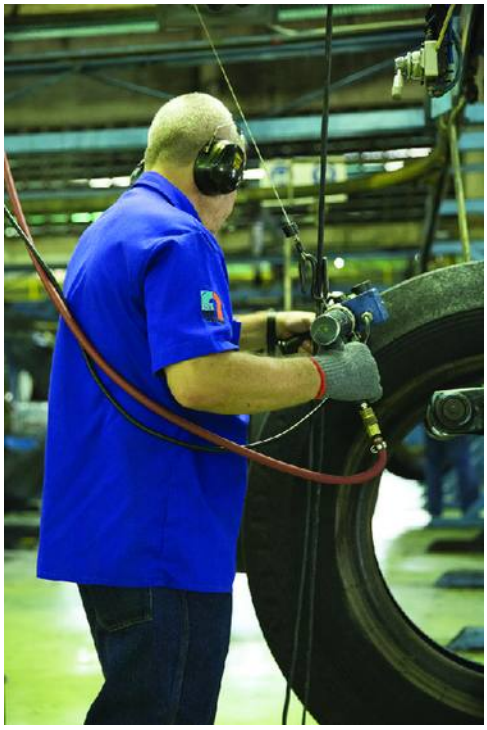


Figure 2.6 Manufacturing firms apply direct labor to raw materials in order to produce the finished goods purchased from retailers. (credit: "work manufactures" by "dodaning0"/Pixabay, CC0)

Manufacturing firms are more complex organizations than merchandising firms and therefore have a larger variety of costs to control. For example, a merchandising firm may purchase furniture to sell to consumers, whereas a manufacturing firm must acquire raw materials such as lumber, paint, hardware, glue, and varnish that they transform into furniture. The manufacturer incurs additional costs, such as direct labor, to convert the raw materials into furniture. Operating a physical plant where the production process takes place also generates costs. Some of these costs are tied directly to production, while others are general expenses necessary to operate the business. Because the manufacturing process can be highly complex, manufacturing firms constantly evaluate their production processes to determine where cost savings are possible.

CONCEPTS IN PRACTICE

Cost Control

Controlling costs is an integral function of all managers, but companies often hire personnel to specifically oversee cost control. As you've learned, controlling costs is vital in all industries, but at Hilton Hotels, they translate this into the position of Cost Controller. Here is an excerpt from one of Hilton's recent job postings.

Position Title: Cost Controller

Job Description: "A Cost Controller will work with all Heads of Departments to effectively control all products that enter and exit the hotel."[1]

Job Requirements:

"As Cost Controller, you will work with all Heads of Departments to effectively control all products that enter and exit the hotel. Specifically, you will be responsible for performing the following tasks to the highest standards:

1 Hilton. "Cost Controller: Job Description." Hosco. https://www.hosco.com/en/job/hilton-istanbul-bomonti-hotel-conference-center/cost-controller

- Review the daily intake of products into the hotel and ensure accurate pricing and quantity of goods received
- Control the stores by ensuring accuracy of inventory and stock control and the pricing of goods received
- Alert relevant parties of slow-moving goods and goods nearing expiry dates to reduce waste and alter product purchasing to accommodate
- Manage cost reporting on a weekly basis
- Attend finance meetings, as required
- Maintain good communication and working relationships with all hotel areas
- Act in accordance with fire, health and safety regulations and follow the correct procedures when required"[2]

As you can see, the individual in this position will interact with others across the organization to find ways to control costs for the benefit of the company. Some of the benefits of cost control include:

- Lowering overall company expenses, thereby increasing net income.
- Freeing up financial resources for investment in research & development of new or improved products, goods, or services
- Providing funding for employee development and training, benefits, and bonuses
- Allowing corporate earnings to be used to support humanitarian and charitable causes

Manufacturing organizations account for costs in a way that is similar to that of merchandising firms. However, as you will learn, there is a significant difference in the calculation of cost of goods sold. Figure 2.7 shows a simplification of the income statement for a manufacturing firm:

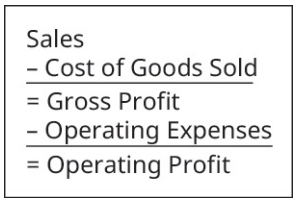


Figure 2.7 Simplified Income Statement for a Manufacturing Firm. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

At first it appears that there is no difference between the income statements of the merchandising firm and the manufacturing firm. However, the difference is in how these two types of firms account for the cost of goods sold. Merchandising firms determine their cost of goods sold by accounting for both existing inventory and new purchases, as shown in the Plum Crazy example. It is typically easy for merchandising firms to calculate their costs because they know exactly what they paid for their merchandise.

Unlike merchandising firms, manufacturing firms must calculate their cost of goods sold based on how much they manufacture and how much it costs them to manufacture those goods. This requires manufacturing firms to prepare an additional statement before they can prepare their income statement. This additional statement is the *Cost of Goods Manufactured* statement. Once the cost of goods manufactured is calculated, the cost is then incorporated into the manufacturing firm's income statement to calculate its cost of goods sold.

One thing manufacturing firms must consider in their cost of goods manufactured is that, at any given time, they have products at varying levels of production: some are finished and others are still process. The cost of goods manufactured statement measures the cost of the goods actually finished during the period, whether or not they were started during that period.

2 Hilton. "Cost Controller: Job Description." Hosco. https://www.hosco.com/en/job/hilton-istanbul-bomonti-hotel-conference-center/cost-controller

Before examining the typical manufacturing firm's process to track cost of goods manufactured, you need basic definitions of three terms in the schedule of Costs of Goods Manufactured: direct materials, direct labor, and manufacturing overhead. **Direct materials** are the components used in the production process whose costs can be identified on a per item-produced basis. For example, if you are producing cars, the engine would be a direct material item. The direct material cost would be the cost of one engine. **Direct labor** represents production labor costs that can be identified on a per item-produced basis. Referring to the car production example, assume that the engines are placed in the car by individuals rather than by an automated process. The direct labor cost would be the amount of labor in hours multiplied by the hourly labor cost. **Manufacturing overhead** generally includes those costs incurred in the production process that are not economically feasible to measure as direct material or direct labor costs. Examples include the department manager's salary, the production factory's utilities, or glue used to attach rubber molding in the auto production process. Since there are so many possible costs that can be classified as manufacturing overhead, they tend to be grouped and then allocated in a predetermined manner to the production process.

Figure 2.8 is an example of the calculation of the Cost of Goods Manufactured for Koeller Manufacturing. It demonstrates the relationship between cost of goods manufactured and cost of goods in progress and includes the three main types of manufacturing costs.

KOELLER MANUFACTURING
Schedule of Cost of Goods Manufactured
For the Month Ended March 31, 2017

Work in Process Inventory (beginning balance)		$ 75,000
Current Manufacturing Costs:		
Direct Material	$15,000	
Direct Labor	25,000	
Manufacturing Overhead	23,000	
Total Manufacturing Costs		63,000
Total Cost of Work in Process		138,000
– Work in Process, ending balance		43,000
Cost of Goods Manufactured		$ 95,000

Figure 2.8 Koeller Manufacturing's Cost of Goods Manufactured. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

As you can see, the manufacturing firm takes into account its work-in-process (WIP) inventory as well as the costs incurred during the current period to finish not only the units that were in the beginning WIP inventory, but also a portion of any production that was started but not finished during the month. Notice that the current manufacturing costs, or the additional costs incurred during the month, include direct materials, direct labor, and manufacturing overhead. Direct materials are calculated as

Materials Inventory (beginning balance)
+ Net Material Purchases
= Materials Available for Use
− Materials Inventory (ending balance)
= Direct Materials Used in Production

All of these costs are carefully tracked and classified because the cost of manufacturing is a vital component of the schedule of cost of goods sold. To continue with the example, Koeller Manufacturing calculated that the cost of goods manufactured was $95,000, which is carried through to the Schedule of Cost of Goods Sold (Figure 2.9).

KOELLER MANUFACTURING Schedule of Cost of Goods Sold For the Month Ended March 31, 2017	
Beginning Finished Goods Inventory	$ 65,000
+ Cost of Goods Manufactured	95,000
Goods Available for Sale	160,000
– Ending Finished Goods Inventory	58,000
Cost of Goods Sold	$102,000

Figure 2.9 Koeller Manufacturing's Cost of Goods Sold. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Now when Koeller Manufacturing prepares its income statement, the simplified statement will appear as shown in Figure 2.10.

KOELLER MANUFACTURING Income Statement For the Month Ended March 31, 2017	
Sales	$214,000
– Cost of Goods Sold	102,000
Gross Profit	112,000
– Operating Expenses	80,000
Operating Income	$ 32,000

Figure 2.10 Koeller Manufacturing's Income Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

So, even though the income statements for the merchandising firm and the manufacturing firm appear very similar at first glance, there are many more costs to be captured by the manufacturing firm. Figure 2.11 compares and contrasts the methods merchandising and manufacturing firms use to calculate the cost of goods sold in their income statement.

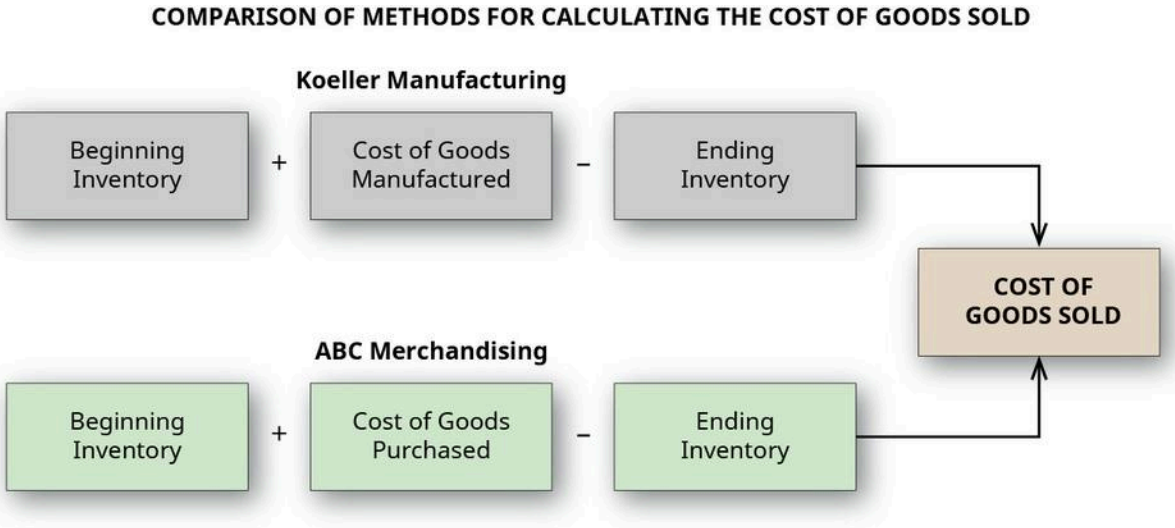


Figure 2.11 Merchandising firms consider the cost of goods purchased, and manufacturing firms consider the cost of goods manufactured in order to determine the cost of goods sold. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

CONCEPTS IN PRACTICE

Calculating Cost of Goods Sold in Manufacturing

Just Desserts is a bakery that produces and sells cakes and pies to grocery stores for resale. Although they are a small manufacturer, they incur many of the costs of a much larger organization. In 2017, they reported these revenue and expenses:

Office rent	$20,000	Sales revenue	$150,000
Office utilities	1,500	Cost of goods sold	70,000
Administrative salaries	35,000	Administrative expenses	12,000

Their income statement is shown in Figure 2.12.

JUST DESSERTS
Income Statement
For the Year Ended December 31, 2017

Sales Revenue		$150,000
Cost of Goods Sold		70,000
Gross Profit		80,000
Administrative Expenses	$12,000	
Administrative Salaries	35,000	
Office Utilities	1,500	
Office Rent	20,000	
Operating Expenses		68,500
Net Income		$ 11,500

Figure 2.12 Just Desserts' Income Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

You'll learn more about the flow of manufacturing costs in Identify and Apply Basic Cost Behavior Patterns. For now, recognize that, unlike a merchandising firm, calculating cost of goods sold in manufacturing firms can be a complex task for management.

Service Organizations

A **service organization** is a business that earns revenue by providing **intangible products**, those that have no physical substance. The service industry is a vital sector of the U.S. economy, providing 65% of the U.S. private-sector gross domestic product and more than 79% of U.S. private-sector jobs.[3] If **tangible products**, physical goods that customers can handle and see, are provided by a service organization, they are considered ancillary sources of revenue. Large service organizations such as airlines, insurance companies, and hospitals incur a variety of costs in the provision of their services. Costs such as labor, supplies, equipment, advertising, and facility maintenance can quickly spiral out of control if management is not careful. Therefore, although their cost drivers are sometimes not as complex as those of other types of firms, cost identification and control are every bit as important in the service industry.

For example, consider the services that a law firm provides its clients. What clients pay for are services such as representation in legal proceedings, contract negotiations, and preparation of wills. Although the true value of these services is not contained in their physical form, they are of value to the client and the source of revenue

3 John Ward. "The Services Sector: How Best to Measure It?" International Trade Administration. Oct. 2010. https://2016.trade.gov/publications/ita-newsletter/1010/services-sector-how-best-to-measure-it.asp. "United States GDP from Private Services Producing Industries." Trading Economics / U.S. Bureau of Economic Analysis. July 2018. https://tradingeconomics.com/united-states/gdp-from-services. "Employment in Services (% of Total Employment) (Modeled ILO Estimate)." International Labour Organization, ILOSTAT database. The World Bank. Sept. 2018. https://data.worldbank.org/indicator/SL.SRV.EMPL.ZS.

to the firm. The managing partners in the firm must be as cost conscious as their counterparts in merchandising and manufacturing firms. Accounting for costs in service firms differs from merchandising and manufacturing firms in that they do not purchase or produce goods. For example, consider a medical practice. Although some services provided are tangible products, such as medications or medical devices, the primary benefits the physicians provide their patients are the intangible services that are comprised of his or her knowledge, experience, and expertise.

Service providers have some costs (or revenue) derived from tangible goods that must be taken into account when pricing their services, but their largest cost categories are more likely to be administrative and personnel costs rather than product costs.

Service Revenue
– Operating Expenses
= Operating Profit

For example, Whichard & Klein, LLP, is a full-service accounting firm with their primary offices in Baltimore, Maryland. With two senior partners and a small staff of accountants and payroll specialists, the majority of the costs they incur are related to personnel. The value of the accounting and payroll services they provide to their clients is intangible in comparison to goods sold by a merchandiser or produced by a manufacturer but has value and is the primary source of revenue for the firm. At the end of 2019, Whichard and Klein reported the following revenue and expenses:

Revenue from services provided	$412,000	Utilities	$11,000
Accounting personnel salaries	210,000	Miscellaneous expenses	7,500
Office expense	35,000	Administrative salaries	45,000
Office equipment	9,000		

Their Income Statement for the period is shown in Figure 2.13.

WHICHARD & KLEIN, LLP
Income Statement
For the Year Ended December 31, 2019

Service Revenue	$412,000
Operating Expenses	
Salaries	210,000
Office Expense	35,000
Office Equipment	9,000
Administrative Salaries	45,000
Utilities	11,000
Miscellaneous	7,500
Total Operating Expenses	317,500
Operating Income	$ 94,500

Figure 2.13 Whichard & Klein's Income Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The bulk of the expenses incurred by Whichard & Klein are in personnel and administrative/office costs, which are very common among businesses that have services as their primary source of revenue.

CONCEPTS IN PRACTICE

Revenue and Expenses for a Law Office

The revenue and expenses for a law firm illustrate how the income statement for a service firm differs from that of a merchandising or manufacturing firm.

Welch & Graham is a well-established law firm that provides legal services in the areas of criminal law, real estate transactions, and personal injury. The firm employs several attorneys, paralegals, and office support staff. In 2017, they reported the following revenue and expenses:

Office rent	$ 20,000	Paralegal salaries	$ 100,000
Office utilities	12,500	Service revenue	1,500,000
Administrative salaries	150,000	Office expenses	12,000
Attorneys' salaries	750,000		

Their income statement is shown in Figure 2.14.

WELCH & GRAHAM, ATTORNEYS AT LAW
Income Statement
For the Year Ended December 31, 2017

Service Revenue		$1,500,000
Operating Expenses		
Administrative Salaries	$ 150,000	
Attorney Salaries	750,000	
Office Expenses	12,000	
Office Rent	20,000	
Paralegal Salaries	100,000	
Office Utilities Expenses	12,500	
Total Operating Expenses		1,044,500
Net Income		$ 455,500

Figure 2.14 Welch & Graham's Income Statement. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

As you can see, the majority of the costs incurred by the law firm are personnel related. They may also incur costs from equipment and materials such computer networks, phone and switchboard equipment, rent, insurance, and law library materials necessary to support the practice, but these costs represent a much smaller percentage of total cost than the administrative and personnel costs.

THINK IT THROUGH

Expanding a Business

Margo is the owner of a small retail business that sells gifts and home decorating accessories. Her business is well established, and she is now considering taking over additional retail space to expand her business to include gourmet foods and gift baskets. Based on customer feedback, she is confident that there is a demand for these items, but she is unsure how large that demand really is. Expanding her business this way will require that she incur not only new costs but also increases in existing costs.

Margo has asked for your help in identifying the impact of her decision to expand in terms of her costs. When discussing these cost increases, be sure to specifically identify those costs that are directly tied to her products and that would be considered overhead expenses.

2.2 Identify and Apply Basic Cost Behavior Patterns

Now that we have identified the three key types of businesses, let's identify cost behaviors and apply them to the business environment. In managerial accounting, different companies use the term *cost* in different ways depending on how they will use the cost information. Different decisions require different costs classified in

different ways. For instance, a manager may need cost information to plan for the coming year or to make decisions about expanding or discontinuing a product or service. In practice, the classification of costs changes as the use of the cost data changes. In fact, a single cost, such as rent, may be classified by one company as a fixed cost, by another company as a committed cost, and by even another company as a period cost. Understanding different cost classifications and how certain costs can be used in different ways is critical to managerial accounting.

ETHICAL CONSIDERATIONS

Institute of Management Accountants and Certified Management Accountant Certification

Managerial accountants provide businesses with clear and direct insight into the monetary effects of any particular operational action under consideration. They are expected to report financial information in a transparent and ethical fashion. The Institute of Management Accountants (IMA) offers the Certified Management Accountant (CMA) certification. IMA members and CMAs agree to uphold a set of ethical principles that includes honesty, fairness, objectivity, and responsibility. Any managerial accountant, even if not an IMA member or certified CMA, should act in accordance with these principles and encourage coworkers to follow ethical principles for reporting financial results and monetary effects of financial decisions related to their organization. The IMA Committee on Ethics encourages organizations and individuals to adopt, promote, and execute business practices consistent with high ethical standards.[4]

Major Cost Behavior Patterns

Any discussion of costs begins with the understanding that most costs will be classified in one of three ways: fixed costs, variable costs, or mixed costs. The costs that don't fall into one of these three categories are hybrid costs, which are examined only briefly because they are addressed in more advanced accounting courses. Because fixed and variable costs are the foundation of all other cost classifications, understanding whether a cost is a fixed cost or a variable cost is very important.

Fixed versus Variable Costs

A **fixed cost** is an unavoidable operating expense that does not change in total over the short term, even if a business experiences variation in its level of activity. Table 2.2 illustrates the types of fixed costs for merchandising, service, and manufacturing organizations.

Examples of Fixed Costs

Type of Business	Fixed Cost
Merchandising	Rent, insurance, managers' salaries
Manufacturing	Property taxes, insurance, equipment leases
Service	Rent, straight-line depreciation, administrative salaries, and insurance

Table 2.2

We have established that fixed costs do not change in total as the level of activity changes, but what about fixed costs on a *per-unit* basis? Let's examine Tony's screen-printing company to illustrate how costs can remain fixed in total but change on a per-unit basis.

4 "Ethics Center." Institute of Management Accountants. https://www.imanet.org/career-resources/ethics-center?ssopc=1

Tony operates a screen-printing company, specializing in custom T-shirts. One of his fixed costs is his monthly rent of $1,000. Regardless of whether he produces and sells any T-shirts, he is obligated under his lease to pay $1,000 per month. However, he can consider this fixed cost on a per-unit basis, as shown in Figure 2.15.

Monthly Rent	Number of T-Shirts Manufactured	Average Rent Cost per T-Shirt
$1,000	200	$5.00
1,000	400	2.50
1,000	600	1.67

Figure 2.15 Individual Rent Cost per T-Shirt Produced. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Tony's information illustrates that, despite the unchanging fixed cost of rent, as the level of activity increases, the per-unit fixed cost falls. In other words, fixed costs remain fixed in total but can increase or decrease on a per-unit basis.

Two specialized types of fixed costs are committed fixed costs and discretionary fixed costs. These classifications are generally used for long-range planning purposes and are covered in upper-level managerial accounting courses, so they are only briefly described here.

Committed fixed costs are fixed costs that typically cannot be eliminated if the company is going to continue to function. An example would be the lease of factory equipment for a production company.

Discretionary fixed costs generally are fixed costs that can be incurred during some periods and postponed during other periods but which cannot normally be eliminated permanently. Examples could include advertising campaigns and employee training. Both of these costs could potentially be postponed temporarily, but the company would probably incur negative effects if the costs were permanently eliminated. These classifications are generally used for long-range planning purposes.

In addition to understanding fixed costs, it is critical to understand variable costs, the second fundamental cost classification. A **variable cost** is one that varies in direct proportion to the level of activity within the business. Typical costs that are classified as variable costs are the cost of raw materials used to produce a product, labor applied directly to the production of the product, and overhead expenses that change based upon activity. For each variable cost, there is some activity that drives the variable cost up or down. A **cost driver** is defined as any activity that causes the organization to incur a variable cost. Examples of cost drivers are direct labor hours, machine hours, units produced, and units sold. Table 2.3 provides examples of variable costs and their associated cost drivers.

Variable Costs and Associated Cost Drivers

	Variable Cost	Cost Driver
Merchandising	Total monthly hourly wages for sales staff	Hours business is open during month
Manufacturing	Direct materials used to produce one unit of product	Number of units produced
Service	Cost of laundering linens and towels	Number of hotel rooms occupied

Table 2.3

Unlike fixed costs that remain fixed in total but change on a per-unit basis, variable costs remain the same per unit, but change in total relative to the level of activity in the business. Revisiting Tony's T-Shirts, Figure 2.16 shows how the variable cost of ink behaves as the level of activity changes.

Cost of Ink per T-Shirt	Number of T-Shirts Produced	Total Variable Cost of Ink
$0.15	2,000	$300
0.15	4,000	600
0.15	6,000	900

Figure 2.16 Variable Costs per Unit. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

As Figure 2.16 shows, the variable cost per unit (per T-shirt) does not change as the number of T-shirts produced increases or decreases. However, the variable costs change in total as the number of units produced increases or decreases. In short, total variable costs rise and fall as the level of activity (the cost driver) rises and falls.

Distinguishing between fixed and variable costs is critical because the **total cost** is the sum of all fixed costs (the **total fixed costs**) and all variable costs (the **total variable costs**). For every unit produced, every customer served, or every hotel room rented, for example, managers can determine their total costs both per unit of activity and in total by combining their fixed and variable costs together. The graphic in Figure 2.17 illustrates the concept of total costs.

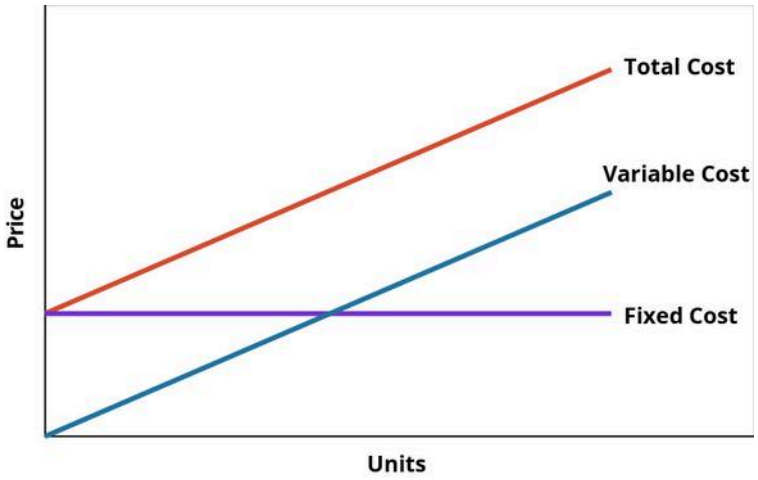


Figure 2.17 Total Cost as the Sum of Total Fixed Costs and Total Variable Costs. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Remember that the reason that organizations take the time and effort to classify costs as either fixed or variable is to be able to control costs. When they classify costs properly, managers can use cost data to make decisions and plan for the future of the business.

CONCEPTS IN PRACTICE

Boeing[5]

If you've ever flown on an airplane, there's a good chance you know Boeing. The Boeing Company generates around $90 billion each year from selling thousands of airplanes to commercial and military customers around the world. It employs around 200,000 people, and it's indirectly responsible for more

5 Attribution: Modification of work by Sharon Kioko and Justin Marlowe. "Cost Analysis." *Financial Strategy for Public Managers.* CC BY 4.0. https://press.rebus.community/financialstrategy/chapter/cost-analysis/

than a million jobs through its suppliers, contractors, regulators, and others. Its main assembly line in Everett, WA, is housed in the largest building in the world, a colossal facility that covers nearly a half-trillion cubic feet. Boeing is, simply put, a massive enterprise.

And yet, Boeing's managers know the exact cost of everything the company uses to produce its airplanes: every propeller, flap, seat belt, welder, computer programmer, and so forth. Moreover, they know how those costs would change if they produced more airplanes or fewer. They also know the price at which they sold each plane and the profit the company made on each sale. Boeing's executives expect their managers to know this information, in real time, if the company is to remain profitable.

Link between Business Decision and Cost Information Utilized

Decision	Cost Information
Discontinue a product line	Variable costs, overhead directly tied to product, potential reduction in fixed costs
Add second production shift	Labor costs, cost of fringe benefits, potential overhead increases (utilities, security personnel)
Open additional retail outlets	Fixed costs, variable operating costs, potential increases in administrative expenses at corporate headquarters

Table 2.4

Average Fixed Costs versus Average Variable Costs

Another way management may want to consider their costs is as average costs. Under this approach, managers can calculate both average fixed and average variable costs. **Average fixed cost (AFC)** is the total fixed costs divided by the total number of units produced, which results in a per-unit cost. The formula is:

$$\text{Average Fixed Cost (AFC)} = \frac{\text{Total Fixed Costs}}{\text{Total Number of Units Produced}}$$

To show how a company would use AFC to make business decisions, consider Carolina Yachts, a company that manufactures sportfishing boats that are sold to consumers through a network of marinas and boat dealerships. Carolina Yachts produces 625 boats per year, and their total annual fixed costs are $1,560,000. If they want to determine an average fixed cost per unit, they will find it using the formula for AFC:

$$\text{AFC} = \frac{\$1{,}560{,}000}{625} = \$2{,}496 \text{ per boat}$$

When they produce 625 boats, Carolina Yachts has an AFC of $2,496 per boat. What happens to the AFC if they increase or decrease the number of boats produced? Figure 2.18 shows the AFC for different numbers of boats.

Number of Boats Produced	Total Fixed Costs	Average Fixed Cost (per boat)
500	$1,560,000	$3,120
625	1,560,000	2,496
700	1,560,000	2,229

Figure 2.18 Average Fixed Costs for Carolina Yachts. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

We see that total fixed costs remain unchanged, but the average fixed cost per unit goes up and down with the number of boats produced. As more units are produced, the fixed costs are spread out over more units, making the fixed cost per unit fall. Likewise, as fewer boats are manufactured, the average fixed costs per unit rises. We can use a similar approach with variable costs.

Average variable cost (AVC) is the total variable costs divided by the total number of units produced, which results in a per-unit cost. Like ATC, we can use this formula:

$$\text{Average Variable Cost (AVC)} = \frac{\text{Total Variable Costs}}{\text{Total Number of Units Produced}}$$

To demonstrate AVC, let's return to Carolina Yachts, which incurs total variable costs of $6,875,000 when they produce 625 boats per year. They can express this as an average variable cost per unit:

$$\text{AVC} = \frac{\$6{,}875{,}000}{625} = \$11{,}000 \text{ per boat}$$

Because average variable costs are the average of all costs that change with production levels on a per-unit basis and include both direct materials and direct labor, managers often use AVC to determine if production should continue or not in the short run. As long as the price Carolina Yachts receives for their boats is greater than the per-unit AVC, they know that they are not only covering the variable cost of production, but each boat is making a contribution toward covering fixed costs. If, at any point, the average variable cost per boat rises to the point that the price no longer covers the AVC, Carolina Yachts may consider halting production until the variable costs fall again.

These changes in variable costs per unit could be caused by circumstances beyond their control, such as a shortage of raw materials or an increase in shipping costs due to high gas prices. In any case, average variable cost can be useful for managers to get a big picture look at their variable costs per unit.

LINK TO LEARNING

Watch the video from Khan Academy that uses the scenario of computer programming to teach fixed, variable, and marginal cost (https://openstax.org/l/50costing) to learn more.

Mixed Costs and Stepped Costs

Not all costs can be classified as purely fixed or purely variable. **Mixed costs** are those that have both a fixed and variable component. It is important, however, to be able to separate mixed costs into their fixed and variable components because, typically, in the short run, we can only change variable costs but not most fixed costs. To examine how these mixed costs actually work, consider the Ocean Breeze hotel.

The Ocean Breeze is located in a resort area where the county assesses an occupancy tax that has both a fixed and a variable component. Ocean Breeze pays $2,000 per month, regardless of the number of rooms rented. Even if it does not rent a single room during the month, Ocean Breeze still must remit this tax to the county. The hotel treats this $2,000 as a fixed cost. However, for every night that a room is rented, Ocean Breeze must remit an additional tax amount of $5.00 per room per night. As a result, the occupancy tax is a mixed cost.

Figure 2.19 further illustrates how this mixed cost behaves.

Number of Rooms Rented per Month (Cost Driver)	Fixed Cost Component ($2,000 per month)	Variable Cost Component ($5 per room)	Total Cost (Fixed + Variable)
0	$2,000	$ 0	$2,000
60	2,000	300	2,300
85	2,000	425	2,425
100	2,000	500	2,500

Figure 2.19 Mixed Costs Example for Ocean Breeze. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Notice that Ocean Breeze cannot control the fixed portion of this cost and that it remains fixed in total, regardless of the activity level. On the other hand, the variable component is fixed per unit, but changes in total based upon the level of activity. The fixed portion of this cost plus the variable portion of this cost combine to make the total cost. As a result, the formula for total cost looks like this:

$$Y = a + bx$$

where Y is the total mixed cost, a is the fixed cost, b is the variable cost per unit, and x is the level of activity.

Graphically, mixed costs can be explained as shown in Figure 2.20.

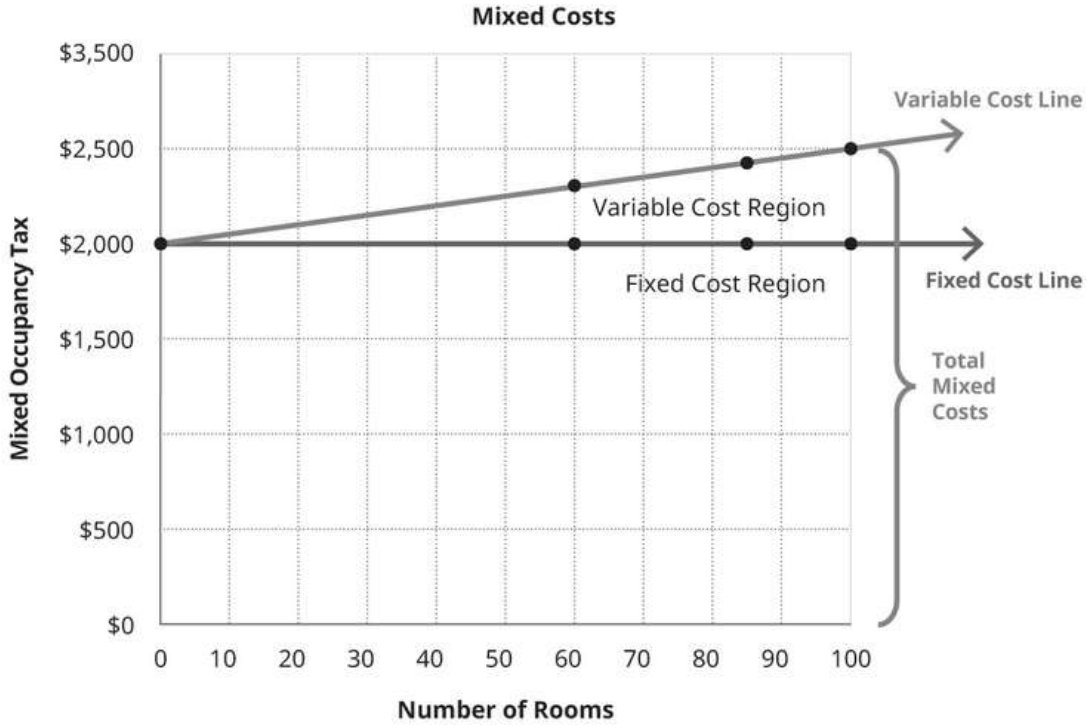


Figure 2.20 Ocean Breeze's Mixed Cost Graph. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The graph shows that mixed costs are typically both fixed and linear in nature. In other words, they will often have an initial cost, in Ocean Breeze's case, the $2,000 fixed component of the occupancy tax, and a variable component, the $5 per night occupancy tax. Note that the Ocean Breeze mixed cost graph starts at an initial $2,000 for the fixed component and then increases by $5 for each night their rooms are occupied.

Some costs behave less linearly. A cost that changes with the level of activity but is not linear is classified as a **stepped cost**. Step costs remain constant at a fixed amount over a range of activity. The range over which these costs remain unchanged (fixed) is referred to as the **relevant range**, which is defined as a specific activity level that is bounded by a minimum and maximum amount. Within this relevant range, managers can

predict revenue or cost levels. Then, at certain points, the step costs increase to a higher amount. Both fixed and variable costs can take on this stair-step behavior. For instance, wages often act as a stepped variable cost when employees are paid a flat salary and a commission or when the company pays overtime. Further, when additional machinery or equipment is placed into service, businesses will see their fixed costs stepped up. The "trigger" for a cost to step up is the relevant range. Graphically, step costs appear like stair steps (Figure 2.21).

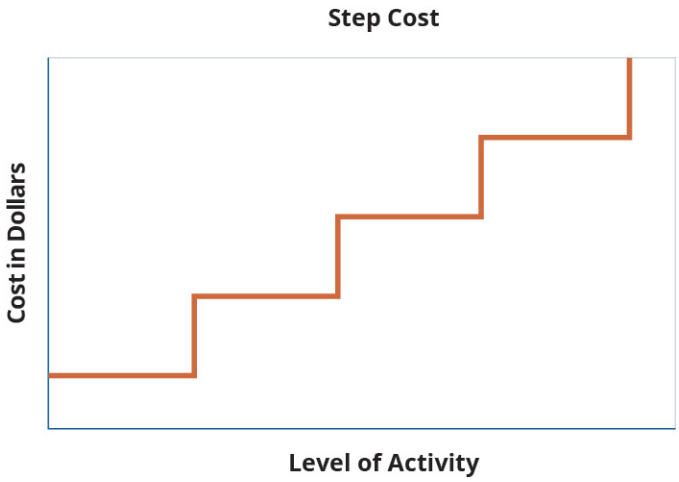


Figure 2.21 Step Cost Graph. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

For example, suppose a quality inspector can inspect a maximum of 80 units in a regular 8-hour shift and his salary is a fixed cost. Then the relevant range for QA inspection is from 0–80 units per shift. If demand for these units increases and more than 80 inspections are needed per shift, the relevant range has been exceeded and the business will have one of two choices:

(1) Pay the quality inspector overtime in order to have the additional units inspected. This overtime will "step up" the variable cost per unit. The advantage to handling the increased cost in this way is that when demand falls, the cost can quickly be "stepped down" again. Because these types of step costs can be adjusted quickly and often, they are often still treated as variable costs for planning purposes.

(2) "Step up" fixed costs. If the company hires a second quality inspector, they would be stepping up their fixed costs. In effect, they will double the relevant range to allow for a maximum of 160 inspections per shift, assuming the second QA inspector can inspect an additional 80 units per shift. The down side to this approach is that once the new QA inspector is hired, if demand falls again, the company will be incurring fixed costs that are unnecessary. For this reason, adding salaried personnel to address a short-term increase in demand is not a decision most businesses make.

Step costs are best explained in the context of a business experiencing increases in activity beyond the relevant range. As an example, let's return to Tony's T-Shirts.

Tony's cost of operations and the associated relevant ranges are shown in Table 2.5.

Tony's T-Shirts Cost Options

	Cost	Type of Cost	Relevant Range
Lease on Screen-Printing Machine	$2,000 per month	Fixed	0–2,000 T-shirts per month
Employee	$10 per hour	Variable	20 shirts per hour
Tony's Salary	$2,500 per month	Fixed	N/A
Screen-Printing Ink	$0.25 per shirt	Variable	N/A
Building Rent	$1,500 per month	Fixed	2 screen-printing machines and 2 employees

Table 2.5

As you can see, Tony has both fixed and variable costs associated with his business. His one screen-printing machine can only produce 2,000 T-shirts per month and his current employee can produce 20 shirts per hour (160 per 8-hour work day). The space that Tony leases is large enough that he could add an additional screen-printing machine and 1 additional employee. If he expands beyond that, he will need to lease a larger space, and presumably his rent would increase at that point. It is easy for Tony to predict his costs as long as he operates within the relevant ranges by applying the total cost equation $Y = a + bx$. So, for Tony, as long as he produces 2,000 or fewer T-shirts, his total cost will be found by $Y = \$6{,}000 + \$0.75x$, where the variable cost of $0.75 is the $0.25 cost of the ink per shirt and $0.50 per shirt for labor ($10 per hour wage/20 shirts per hour). As soon as his production passes the 2,000 T-shirts that his one employee and one machine can handle, he will have to add a second employee and lease a second screen-printing machine. In other words, his fixed costs will rise from $6,000 to $8,000, while his variable costs will increase in total because he has two workers now, his variable cost per T-shirt will stay the same (ink plus hourly rate to make one shirt). Thus, his new cost equation is $Y = \$8{,}000 + \$0.75x$ until he "steps up" again and adds a third machine *and* moves to a new location with a presumably higher rent. Let's take a look at this in chart form to better illustrate the "step" in cost Tony will experience as he steps past 2,000 T-shirts.

Tony's cost information is shown in the chart for volume between 500 and 4,000 shirts.

Number of T-Shirts	Total Cost (rounded)
500	$ 6,375
750	6,563
1,000	6,750
1,250	6,938
1,500	7,125
1,750	7,313
2,000	7,500
2,250	9,688
2,500	9,875
2,750	10,063
3,000	10,250
3,250	10,438
3,500	10,625
3,750	10,813

When presented graphically, notice what happens when Tony steps outside of his original relevant range and has to add a second employee and a second screen-printing machine:

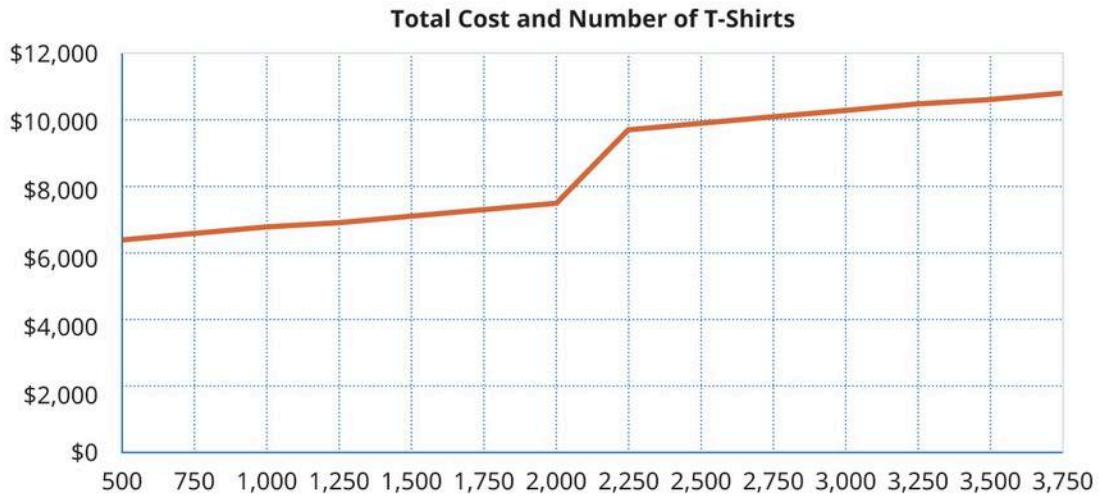


Figure 2.22 Stepped Variable Costs for Tony's T-Shirts. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

It is important to remember that even though Tony's costs stepped up when he exceeded his original capacity (relevant range), the *behavior* of the costs did not change. His fixed costs still remained fixed in total and his total variable cost rose as the number of T-shirts he produced rose. Table 2.6 summarizes how costs behave within their relevant ranges.

Summary of Fixed and Variable Cost Behaviors

Cost	In Total	Per Unit
Variable Cost	Changes in response to the level of activity	Remains fixed per unit regardless of the level of activity
Fixed Cost	Does not change with the level of activity, within the relevant range, but does change when the relevant range changes	Changes based upon activity within the relevant range: increased activity decreases per-unit cost; decreased activity increases per-unit cost

Table 2.6

Product versus Period Costs

Many businesses can make decisions by dividing their costs into fixed and variable costs, but there are some business decisions that require grouping costs differently. Sometimes companies need to consider how those costs are reported in the financial statements. At other times, companies group costs based on functions within the business. For example, a business would group administrative and selling expenses by the period (monthly or quarterly) so that they can be reported on an Income Statement. However, a manufacturing firm may carry product costs such as materials from one period to the other in order to have the costs "travel" with the units being produced. It is possible that both the selling and administrative costs and materials costs have both fixed and variable components. As a result, it may be necessary to analyze some fixed costs together with some variable costs. Ultimately, businesses strategically group costs in order to make them more useful for decision-making and planning. Two of the broadest and most common grouping of costs are product costs and period costs.

Product costs are all those associated with the acquisition or production of goods and products. When products are purchased for resale, the cost of goods is recorded as an asset on the company's balance sheet. It is not until the products are sold that they become an expense on the income statement. By moving product costs to the expense account for the cost of goods sold, they are easily matched to the sales revenue income account. For example, Bert's Bikes is a bicycle retailer who purchases bikes from several wholesale distributors and manufacturers. When Bert purchases bicycles for resale, he places the cost of the bikes into his inventory account, because that is what those bikes are—his inventory available for sale. It is not until someone purchases a bike that it creates sales revenue, and in order to fulfill the requirements of double-entry accounting, he must match that income with an expense: the cost of goods sold (Figure 2.23).

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold	25,000	
	Finished Goods Inventory		25,000
	To record the cost of products sold		

Figure 2.23 Journal Entry for Cost of Goods Sold. Product costs are collected in the finished goods inventory, where they remain until the goods are sold. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Some product costs have both a fixed and variable component. For example, Bert purchases 10 bikes for $100 each. The distributor charges $10 per bike for shipping for 1 to 10 bikes but $8 per bike for 11 to 20 bikes. This shipping cost is fixed per unit but varies in total. If Bert wants to save money and control his cost of goods sold, he can order an 11th bike and drop his shipping cost by $2 per bike. It is important for Bert to know what is fixed and what is variable so that he can control his costs as much as possible.

What about the costs Bert incurs that are not product costs? **Period costs** are simply all of the expenses that are not product costs, such as all selling and administrative expenses. It is important to remember that period costs are treated as expenses in the period in which they occur. In other words, they follow the rules of accrual accounting practice by recognizing the cost (expense) in the period in which they occur regardless of when the cash changes hands. For example, Bert pays his business insurance in January of each year. Bert's annual insurance premium is $10,800, which is $900 per month. Each month, Bert will recognize 1/12 of this insurance cost as an expense in the period in which it is incurred (Figure 2.24).

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Insurance Expense	900	
	Prepaid Insurance		900
	To recognize current period insurance expense		

Figure 2.24 Journal Entry for Insurance Expense. Bert applies 1/12 of the prepaid insurance premium per month to the expense account in order to match period costs with period revenues. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Why is it so important for Bert to know which costs are product costs and which are period costs? Bert may have little control over his product costs, but he maintains a great deal of control over many of his period costs. For this reason, it is important that Bert be able to identify his period costs and then determine which of them are fixed and which are variable. Remember that fixed costs are fixed over the relevant range, but variable costs change with the level of activity. If Bert wants to control his costs to make his bike business more profitable, he must be able to differentiate between the costs he can and cannot control.

Just like a merchandising business such as Bert's Bikes, manufacturers also classify their costs as either product costs or period costs. For a manufacturing business, product costs are the costs associated with making the product, and period costs are all other costs. For the purposes of external reporting, separating costs into period and product costs is not all that is necessary. However, for management decision-making activities, refinement of the types of product costs is helpful.

In a manufacturing firm, the need for management to be aware of the types of costs that make up the cost of a product is of paramount importance. Let's look at Carolina Yachts again and examine how they can classify the product costs associated with building their sportfishing boats. Just like automobiles, every year, Carolina Yachts makes changes to their boats, introducing new models to their product line. When the engineers begin to redesign boats for the next year, they must be careful not to make changes that would drive the selling price of their boats too high, making them less attractive to the customer. The engineers need to know exactly what the addition of another feature will do to the cost of production. It is not enough for them to get total product cost data; instead, they need specific information about the three classes of product costs: materials, labor, and overhead.

As you've learned, direct materials are the raw materials and component parts that are directly economically traceable to a unit of production.

Table 2.7 provides some examples of direct materials.

Examples of Direct Materials

Manufacturing Business	Product	Direct Materials
Bakery	Birthday cakes	Flour, sugar, eggs, milk
Automobile manufacturer	Cars	Glass, steel, tires, carpet
Furniture manufacturer	Recliners	Wood, fabric, cotton batting

Table 2.7

In each of the examples, managers are able to trace the cost of the materials directly to a specific unit (cake, car, or chair) produced. Since the amount of direct materials required will change based on the number of units produced, direct materials are almost always classified as a variable cost. They remain fixed per unit of production but change in total based on the level of activity within the business.

It takes more than materials for Carolina Yachts to build a boat. It requires the application of labor to the raw materials and component parts. You've also learned that direct labor is the work of the employees who are directly involved in the production of goods or services. In fact, for many industries, the largest cost incurred in the production process is labor. For Carolina Yachts, their direct labor would include the wages paid to the carpenters, painters, electricians, and welders who build the boats. Like direct materials, direct labor is typically treated as a variable cost because it varies with the level of activity. However, there are some companies that pay a flat weekly or monthly salary for production workers, and for these employees, their compensation could be classified as a fixed cost. For example, many auto mechanics are now paid a flat weekly or monthly salary.

While in the example Carolina Yachts is dependent upon direct labor, the production process for companies in many industries is moving from human labor to a more automated production process. For these companies, direct labor in these industries is becoming less significant. For an example, you can research the current production process for the automobile industry.

The third major classification of product costs for a manufacturing business is overhead. Manufacturing overhead (sometimes referred to as *factory overhead*) includes all of the costs that a manufacturing business incurs, other than the variable costs of direct materials and direct labor required to build products. These overhead costs are not directly attributable to a specific unit of production, but they are incurred to support the production of goods. Some of the items included in manufacturing overhead include supervisor salaries, depreciation on the factory, maintenance, insurance, and utilities. It is important to note that manufacturing overhead does not include any of the selling or administrative functions of a business. For Carolina Yachts, costs like the sales, marketing, CEO, and clerical staff salaries will not be included in the calculation of manufacturing overhead costs but will instead be allocated to selling and administrative expenses.

As you have learned, much of the power of managerial accounting is its ability to break costs down into the smallest possible trackable unit. This also applies to manufacturing overhead. In many cases, businesses have a need to further refine their overhead costs and will track indirect labor and indirect materials.

When labor costs are incurred but are not directly involved in the active conversion of materials into finished products, they are classified as **indirect labor** costs. For example, Carolina Yachts has production supervisors who oversee the manufacturing process but do not actively participate in the construction of the boats. Their wages generally support the production process but cannot be traced back to a single unit. For this reason, the production supervisors' salary would be classified as indirect labor. Similar to direct labor, on a product or department basis, indirect labor, such as the supervisor's salary, is often treated as a fixed cost, assuming that

it does not vary with the level of activity or number of units produced. However, if you are considering the supervisor's salary cost on a per unit of production basis, then it could be considered a variable cost.

Similarly, not all materials used in the production process can be traced back to a specific unit of production. When this is the case, they are classified as **indirect material** costs. Although needed to produce the product, these indirect material costs are not traceable to a *specific* unit of production. For Carolina Yachts, their indirect materials include supplies like tools, glue, wax, and cleaning supplies. These materials are required to build a boat, but management cannot easily track how much of a bottle of glue they use or how often they use a particular drill to build a specific boat. These indirect materials and their associated cost represent a small fraction of the total materials needed to complete a unit of production. Like direct materials, indirect materials are classified as a variable cost since they vary with the level of production. Table 2.8 provides some examples of manufacturing costs and their classifications.

Examples of Classifications of Manufacturing Costs

Cost	Classification	Fixed or Variable
Production supervisor salary	Indirect labor	Fixed
Raw materials used in production	Direct materials	Variable
Wages of production employees	Direct labor	Variable
Straight-line depreciation on factory equipment	General manufacturing overhead	Fixed
Glue and adhesives	Indirect materials	Variable

Table 2.8

Prime Costs versus Conversion Costs

In certain production environments, once a business has separated the costs of the product into direct materials, direct labor, and overhead, the costs can then be gathered into two broader categories: prime costs and conversion costs. **Prime costs** are the direct material expenses and direct labor costs, while **conversion costs** are direct labor and general factory overhead combined. Please note that these two categories of costs are examples of cost categories where a particular cost can be included in both. In this case, direct labor is included in both prime costs and conversion costs.

These cost classifications are common in businesses that produce large quantities of an item that is then packaged into smaller, sellable quantities such as soft drinks or cereal. In these types of production environments, it easier to lump the costs of direct labor and overhead into one category, since these costs are what are needed to convert raw materials into a finished product. This method of costing is termed *process costing* and is covered in Process Costing.

Although it seems as if there are many classifications or labels associated with costs, remember that the purpose of cost classification is to assist managers in the decision-making process. Since this type of data is not used for external reporting purposes, it is important to understand that (1) a single cost can have many different labels; (2) the terms are used independently, not simultaneously; and (3) each classification is important to understand in order to make business decisions. Figure 2.25 uses some example costs to demonstrate these principles.

Cost	Fixed	Variable	Mixed	Step	Period	Product			Prime	Conversion
						DM	DL	OH		
Rent on production facility	√						OH			√
Plant supervisor salary	√						OH			√
Raw materials		√					DM		√	
Administrative salaries	√				√					
Commissioned sales staff		√	√		√					
Delivery truck			√		√					
Advertising	√				√					
Plant utilities			√				OH			√
Income tax		√		√	√					

Figure 2.25 Classification Based on Cost Function. Costs can fall into more than one category, sometimes making the process of cost identification complex. DM, direct materials; DL, direct labor; OH, overhead. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Effects of Changes in Activity Level on Unit Costs and Total Costs

We have spent considerable time identifying and describing the various ways that businesses categorize costs. However, categorization itself is not enough. It is important not only to understand the categorization of costs but to understand the relationships between changes in activity levels and the changes in costs in total. It is worth repeating that when a cost is considered to be fixed, that cost is only fixed for the relevant range. Once the boundary of the relevant range has been reached or moved beyond, fixed costs will change and then remain fixed for the new relevant range. Remember that, within a relevant range of activity, where the relevant range refers to a specific activity level that is bounded by a minimum and maximum amount, total fixed costs are constant, but costs change on a per-unit basis. Let's examine an example that demonstrates how changes in activity can affect costs.

ETHICAL CONSIDERATIONS

Cost Accounting Helps Reduce Fraud and Promotes Ethical Behavior

Managerial and related cost accounting systems assist managers in making ethical and sound business decisions. Managerial accountants implement accounting reporting systems to minimize or prevent fraud and promote ethical decision-making. For example, tracking changes in costing activity and ensuring that activity remains in a relevant range, helps ensure that an organization's business activity is properly bounded within a reasonable range of expense. If the minimum or maximum expense range is exceeded, this can indicate that management is acting without authority or is pursuing unauthorized activities. Excessive costs may even be a red flag that possible fraud is occurring. Cost accounting helps ensure that financial costs are within an acceptable range and helps an organization make reliable forward-looking financial decisions.

Comprehensive Example of the Effect on Changes in Activity Level on Costs

Pat is planning a three-day ski trip on his spring break after he works on a Habitat for Humanity project in Dallas. The costs for the trip are as follows:

Car rental (up to five passengers)	$200
Condo rental (up to five occupants)	400
Gasoline	120
Food (per day)	40
Lift tickets (per day)	30

He is considering his costs for the trip if he goes alone, or if he takes one, two, three, or four friends. However, before he can begin his analysis, he needs to consider the characteristics of the costs. Some of the costs will stay the same no matter how many people go, and some of the costs will fluctuate, based on the number of participants.

Those costs that do not change are the fixed costs. Once you incur a fixed cost, it does not change within a given range. For example, Pat can take up to five people in one car, so the cost of the car is fixed for up to five people. However, if he took more friends, then he would need more cars. The condo rental and the gasoline expenses would also be considered fixed costs, because they are not going to change in the reference range.

The costs that do change as the number of participants change are the variable costs. The food and lift ticket expenses are examples of variable costs, since they fluctuate based upon the number of participants and the number of days of activities.

In analyzing the costs, Pat also needs to consider the total costs and average costs. The analysis will calculate the average fixed costs, the total fixed costs, the average variable costs, and the total variable costs.

In the analysis of total costs versus average costs, both total and average fixed costs will stay the same and total and average variable costs will change. Here are the total fixed costs:

Car rental	$200
Condo	400
Gasoline	120
Total fixed costs	$720

The total fixed costs for the trip will be $720.00, no matter whether Pat goes alone or takes up to 4 friends. However, the average fixed costs will be the total fixed costs divided by the number of participants. The average fixed cost could range from $720 (720/1) to $144 (720/5).

Here are the variable costs:

Food (per day)	$40
Lift tickets (per day)	30
Average variable cost (per day)	$70

The average variable cost will be $70.00 per person per day, no matter how many people go on the trip. However, the total variable costs will range from $70.00, if Pat goes alone, to $350.00, if five people go. Figure 2.26 shows the relationships of the various costs, based on the number of participants.

Number of Skiers	Average Variable Cost	Total Variable Cost	Total Fixed Cost	Average Fixed Cost	Average Cost per Skier
1	$210	$ 210	$720	$720	$930
2	210	420	720	360	570
3	210	630	720	240	450
4	210	840	720	180	390
5	210	1,050	720	144	354

Figure 2.26 Comprehensive Ski Trip Cost Classification. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Looking at this analysis, it is clear that, if there is an activity that you think that you cannot afford, it can become less expensive if you are creative in your cost-sharing techniques.

YOUR TURN

Spring Break Trip Planning

Margo is planning an 8-day spring break trip from Atlanta, Georgia, to Tampa, Florida, leaving on Sunday and returning the following Sunday. She has located a condominium on the beach and has put a deposit down on the unit. The rental company has a maximum occupancy for the condominium of seven adults. There is an amusement park that she plans to visit. She is going to use her parents' car, an SUV that can carry up to six people and their luggage. The SUV can travel an average of 20 miles per gallon, the total distance is approximately 1,250 miles (550 miles each way plus driving around Tampa every day), and the average price of gas is $3 per gallon. A season pass for an amusement park she wants to visit is $168 per person. Margo estimates spending $40 per day per person for food. She estimates the costs for the trip as follows:

Condo Rental	$1,400
Gasoline	188
Food (per person per day)	40
Amusement Park Season Pass (per person)	168

Now that she has cost estimates, she is trying to decide how many of her friends she wants to invite. Since the car can only seat six people, Marg made a list of five other girls to invite. Use her data to answer the following questions and fill out the cost table:

1. What are the total variable costs for the trip?
2. What are the average variable costs for the trip?
3. What are the total fixed costs for the trip?
4. What are the average fixed costs for the trip?
5. What are the average costs per person for the trip?
6. What would the trip cost Margo if she were to go alone?

Number of People, Including Margo	Total Variable Cost	Average Variable Cost	Total Fixed Cost	Average Fixed Cost	Average Cost per Person
1					
2					
3					
4					
5					
6					

7. What additional costs would be incurred if a seventh girl was invited on the trip? Would this be a wise decision (from a cost perspective)? Why or why not?
8. Which cost will *not* be affected if a seventh girl was invited on the trip?

Solution

Number of People, Including Margo	Total Variable Cost	Average Variable Cost	Total Fixed Cost	Average Fixed Cost	Average Cost per Person
1	$488	$488	$1,588	$1,588	$2,076
2	976	488	1,588	794	1,282
3	1,464	488	1,588	529	1,017
4	1,952	488	1,588	397	885
5	2,440	488	1,588	318	806
6	2,928	488	1,588	265	753

Answers will vary. All responses should recognize that there is no room in the car for the seventh girl and her luggage, although the condominium will accommodate the extra person. This means they will have to either find a larger vehicle and incur higher gas expenses or take a second car, which will at least double the fixed gas cost.

2.3 Estimate a Variable and Fixed Cost Equation and Predict Future Costs

Sometimes, a business will need to use cost estimation techniques, particularly in the case of mixed costs, so that they can separate the fixed and variable components, since only the variable components change in the short run. Estimation is also useful for using current data to predict the effects of future changes in production on total costs. Three estimation techniques that can be used include the scatter graph, the high-low method, and regression analysis. Here we will demonstrate the scatter graph and the high-low methods (you will learn the regression analysis technique in advanced managerial accounting courses.

Functions of Cost Equations

The cost equation is a linear equation that takes into consideration total fixed costs, the fixed component of mixed costs, and variable cost per unit. Cost equations can use past data to determine patterns of past costs that can then project future costs, or they can use estimated or expected future data to estimate future costs. Recall the mixed cost equation:

$$Y = a + bx$$

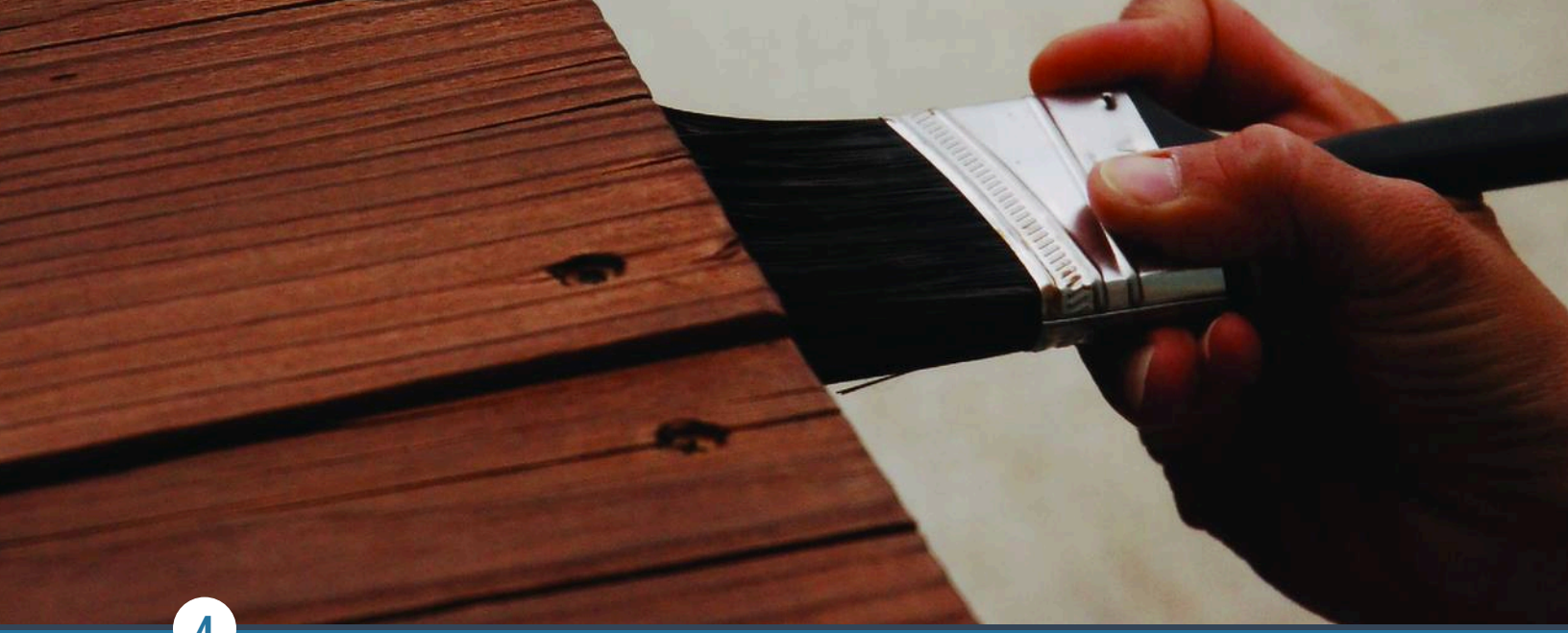
where Y is the total mixed cost, a is the fixed cost, b is the variable cost per unit, and x is the level of activity.

Let's take a more in-depth look at the cost equation by examining the costs incurred by Eagle Electronics in the manufacture of home security systems, as shown in Table 2.9.

Cost Information for Eagle Electronics

Cost Incurred	Fixed or Variable	Cost
Lease on manufacturing equipment	Fixed	$50,000 per year
Supervisor salary	Fixed	$75,000 per year

Table 2.9



4

Job Order Costing

Figure 4.1 Hallie Refinishing Furniture. Companies can generally choose from two systems—job order costing or process costing—to account for the costs involved in making a product. Job order costing is the optimal decision when costs are readily assigned to the individual product. Managers should consider their options to select the best accounting system for their company's production and pricing. (credit: modification of "120425-staining-wood-hand-brush" by r. nial bradshaw/Flickr, CC BY 2.0)

Chapter Outline

- LO **4.1** Distinguish between Job Order Costing and Process Costing
- LO **4.2** Describe and Identify the Three Major Components of Product Costs under Job Order Costing
- LO **4.3** Use the Job Order Costing Method to Trace the Flow of Product Costs through the Inventory Accounts
- LO **4.4** Compute a Predetermined Overhead Rate and Apply Overhead to Production
- LO **4.5** Compute the Cost of a Job Using Job Order Costing
- LO **4.6** Determine and Dispose of Underapplied or Overapplied Overhead
- LO **4.7** Prepare Journal Entries for a Job Order Cost System
- LO **4.8** Explain How a Job Order Cost System Applies to a Nonmanufacturing Environment

Why It Matters

Hallie graduated from college last year and moved to Tempe, Arizona, to begin her career. Before moving, she purchased a secondhand dresser for $35 and spent $25 on refinishing materials. After two hours of work, she posted a picture of the dresser on social media, and a friend offered her $100 to refinish another dresser exactly the same way.

Fortunately, Hallie understands cost accounting and knew she needed to calculate the cost to refinish another dresser. She found a similar dresser for $65. She knows that the refinishing materials will cost $25, and thus before adding in any cost for labor she is already at a cost of $90, without considering any overhead, such as electricity to run her sander.

Hallie estimated that her labor costs should be $20 per hour. The total cost then would be $130, and accepting less would mean accepting less for her labor. For a business in this situation, agreeing to the $100 offer would be considered a loss. If Hallie accepts the $100 price before checking her costs, she would have received only $10 for her labor (the sales price of $100 less the $90 cost of the dresser and materials).

Hallie didn't know if she would lose a potential customer by raising the price, so she found a different style dresser costing $25. A sales price of $100 would be fair with the two hours to refinish at $20 per hour and a materials cost of $25. She offered her friend the original style dresser for $130 or the alternate style dresser for $100.

As this example illustrates, it was essential for Hallie to know the cost to complete her project. It is also essential for all types and sizes of organizations to know the costs to complete their project. Manufacturing organizations need to know the costs of production, retail organizations need to know the cost to sell their products, and service organizations need to know the cost of providing their services. Management strives to eliminate unnecessary costs and needs to know the costs associated with using large pieces of equipment as well as seemingly insignificant office supplies. Cost accounting involves measuring and reporting the cost of production or service, while also providing data to determine the cost of the individual unit produced.

4.1 Distinguish between Job Order Costing and Process Costing

Pet Smart, H&R Block, Chili's, and Marshalls are popular chains often found at the same shopping center, even though they are very different businesses. Although they have a retail store, the Pet Smart Corporation also manufactures large volumes of its own products, whereas H&R Block prepares taxes for individual customers. Chili's prepares food, and its wait staff provides a service, whereas Marshalls sells a variety of products at retail.

The management of each business relies on knowing each cost when making decisions, such as setting the sales price, planning production and staffing schedules, and ordering materials. Although these companies share a common location, which suggests similar rental costs, all the other costs vary significantly. Because of these cost differences, each company must have a system for gathering its cost data. For example, Pet Smart manufactures Great Choice squeaker balls in large batches and collects cost data through a process costing system. A process costing system is often used to trace and determine production costs when similar products or services are provided. The concept and mechanics of a process costing system are addressed in Process Costing.

Since a typical tax return can vary significantly from one taxpayer to the next, H&R Block provides a service that they customize for each customer. Its cost data are collected via a job order cost system, which is designed to allow for individualized products or services.

Marshalls does not produce a product yet still needs a system to assign overhead costs to the products it sells. (Overhead was addressed in Building Blocks of Managerial Accounting.) And while Chili's has the same nationwide menu, it needs a system to collect the costs for each menu item within each location.

While companies may choose different cost accounting systems, each system must be capable of accumulating the costs incurred and allocating the costs to the product. Each costing system also requires the ability to obtain and analyze the cost data, and the more detailed the information needed, the higher the cost of collecting the data. The choice of cost accumulation system depends on the variety and type of products or services sold, or the type of manufacturing processes employed. The system used should be determined by weighing the cost of collecting the data and the benefit of having that information.

Companies use different costing systems for determining the cost of custom products than they do for determining the cost of mass-produced products. When products are custom ordered, knowing the cost of the materials, labor, and overhead is critical to determining the sales price. As an easy example, think of a tailor who alters, repairs, and makes custom clothes for customers. If a customer orders a custom-made suit, the specific fabric, detail of any special features, and the time involved in sewing are all factors that will determine the total cost and, therefore, the selling price of the garment. Each component of the cost of producing the clothing will be tracked as it occurs, thus improving the accuracy of determining the price. However, in mass production, wherein one batch leads to a second batch, stopping the process to properly identify the materials, labor, and overhead costs used for each batch does not provide enough valuable data to justify

determining the individual costing of each product. For example, in the case of a mass-produced clothing item, such as jeans, a company like Levi's will track costs for a batch of jeans rather than for a pair of jeans. Levi's had over $4.9 billion in revenue in 2017 generated from the many different styles and brands of clothing items they produce and sell. It would be difficult, and not cost effective, to track the cost of each individual clothing item; rather, it is more efficient to track the costs in each phase of the clothing-making process. Levi's can then accumulate the costs of the phases of production to determine the total cost of production for a batch and allocate those costs over the number of pairs of jeans made. This process allows them to determine the cost of each item.

Even retail companies need to know the cost of the purchased products before the sales price is set. While it seems simple to think of the sales price as the purchase price plus a markup, determining the markup costs needs to be an accurate process in order to ensure the sale price is higher than the product cost. To properly capture the information necessary for decision-making, there are different costing systems that track costs in order to determine sales prices, and to measure profits and manufacturing efficiency.

As previously mentioned, the two traditional types of costing systems are job order costing and process costing. Each anticipates or determines unit costs of products being manufactured and/or services being provided prior to year-end. Companies may decide to use only one or a combination of methods. This chapter examines job order costing and demonstrates how it differs from process costing. Process Costing and other costing systems (Activity-Based, Variable, and Absorption Costing) are covered in other chapters.

In this chapter, you will also learn the terminology used to track costs within the job order cost system and how to segregate and aggregate these costs to determine the costs of production in a job order costing environment. You will also learn how to record these job costs and where they appear on financial statements.

Job Order Costing versus Process Costing

Job order costing is an accounting system that traces the individual costs directly to a final job or service, instead of to the production department. It is used when goods are made to order or when individual costs are easy to trace to individual jobs, assuming that the additional information provides value. In these circumstances, the individual costs are easy to trace to the individual jobs.

For example, assume that a homeowner wants to have a custom deck added to her home. Also assume that in order to fit her lot's topography and her anticipated uses for the addition, she needs a uniquely designed deck. Her contractor will design the deck, price the necessary components (in this case, the direct materials, direct labor, and overhead), and construct it.

The final cost will be unique to this project. If another homeowner wanted the contractor to construct a deck, the contractor would go through the same design and pricing process, and you would expect that the design and costs would not be the same as those of the deck in the first example, since the decks would differ from one another.

The job order costing method also works well for companies such as movie production companies, print service providers, advertising agencies, building contractors, accounting firms, consulting entities, and repair service providers. For example, *Star Wars: The Last Jedi* is believed to have cost $200 million to produce, whereas *Logan* only cost $97 million. The production processes for both films differed significantly, so that the accumulated costs for each job also differed significantly. Both were made in 2017.[1]

In contrast, process costing is used when the manufacturing process is continuous, so it is difficult to establish how much of each material is used and exactly how much time is invested in each unit of finished product. Therefore, in process costing, costs are accounted for by the production process or production department instead of by the product or by the job. This method works well for manufacturers of products such as Titleist golf balls, Kellogg's cereal, Turkey Hill ice cream, CITGO gasoline, Dow Chemicals, or Sherwin Williams' paints.

1 "Production Costs and Global Box Office Revenue of Star Wars Movies from 1977 to 2018 (in million U.S. dollars)." The Statistics Portal. https://www.statista.com/statistics/311356/star-wars-production-costs-box-office-revenue/.

However, process costing is not limited to basic manufacturing activities: It can also be used in the manufacturing of more complex items, such as small engines. A process costing system assigns costs to each department as the costs are incurred, and the costs to produce one unit are calculated based on the information from the production department. Unit costs are determined after total production costs are determined.

One factor that can complicate the choice between job order costing and process costing is the growth of automation in the production process, which typically is accompanied by a reduction in direct labor. The cost of the increase in equipment (typically reflected as a depreciation expense) is allocated to overhead, while the decreased need for labor usually reduces the direct labor cost. Because of these issues, some companies choose a hybrid system, using process costing to account for mass producing a part and using job order costing to account for assembling some of those individual parts into a custom product. Table 4.1 summarizes the use of these two systems.

Job Order and Process Cost Systems

	Job Order Cost System	Process Cost System
Product type	Custom order	Mass production
Examples	Signs, buildings, tax returns	Folding tables, toys, buffet restaurants
Cost accumulation	Job lot	Accumulated per process
Work in process inventory	Individual job cost sheets	Separate work in process inventory department
Record keeping	Individual job cost sheets	Production cost report

Table 4.1 This table shows some of the differences between job order costing and process costing.

To illustrate how a company can determine whether to use job order costing or process costing, consider the cost accounting options for a local restaurant. Macs & Cheese makes specialty macaroni and cheese, and the company wants to erect a special sign on an already constructed billboard outside a stadium. It wants to use this space to target stadium customers; thus, the company wants a sign built specifically for that site. Dinosaur Vinyl is secured as the sign manufacturer and would use job order costing to account for the associated manufacturing costs because of the unique nature of the sign, including the art work involved. However, if Macs & Cheese was designing a costing system for the specialty food product they market, they typically would use a process costing approach because their product is made and marketed in homogeneous, similar batches.

LINK TO LEARNING

Both job order and process costing are used to track costs with a goal of improved cost measurement efficiency. Bar codes can help with either type of costing. Verified Label, Print & Promotions, Inc. provides a list of how barcodes contribute to efficiency in cost tracking (https://openstax.org/l/50barcode) for more information.

Organization of Flow of Goods through Production

Regardless of the costing method used (job order costing, process costing, or another method), manufacturing companies are generally similar in their organizational structure and have a similar flow of goods through production. The diagram in Figure 4.2 shows a partial organizational chart for sign manufacturer Dinosaur Vinyl. The CEO has several direct reporting units—Financing, Production, Information Technology, Marketing, Human Resources, and Maintenance—each with a director responsible for several departments.

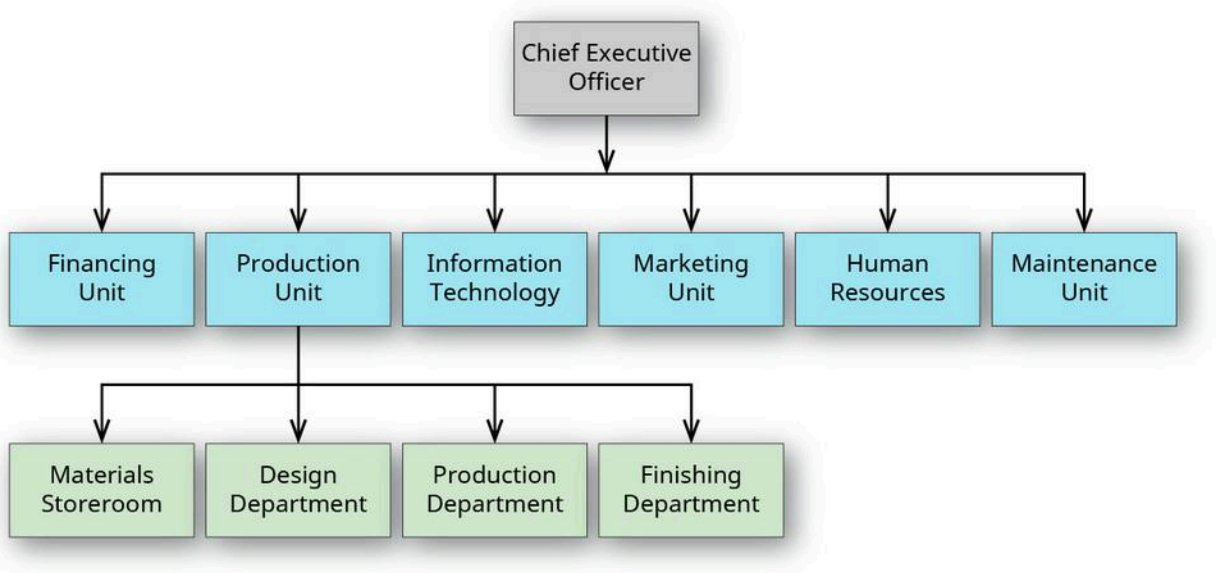


Figure 4.2 Organizational Chart for a Manufacturing Company. The different units within Dinosaur Vinyl illustrate the two main cost categories of a manufacturing company: manufacturing costs (the production unit), and selling and administrative costs. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The diagram also shows the departments that report to the production unit director and gives an indication as to the flow of goods through production. The flow of goods through production is more evident in Figure 4.3, which depicts Dinosaur Vinyl as a simple factory with three stages of production.

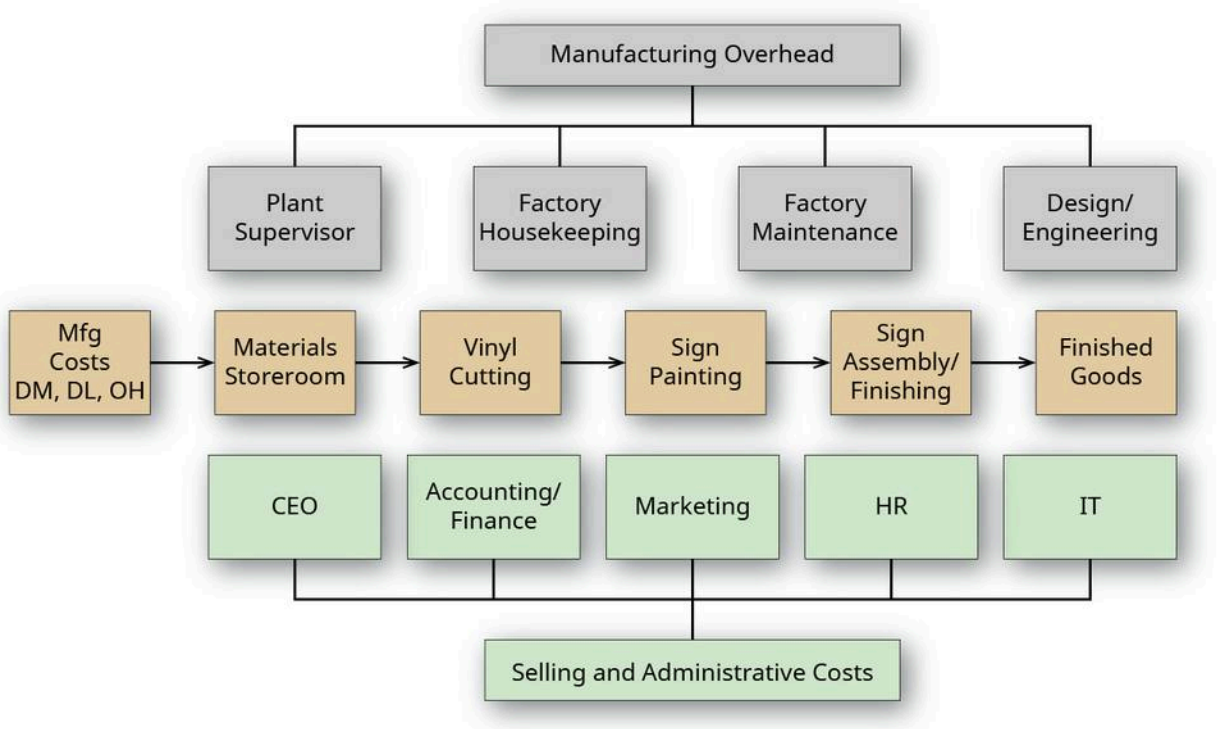


Figure 4.3 Factory Layout for Dinosaur Vinyl. The flow of goods and areas of manufacturing versus administrative costs are more easily seen by looking at the factory layout for Dinosaur Vinyl. The departments across the bottom represent administrative costs, whereas the departments in the middle represent manufacturing costs—although Design/Engineering can sometimes be considered part of administrative costs, depending on how management chooses to categorize those costs. The departments listed across the top represent examples of manufacturing overhead. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Raw materials are stored in the materials storeroom and delivered to the appropriate production department—cutting, painting, or assembly/finishing. The design department uses direct labor to create the design specifications, and, when completed, it sends them to the production department. The production department uses the material and design specifications and adds additional labor to create the sign. The sign is transferred to the finishing department for final materials and labor, before the sign is installed or delivered to the customer.

Manufacturing Costs

In a manufacturing environment, the **manufacturing costs** are also called *product costs* and include all expenses used to manufacture the product: direct materials, direct labor, and manufacturing overhead. To review these costs, see Building Blocks of Managerial Accounting. The total of these costs becomes the cost of ending inventory and later becomes the cost of goods sold when the product is sold. Both job order costing and process costing use categorized cost information to make decisions and evaluate the effectiveness of the cost tracking process. Because of the difference in how each of the two costing systems track costs, different terminology is used. Thus, it is important to separate product costs from period costs, and it is sometimes important to separate product costs into **prime costs** and **conversion costs**. Prime costs are costs that include the primary (or direct) product costs: direct materials and direct labor. Conversion costs are costs that include the expenses necessary to convert direct materials into a finished product: direct labor and manufacturing overhead. Their relationship is shown in Figure 4.4.

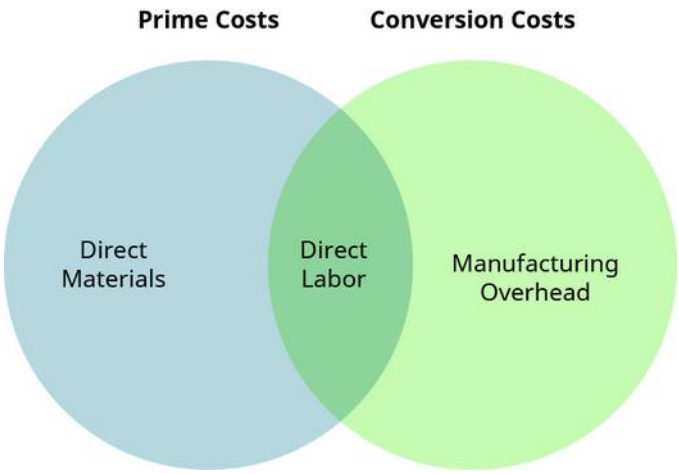


Figure 4.4 Manufacturing Costs. Management sometimes needs additional information to make decisions and needs the costs categorized as prime costs or conversion costs. Prime costs and conversion costs are not included together as direct labor is included in both categories. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Job order costing systems assign costs directly to the product by assigning direct materials and direct labor to the work in process (WIP) inventory. As you learned in Building Blocks of Managerial Accounting, direct materials are the components that can be directly traced to the products produced, whereas direct labor is the labor cost that can be directly traced to the products produced.

Material and labor costs that cannot be traced directly to the product produced are included in the overhead costs that are allocated in the production costing process. Overhead is applied to each product based on an activity base, which will be explained in Compute a Predetermined Overhead Rate and Apply Overhead to Production.

The assignment of direct materials and direct labor to each production unit illustrates the job order costing system's focus on prime costs, in contrast to the process costing system, which assigns costs to the department and focuses on direct materials and conversion costs, which are composed of a combination of direct labor and overhead. Process costs will be demonstrated in Process Costing.

Selling and Administrative Costs

Selling and administrative costs (S&A) are period costs, and these costs are expensed as incurred, instead of being included in the product's costs, as they move through the relevant inventory accounts. A **period cost** is a cost tied to a specific time period, such as a month, quarter, or year, instead of being associated with a particular job order. For example, if a company paid an insurance company $12,000 for one year's liability insurance coverage, the first month's expense would be $1,000. This expense would not be related to a particular job order, but instead would be classified as a period cost, and in this case recorded monthly as an administrative expense. Selling costs are the expenses related to the promotion and sale of the company's products, whereas administrative costs are the expenses related to the operations of the company. The S&A costs are considered period costs because they include costs of departments not directly associated with manufacturing but necessary to operate the business. Some examples include research and development costs, marketing costs, sales commissions, administration building rent, the CEO's salary expense, and accounting, payroll, and IT department expenses.

YOUR TURN

Maria's Market

A grocery store's analysis of a recent customer survey finds an increasing number of customers interested in being able to custom-order meals to go. Maria sees this as an opportunity to enter a niche market for

busy families or individuals who want home-cooked meals with a variety of options and combinations, but who have little time. Maria already has an expansive deli, bakery, and prepared foods section in the store and sees this opportunity as a viable option to increase sales and its customer base. With meals to go, customers can choose from an array of options and can indicate the quantity of each item and the time of pickup. The customer simply pulls up in a designated spot at Maria's and the food is brought to their car, packaged, and ready to take home to enjoy.

What type of costing system will work best for the Maria's Market? What sales price information, cost information, and other options are important to this decision?

Solution

A job order cost system will work well for this store. In addition to specific price and cost, these are other important considerations.

- The optimal sales price should be set to encourage customers to purchase the meals.
- The materials, labor, and overhead cost should be considered for each meal option.
  - Direct material costs may include the cost of the protein, grain, and vegetable option, as well as the cost of the packing containers.
  - The direct labor cost is for employees who are directly involved in preparing the meals.
  - Manufacturing overhead includes the cost of gloves used when preparing the meals, the cost of employees who support but are not directly involved in preparing the meals, and the cost to operate the oven.
- The cost of the various meal options should all be less than the sales price.
- The meal options should change to take advantage of seasonal items.
- There may be a need to vary the sales price, depending on the combinations selected.

Recording Costs in Job Order Costing versus Process Costing

Both job order costing and process costing track the costs of materials, labor, and overhead as components of virtually all products. The process of production does not change because of the costing method: The costing method is chosen based on the process of production and is intended to provide the most accurate representation of the costs incurred in the production process.

Maintaining accounting records for each system has its advantages. A job order costing system uses a **job cost sheet** to keep track of individual jobs and the direct materials, direct labor, and overhead associated with each job. The focus of a job order costing system is tracking costs per job, since each job is unique and therefore has different costs relative to other jobs. Maintaining this information is typically more expensive than process costing, and it is often used for the production of smaller, more individualized jobs because the benefit of knowing the cost of each product outweighs the additional cost of maintaining a job order costing system.

In contrast, a process costing system does not need to maintain the cost for individual jobs because the jobs use a continual system of production, and the items are typically not significantly unique but instead are basically equivalent. The accounting emphasis is in keeping records for the individual departments, which is useful for large batches or runs. Process costing is the optimal system to use when the production process is continuous and when it is difficult to trace a particular input cost to an individual product. Process costing systems assign costs to each department as the costs are incurred. The costs to produce one unit are calculated, based on the information from the production department. Therefore, the focus of process costing systems is on measuring and assigning the conversion costs to the proper department in order to best determine the cost of individual units.

Under either costing method, accounting theory explains why it is important to understand when costs

become expenses. A primary reason for separating production costs from other company expenses is the **expense recognition principle**, which requires costs to be expensed when they match the revenue being earned and to separate the costs of production from other costs for the proper timing of recognition of expenses. Think about measuring the profit from the sale of an item, say a TV, in a nonmanufacturing environment. It is logical to subtract the costs associated with buying the TV in order to determine the profit, before applying other costs from that sale. Suppose the TV was purchased as inventory by the store in January and sold to a customer in March. This requires that the cost of the TV not be recorded as an expense (cost of goods sold) until March, when the sale from the TV is recorded, thus matching the revenue with the expense. Until that time, the TV and its cost are considered inventory. This same idea applies to the manufacturing process.

Per the expense recognition principle, product costs—the direct materials, direct labor, and manufacturing overhead incurred to produce the job—are expensed on the income statement for the period of the sale as cost of goods sold when the completed job is sold. If the products are not sold, their costs remain in ending inventory.

Prior to the sale of the product, separating production costs and assigning them to the product results in these costs remaining with the inventory. Until they are sold, the costs incurred are reflected in an assortment of inventory accounts, such as raw materials inventory, work in process inventory, and finished goods inventory.

In contrast, period costs are not directly related to the production process and are expensed during the period in which they are incurred. This approach matches administrative and other expenses shown on the income statement in the same period in which the company earns income.

4.2 Describe and Identify the Three Major Components of Product Costs under Job Order Costing

In order to set an appropriate sales price for a product, companies need to know how much it costs to produce an item. Just as a company provides financial statement information to external stakeholders for decision-making, they must provide costing information to internal managerial decision makers. Virtually every tangible product has direct materials, direct labor, and overhead costs that can include indirect materials and indirect labor, along with other costs, such as utilities and depreciation on production equipment. To account for these and inform managers making decisions, the costs are tracked in a cost accounting system.

While the flow of costs is generally the same for all costing systems, the difference is in the details: Product costs have material, labor, and overhead costs, which may be assessed differently. In most production facilities, the raw materials are moved from the raw materials inventory into the work in process inventory. The work in process involves one or more production departments and is where labor and overhead convert the raw materials into finished goods. The movement of these costs through the work in process inventory is shown in Figure 4.5.

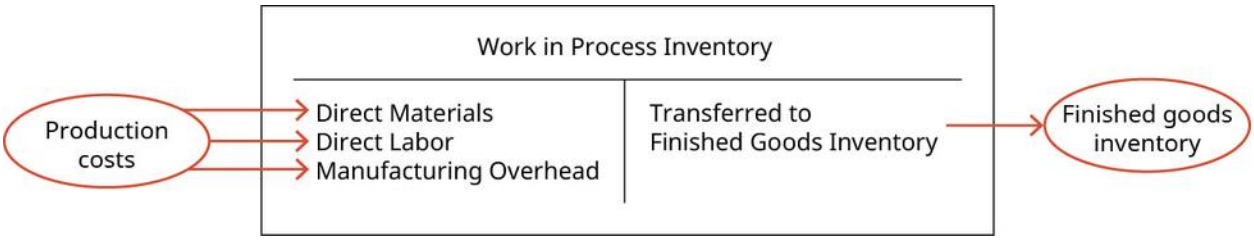


Figure 4.5 Work in Process Inventory. Direct materials, direct labor, and manufacturing overhead enter the work in process inventory as the costs associated with the products that are in production. Once the products are completed, their costs are transferred to the finished goods inventory. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

At this stage, the completed products are transferred into the finished goods inventory account. When the product is sold, the costs move from the finished goods inventory into the cost of goods sold.

While many types of production processes could be demonstrated, let's consider an example in which a contractor is building a home for a client. The accounting system will track direct materials, such as lumber, and direct labor, such as the wages paid to the carpenters constructing the home. Along with these direct materials and labor, the project will incur manufacturing overhead costs, such as indirect materials, indirect labor, and other miscellaneous overhead costs. Samples of these costs include indirect materials, such as nails, indirect labor, such as the supervisor's salary, assuming that the supervisor is overseeing several projects at the same time, and miscellaneous overhead costs such as depreciation on the equipment used in the construction project.

As direct materials, direct labor, and overhead are introduced into the production process, they become part of the work in process inventory value. When the home is completed, the accumulated costs become part of the finished goods inventory value, and when the home is sold, the finished goods value of the home becomes the cost of goods sold. Figure 4.6 illustrates the flow of these costs through production.

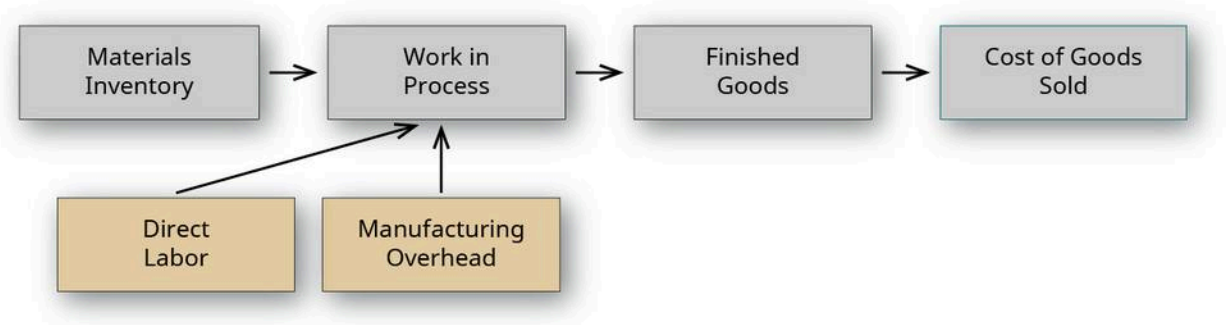


Figure 4.6 Flow of Materials from Raw Materials to Finished Goods. Accounting methods track a product's material, labor, and overhead costs, as it moves through production. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The three general categories of costs included in manufacturing processes are direct materials, direct labor, and overhead. Note that there are a few exceptions, since some service industries do not have direct material costs, and some automated manufacturing companies do not have direct labor costs. For example, a tax accountant could use a job order costing system during tax season to trace costs. The one major difference between the home builder example and this one is that the tax accountant will not have direct material costs to track. The few assets used will typically be categorized as overhead.

A benefit of knowing the production costs for each job in a job order costing system is the ability to set appropriate sales prices based on all the production costs, including direct materials, direct labor, and overhead. The unique nature of the products manufactured in a job order costing system makes setting a price even more difficult. For each job, management typically wants to set the price higher than its production cost. Even if management is willing to price the product as a loss leader, they still need to know how much money will be lost on each product. To achieve this, management needs an accounting system that can accurately assign and document the costs for each product.

If you're not familiar with the concept of a loss leader, a simple example might help clarify the concept. A **loss leader** is a product that is sold at a price that is often less than the cost of producing it in order to entice you to buy accessories that are necessary for its use. For example, you might pay $50 or $60 for a printer (for which the producer probably does not make any profit) in order to then sell you extremely expensive printer cartridges that only print a few pages before they have to be replaced. However, even pricing a product as a loss leader requires analysis of the three categories of costs: direct materials, direct labor, and overhead.

Direct Materials

Direct materials are those materials that can be directly traced to the manufacturing of the product. Some examples of direct materials for different industries are shown in Table 4.2. In order to respond quickly to production needs, companies need raw materials inventory on hand. While production volume might change,

management does not want to stop production to wait for raw materials to be delivered. Further, a company needs raw materials on hand for future jobs as well as for the current job. The materials are sent to the production department as it is needed for production of the products.

Common Direct Materials by Industry

Industry	Direct Materials
Automotive	Iron, aluminum, glass, rubber
Cell phones	Glass, various metals, plastic
Furniture	Wood, leather, vinyl
Jewelry	Gold, silver, diamonds, rubies
Pharmaceuticals	Natural or synthetic biological ingredients

Table 4.2

Each job begins when raw materials are put into the work in process inventory. When the materials are requested for production, a materials requisition slip is completed and shows the exact items and quantity requested, along with the associated cost. The completed form is signed by the requestor and approved by the manager responsible for the budget.

Returning to the example of Dinosaur Vinyl's order for Macs & Cheese's stadium sign, Figure 4.7 shows the **materials requisition form** for Job MAC001. This form indicates the quantity and specific items to be put into the work in process. It also transfers the cost of those items to the work in process inventory and decreases the raw materials inventory by the same amount. The raw materials inventory department maintains a copy to document the change in inventory levels, and the accounting department maintains a copy to properly assign the costs to the particular job.

MATERIALS REQUISITION FORM
Dinosaur Vinyl, Inc.

Material Requisition No.: 3392
Job No.: MAC001
Date of Request: 4/5/2017
Date Needed: 4/5/2017

Description	Quantity	Unit Cost	Total Cost
Raw materials inventory: Vinyl	1	$300	$300
Raw materials inventory: Black ink	2	50	100
Raw materials inventory: Red ink	1	60	60
Raw materials inventory: Gold ink	1	60	60
Raw materials inventory: Grommets	12	10	120
Raw materials inventory: Framing wood	40	1.50	60
			$700

Requested by: john ming Date: 4/5/17

Authorized by: Isla Clark Date: 4/5/17

Figure 4.7 Materials Requisition Form for Job MAC001. The materials requisition form allows different departments to track and account for the direct materials needed to manufacture the product. (attribution: Copyright Rice University, OpenStax, under CC BY-

NC-SA 4.0 license)

Dinosaur Vinyl has a beginning inventory of $1,000 in *raw materials: vinyl*, and $300 in each of its ink inventories: *raw materials: black ink*, *raw materials: red ink*, and *raw materials: gold ink*. In order to have enough inventory on hand for all of its jobs, it purchases $10,000 in vinyl and $500 in black ink. The T-accounts in Figure 4.8 show the stated beginning debit balances. An additional $10,000 of vinyl and $500 of black ink were then purchased for anticipated use, providing the demonstrated final account balances. The red ink and gold ink balances did not change, since no additional quantities were purchased.

The beginning balances and purchases in each of these accounts are illustrated in Figure 4.8.

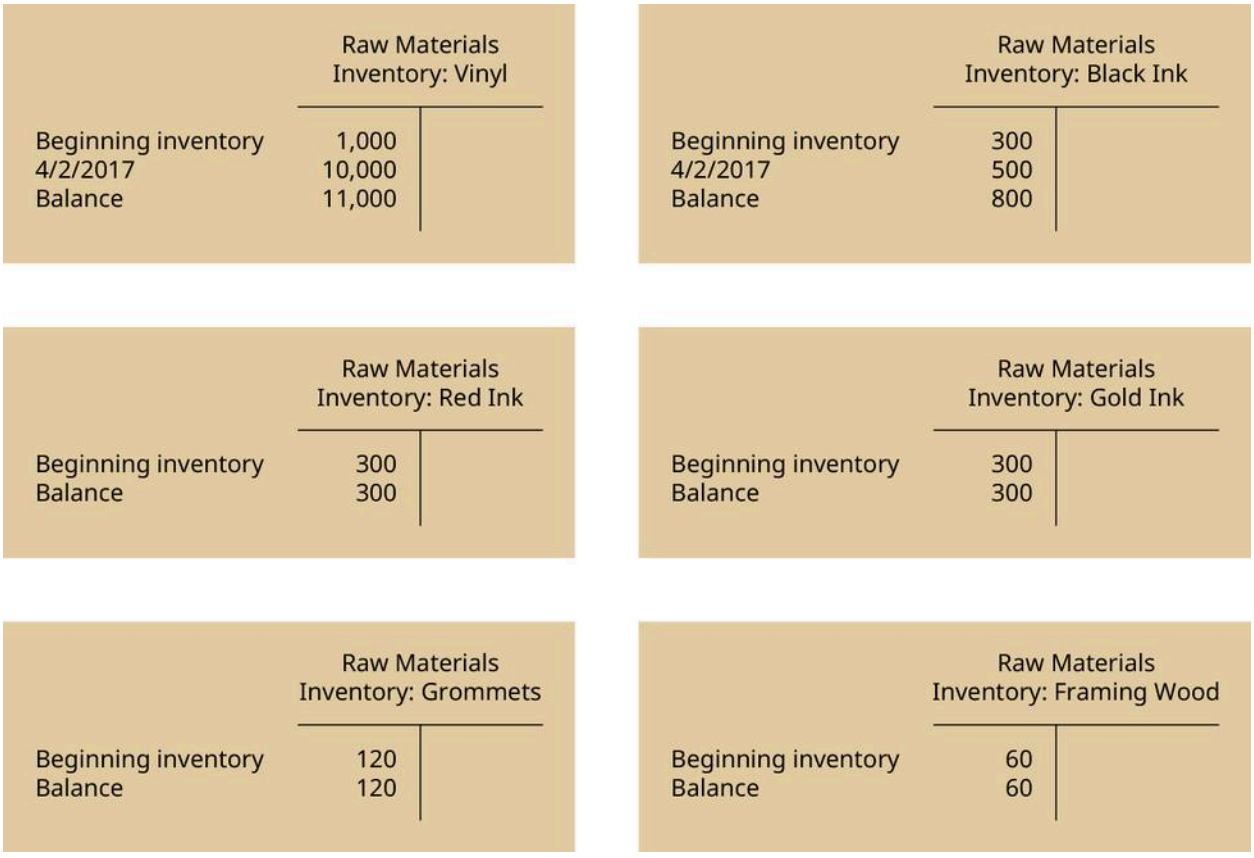


Figure 4.8 Beginning Balances and Purchases. These T-accounts show the balances for the raw materials inventory. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Traditional billboards with the design printed on vinyl include direct materials of vinyl and printing ink, plus the framing materials, which consist of wood and grommets. The typical billboard sign is 14 feet high by 48 feet wide, and Dinosaur Vinyl incurs a vinyl cost of $300 per billboard. The price for the ink varies by color. For this job, Dinosaur Vinyl needs two units of black ink at a cost of $50 each, one unit of red ink and one unit of gold ink at a cost of $60 each, twelve grommets at a cost of $10 each, and forty units of wood at a cost of $1.50 per unit. The total cost of direct materials is $700, as shown in Figure 4.9.

Item	Units	Cost per Unit	Item Cost	Total Cost
Production Department				
Vinyl	1	$ 300	$300	
Black ink	2	50	100	
Red ink	1	60	60	
Gold ink	1	60	60	$520
Finishing Department				
Grommets	12	$ 10	$120	
Framing wood	40	1.50	60	180
Total Direct Materials				$700

Figure 4.9 Direct Materials Needed for Job MAC001. The costs for direct materials needed by both the production and finishing departments are shown. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Some items are more difficult to measure per unit, such as adhesives and other materials not directly traceable to the final product. Their costs are assigned to the product as part of manufacturing overhead as indirect materials.

When Dinosaur Vinyl requests materials to complete Job MAC001, the materials are moved from raw materials inventory to work in process inventory. We will use the beginning inventory balances in the accounts that were provided earlier in the example. The requisition is recorded on the job cost sheet along with the cost of the materials transferred. The costs assigned to job MAC001 are $300 in vinyl, $100 in black ink, $60 in red ink, and $60 in gold ink. During the finishing stages, $120 in grommets and $60 in wood are requisitioned and put into work in process inventory. The costs are tracked from the materials requisition form to the work in process inventory and noted specifically as part of Job MAC001 on the preceding job order cost sheet. The movement of goods is illustrated in Figure 4.10.

Raw Materials Inventory: Vinyl		
Beginning inventory	1,000	
4/2/2017	10,000	
4/2/2017		300
Balance	10,700	

Raw Materials Inventory: Grommets		
Beginning inventory	300	
4/14/2017		120
Balance	180	

Raw Materials Inventory: Black Ink		
Beginning inventory	300	
4/2/2017	500	
4/2/2017		100
Balance	700	

Raw Materials Inventory: Framing Wood		
Beginning inventory	300	
4/14/2017		60
Balance	240	

Raw Materials Inventory: Red Ink		
Beginning inventory	300	
4/2/2017		60
Balance	240	

Work in Process Inventory		
4/2/2017	520	
4/14/2017	180	

Raw Materials Inventory: Gold Ink		
Beginning inventory	300	
4/2/2017		60
Balance	240	

Figure 4.10 Movement of Goods. These T-accounts illustrate the tracking of costs from raw materials inventory to the work in process inventory as the product moves through the manufacturing process. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Each of the T-accounts traces the movement of the raw materials from inventory to work in process. The vinyl and ink were used first to print the billboard, and then the billboard went to the finishing department for the grommets and frame, which were moved to work in process after the vinyl and ink. The final T-account shows the total cost for the raw materials placed into work in process on April 2 (vinyl and ink) and on April 14 (grommets and wood). The journal entries to reflect the flow of costs from raw materials to work in process to finished goods are provided in the section describing how to Prepare Journal Entries for a Job Order Cost System.

Direct Labor

Direct labor is the total cost of wages, payroll taxes, payroll benefits, and similar expenses for the individuals who work directly on manufacturing a particular product. The direct labor costs for Dinosaur Vinyl to complete Job MAC001 occur in the production and finishing departments. In the production department, two individuals each work one hour at a rate of $15 per hour, including taxes and benefits. The finishing department's direct labor involves two individuals working one hour each at a rate of $18 per hour. Figure 4.11 shows the direct labor costs for Job MAC001.

Item	Hours	Rate per Hour	Item Cost	Total Cost
Production Department				
Material Handler	1	$15	$15	
Print Technician	1	15	15	$30
Finishing Department				
Production Assistant	1	$18	$18	
Production Assistant	1	18	18	36
Total Direct Labor				$66

Figure 4.11 Direct Labor for Job MAC001. Labor costs account for individuals working directly on the product. Individuals whose contributions are indirect will be tracked under manufacturing overhead. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Job MAC001 is also manufactured with the work of individuals whose contributions cannot be directly traced to the product: These indirect labor costs are assigned to the product as part of manufacturing overhead.

A company can use various methods to trace employee wages to specific jobs. For example, employees may fill out **time tickets** that include job numbers and time per job, or workers may scan bar codes of specific jobs when they begin a job task. Figure 4.12 shows what time tickets might look like on Job MAC001. Please note that in the employee time tickets that are displayed, each employee worked on more than one job. However, we are only going to track the expenses for Job MAC001.

EMPLOYEE TIME TICKET

Employee Name: Renee Chelsea

Department: Material Handler

Employee ID: 12842

Date: 4/5/2017

Job No.	Hours Worked	Hourly Rate	Total Cost
MAC001	1	$15	$ 15
POR143	7	15	105
			$120

Autho

EMPLOYEE TIME TICKET

Employee Name: Raymond Santiago

Department: Print Technician

Employee ID: 23133

Date: 4/5/2017

Job No.	Hours Worked	Hourly Rate	Total Cost
MAC001	1	$15	$ 15
TJR441	4	15	60
POR143	3	15	45
			$120

Autho

EMPLOYEE TIME TICKET

Employee Name: Rani Fina

Department: Finishing/Assembly

Employee ID: 13353

Date: 4/15/2017

Job No.	Hours Worked	Hourly Rate	Total Cost
MAC001	2	$18	$ 36
TJR441	5	18	90
POR143	1	18	18
			$144

Authorized by: _____ Date: _____

Figure 4.12 Employee Time Tickets. Time tickets (or time cards) are one method a company can use to track direct labor costs per individual and per job. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

When the accounting department processes time tickets, the costs are assigned to the individual jobs, resulting in labor costs being recorded on the work in process inventory, as shown in Figure 4.13.

	Work in Process Inventory
4/2/2017	520
4/5/2017	30
4/14/2017	180
4/15/2017	36

Figure 4.13 Costs Accounted for in the Work in Process Inventory. The direct material costs of $520 and $180, and the direct labor costs of $30 and $36 assigned to Job MAC001 are shown. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Manufacturing Overhead

Recall that the costs of a manufactured item are direct materials, direct labor, and manufacturing overhead. Costs that support production but are not direct materials or direct labor are considered overhead. **Manufacturing overhead** has three components: indirect materials, indirect labor, and overhead.

Indirect Materials

Indirect material costs are derived from the goods not directly traced to the finished product, like the sign adhesive in the Dinosaur Vinyl example. Tracking the exact amount of adhesive used would be difficult, time consuming, and expensive, so it makes more sense to classify this cost as an indirect material.

Indirect materials are materials used in production but not traced to specific products because the net informational value from the time and effort to trace the cost to each individual product produced is impossible or inefficient. For example, a furniture factory classifies the cost of glue, stain, and nails as indirect materials. Nails are often used in furniture production; however, one chair may need 15 nails, whereas another may need 18 nails. At a cost of less than one cent per nail, it is not worth keeping track of each nail per product. It is much more practical to track how many pounds of nails were used for the period and allocate this cost (along with other costs) to the overhead costs of the finished products.

Indirect Labor

Indirect labor represents the labor costs of those employees associated with the manufacturing process, but whose contributions are not directly traceable to the final product. These would include the costs of the factory floor supervisor, the factory housekeeping staff, and factory maintenance workers. For Dinosaur Vinyl, for example, labor costs for the technician who maintains the printers would be indirect labor. It would be too time consuming to determine how much of the technician's time is attributable to each sign being produced. It makes much more sense to classify that labor expense as indirect labor.

It is important to understand that the allocation of costs may vary from company to company. What may be a direct labor cost for one company may be an indirect labor cost for another company or even for another department within the same company. Deciding whether the expense is direct or indirect depends on its task. If the employee's work can be directly tied to the product, it is direct labor. If it is tied to the factory but not to the product, it is indirect labor. If it is tied to the marketing department, it is a sales and administrative expense, and not included in the cost of the product. For example, salaries of factory employees assembling parts are direct labor, salaries of factory employees performing maintenance are indirect labor, and salaries of employees in the marketing department are sales and administration expenses.

Overhead

The last category of manufacturing overhead is the overhead itself. These costs are necessary for production but not efficient to assign to individual product production. Examples of typical overhead costs are production facility electricity, warehouse rent, and depreciation of equipment.

But note that while production facility electricity costs are treated as overhead, the organization's administrative facility electrical costs are not included as overhead costs. Instead, they are treated as period costs, as office rent or insurance would be.

When both administrative and production activities occur in a common building, the production and period costs would be allocated in some predetermined manner. For example, if a 10,000 square foot building were physically allocated at 4,000 square feet for administrative purposes and 6,000 square feet for production, a company might allocate its annual $30,000 property tax expense on a 40%/60% basis, or $12,000 as a period cost for the administrative offices and a production (overhead) cost of $18,000.

LINK TO LEARNING

Do you know of a restaurant that was doing really well until it moved into a larger space? Often this happens because the owners thought their profits could handle the costs of the increased space. Unfortunately, they were not really aware of the production costs. Keeping track of product costs is critical for pricing and cost control. Read advice from restaurant owner John Gutekanst about the importance of understanding food costs (https://openstax.org/l/50pizza) and his approach to account for these in his pizzeria.

Accounting for Manufacturing Overhead

In all costing systems, the expense recognition principle requires costs to be recorded in the period in which they are incurred. The costs are expensed when matched to the revenue with which they are associated; this is commonly referred to as *having the expenses follow the revenues*. This explains why raw material purchases are not assigned to the job until the materials are requested. When companies use an inventory account, the product costs are expensed when the inventory is sold. It is common to have an item produced in one year, such as 2017, and expensed as cost of goods sold in a later year, such as 2018. In addition to the previously mentioned *revenue recognition* treatment, this treatment is justified under GAAP's *matching principle*. If the inventory has not been sold, the company has an inventory asset rather than an expense.

The expense recognition principle also applies to manufacturing overhead costs. The manufacturing overhead is an expense of production, even though the company is unable to trace the costs directly to each specific job. For example, the electricity needed to run production equipment typically is not easily traced to a particular product or job, yet it is still a cost of production. As a cost of production, the electricity—one type of manufacturing overhead—becomes a cost of the product and part of inventory costs until the product or job is sold. Fortunately, the accounting system keeps track of the manufacturing overhead, which is then applied to each individual job in the overhead allocation process.

ETHICAL CONSIDERATIONS

Ethical Job Order Costing

Job order costing requires the assignment of direct materials, direct labor, and overhead to each production unit. The primary focus on costs allows some leeway in recording amounts because the accountant assigns the costs. When jobs are billed on a cost-plus-fee basis, management may be tempted to overcharge the cost of the job. Cost-based contracts may include a guaranteed maximum, time and materials, or cost reimbursable contract. An example is the design and delivery of a corporate training program. The training company may charge for the hours worked by instructors in preparation and delivery of the course, plus a fee for the course materials.

One major issue in all of these contracts is adding too much overhead cost and fraudulent invoicing for unused materials or unperformed work by subcontractors. Management might be tempted to direct the accountant to avoid the appearance of going over the original estimate by manipulating job order costing. It is the accountant's job to ensure that the amounts recorded in the accounting system fairly represent the economic activity of the company, and the fair and proper allocation of costs.

Managers use the information in the manufacturing overhead account to estimate the overhead for the next fiscal period. This estimated overhead needs to be as close to the actual value as possible, so that the allocation of costs to individual products can be accurate and the sales price can be properly determined.

Properly allocating overhead to the individual jobs depends on finding a cost driver that provides a fair basis for the allocation. A **cost driver** is a production factor that causes a company to incur costs. An example would be a bakery that produces a line of apple pies that it markets to local restaurants. To make the pies requires that the bakery incur labor costs, so it is safe to say that pie production is a cost driver. It should also be safe to assume that the more pies made, the greater the number of labor hours experienced (also assuming that direct labor has not been replaced with a greater amount of automation). We assume, in this case, that one of the marketing advantages that the bakery advertises is 100% handmade pastries.

In traditional costing systems, the most common activities used as cost drivers are direct labor in dollars, direct labor in hours, or machine hours. Often in the production process, there is a correlation between an increase in the amount of direct labor used and an increase in the amount of manufacturing overhead incurred. If the company can demonstrate such a relationship, they then often allocate overhead based on a formula that reflects this relationship, such as the upcoming equation. In the case of the earlier bakery, the company could determine an overhead allocation amount based on each hour of direct labor or, in other cases, based on the ratio of anticipated total direct labor costs to total manufacturing overhead costs.

For example, assume that the company estimates total manufacturing overhead for the year to be $400,000 and the direct labor costs for the year to be $200,000. This relationship would lead to $2.00 of applied overhead for each $1.00 of direct labor incurred. The manufacturing overhead cost can be calculated and applied to each specific job, based on the direct labor costs. The formula that represents the overhead allocation relationship is shown, and it is the formula for overhead allocation:

$$\frac{\text{Estimated Annual Overhead Costs (\$)}}{\text{Expected Annual Activity (DL \$)}} = \text{Overhead Allocation Rate}$$

For example, Dinosaur Vinyl determined that the direct labor cost is the appropriate driver to use when establishing an overhead rate. The estimated annual overhead cost for Dinosaur Vinyl is $250,000. The total direct labor cost is estimated to be $100,000, so the allocation rate is computed as shown:

$$\frac{\text{Estimated Annual Overhead Costs (\$250,000)}}{\text{Expected Annual Activity (\$100,000)}} = \text{\$2.50 per \$1.00 Direct Labor Expense}$$

Since the direct labor expense for MAC001 is $66, the overhead allocated is $66 times the overhead application rate of $2.50 per direct labor dollar, or $165, as shown:

$$\text{Overhead Allocated} = \text{\$66 (Direct Labor)} \times \text{\$2.50 (Overhead Application Rate)} = \text{\$165}$$

Figure 4.14 shows the journal entry to record the overhead allocation.

JOURNAL			
Date	**Account**	**Debit**	**Credit**
4/18/17	Work in Process Inventory Manufacturing Overhead *To assign overhead to Job MAC001*	165	165

Work in Process Inventory	
4/2/2017	520
4/5/2017	30
4/14/2017	180
4/15/2017	36
4/18/2017	165

Figure 4.14 Overhead Allocation. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

THINK IT THROUGH

Franchise or Unique Venture?

You are deciding whether to purchase a pizza franchise or open your own restaurant specializing in pizza. List the expenses necessary to sell pizza and identify them as a fixed cost or variable cost; as a manufacturing cost or sales and administrative costs; and as a direct materials, direct labor, or overhead. For each overhead item, state whether it is an indirect material expense, indirect labor expense, or other. For each cost, identify its origination in a job order costing environment.

4.3 Use the Job Order Costing Method to Trace the Flow of Product Costs through the Inventory Accounts

Job order costing can be used for many different industries, and each industry maintains records for one or more inventory accounts. The manufacturing industry keeps track of the costs of each inventory account as the product is moved from raw materials inventory into work in process, through work in process, and into the finished goods inventory.

Conversely, typical companies in the merchandising industry sell products they do not manufacture and purchase their inventory in an already completed state. It is relatively easy to keep track of the inventory cost for a merchandising company through its application of first-in/first-out (FIFO), last-in/last-out (LIFO), weighted average, or specific identification inventory techniques on the unsold items. The primary difference in the four methods is the valuation of the cost of goods sold and the remaining ending inventory valuation, assuming that the company did not sell 100% of the inventory that they had available for sale during a given period. Companies are allowed to choose the method that they feel best represents their cost flows through their cost of goods sold and their ending inventory balances.

Not all service companies have inventory, and those companies do not have direct materials nor do they consider their work in process their inventory, since their final product is often an intangible asset, such as a legal document or tax return. Regardless of whether the service has inventory accounts, service companies all keep track of the direct labor and overhead costs incurred while completing each job in progress.

Inventory is an asset reported on the balance sheet, and each company needs to maintain accurate records for the cost of each type of inventory: raw materials inventory, work in process inventory, and finished goods inventory. All three costs are computed in a similar manner. You can see in Figure 4.15 that the general format is the same for maintaining all accounts, whether the company uses a job order, process, or hybrid cost

system.

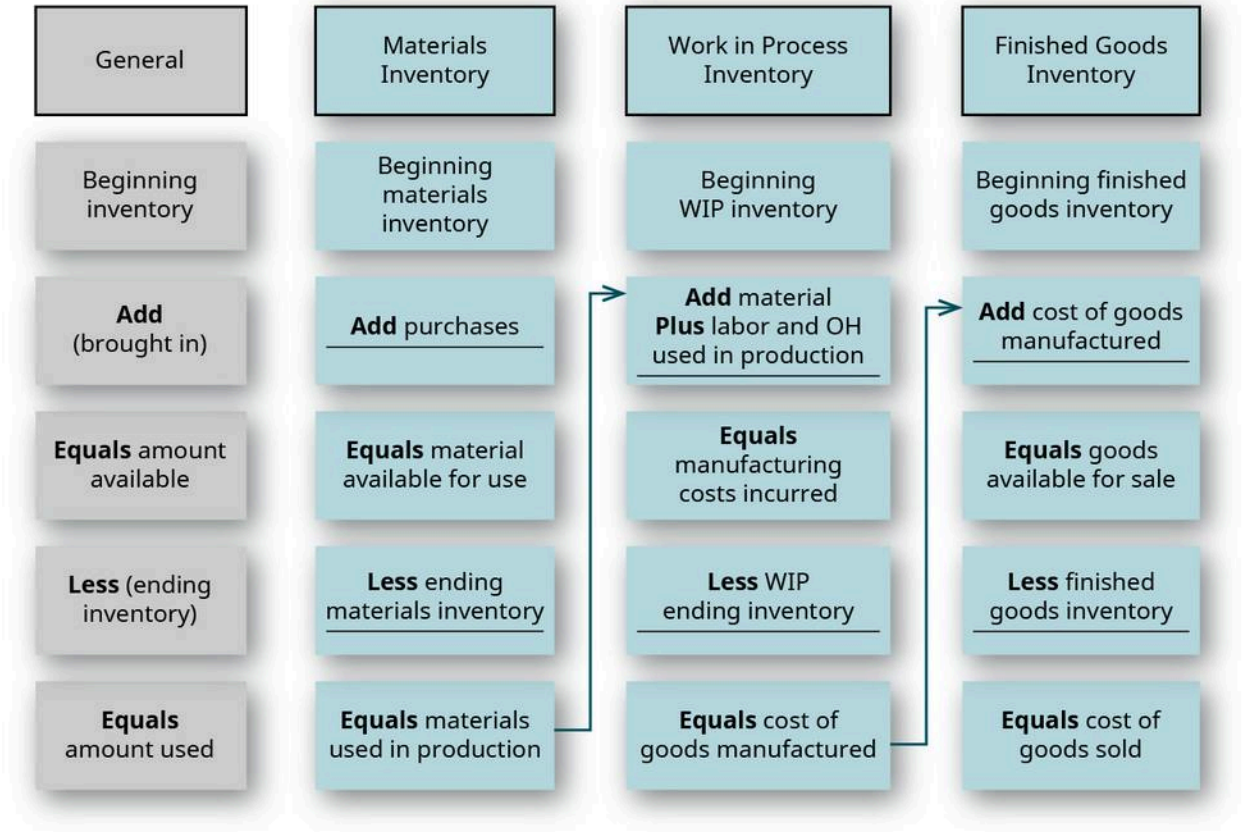


Figure 4.15 Cost of Inventory Accounts. Inventory is accounted for across the flow through production. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Each inventory account starts with a beginning balance at the start of an accounting period. During the period, if additional inventory is purchased, the new inventory amount is added to the beginning balance to calculate the total inventory available for use or sale. The ending inventory balance at the end of the accounting period can then be subtracted from the inventory available for use, and the total represents the cost of the inventory used during the period.

For example, if the beginning inventory balance were $400, and the company bought an additional $1,000, it would have $1,400 of inventory available for use. If the ending inventory balance were $500, the amount of inventory used during the period would be $900 ($400 + $1,000 = $1,400 – $500 = $900).

Raw Materials Inventory

Raw materials inventory is the total cost of materials that will be used in the production process. Usually, several accounts make up the raw materials inventory, and these can be actual accounts or accounts subsidiary to the general raw materials inventory account. In our example, Dinosaur Vinyl has several raw materials accounts: vinyl, red ink, black ink, gold ink, grommets, and wood.

Within the raw materials inventory account, purchases increase the inventory, whereas raw materials sent into production reduce it. It is easy to reconcile the amount of ending inventory and the cost of direct materials used in production, since the materials requisition form (Figure 4.7) keeps track of the inventory requested and sent into each specific job. Since the costs are transferred with production, the calculation shows the amount of materials used in production:

Beginning materials inventory
+ Net materials purchased
= Materials available for use
− Ending raw materials inventory
= Materials used in production

Note: this equation can be easily modified with algebra to compute the cost of the ending raw materials inventory.

Materials available for use
− Materials used in production
= Ending raw materials inventory

Work in Process Inventory

In a job order cost system, the balance in the work in process inventory account is continually updated as job costs are recorded and is the total of all unfinished jobs, as shown on the individual job cost sheets.

The production cycle is a continuous cycle that begins with raw materials being transferred to work in process, moving through production, and ending as finished goods inventory. Typically, as goods are being produced, additional jobs are being started and finished, and the work in process inventory includes unit costs of jobs still in production at the end of the accounting period. At the end of the accounting cycle, there will be jobs that remain unfinished in the production cycle, and these represent the work in process inventory. The costs on the job order cost sheet help reconcile the cost of the items transferred to the finished goods inventory and the cost of the work in process inventory.

For example, Dinosaur Vinyl has completed Job MAC001. The total cost of $931 is transferred to the finished goods inventory:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Finished Goods Inventory	931	
	Work in Process Inventory		931
	To transfer Job MAC001 to Finished Goods		

At this point, we need to examine an important component of the costing process. The **cost of goods manufactured** (COGM) is the costs of all of the units that a company completed and transferred to the finished goods inventory during an accounting period. Obviously, the cost of goods manufactured is not just a single number that can be pulled from one location. We have to look at all costs included in the manufacturing process to determine the cost of goods manufactured. The calculation begins with the beginning balance in the work in process inventory, incorporates the new production costs incurred during the current period (typically a year), and then subtracts the ending balance in the work in process inventory since these costs will be included in the subsequent accounting period's cost of goods manufactured, as shown:

Beginning work in process inventory
+ Current manufacturing costs
  Direct materials used in production
  Direct labor
  Overhead
= Manufacturing costs incurred
− Ending work in process inventory
= Cost of goods manufactured during the period (completed)

Note: this equation can be easily modified with algebra to compute the cost of the ending work in process inventory.

Manufacturing costs incurred
− Cost of goods manufactured
= Ending work in process inventory

Finished Goods Inventory

After each job has been completed and overhead has been applied, the product is transferred to the finished

goods inventory where it stays until it is sold. As each job is transferred, the costs are summarized and transferred as well, and the job cost sheet is completed to show the actual production cost of the product and the sales price of the items produced.

A job order cost system continually updates each job cost sheet as materials, labor, and overhead are added. As a result, all inventory accounts are constantly maintained. The materials inventory balance is continually updated, as materials are purchased and requisitioned for individual jobs. The work in process inventory and finished goods inventory are master accounts, and their balances are determined by adding the total of the job cost sheets. The total of the incomplete jobs becomes the total work in process inventory, and the total of the completed and unsold jobs becomes the total of the finished goods inventory.

Similar to the raw materials and work in process inventories, the cost of goods sold can be calculated as shown:

Beginning finished goods
+ Cost of goods manufactured
= Goods available for sale
− Ending finished goods inventory
= Cost of goods sold

Note: this equation can be easily modified with algebra to compute the cost of the finished goods inventory.

Cost of goods available for sale
− Cost of goods sold
= Ending finished goods inventory

Cost of Goods Sold

The **cost of goods sold** is the manufacturing cost of the items sold during the period. It is calculated by adding the beginning finished goods inventory and the cost of goods manufactured to arrive at the cost of goods available for sale. The cost of goods available for sale less the ending inventory results in the cost of goods sold.

In our example, when the sale has occurred, the goods are transferred to the buyer, and the product is transferred from the finished goods inventory to the cost of goods sold. A corresponding entry is also made to record the sale. Dinosaur Vinyl's sales price for Job MAC001 was $2,000, and its cost of goods sold was $931:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold	931	
	Finished Goods Inventory		931
	To transfer sold Job MAC001		
	Accounts Receivable	2,000	
	Sales		2,000
	To record sale of Job MAC001		

Figure 4.16 shows the flow to cost of goods sold.

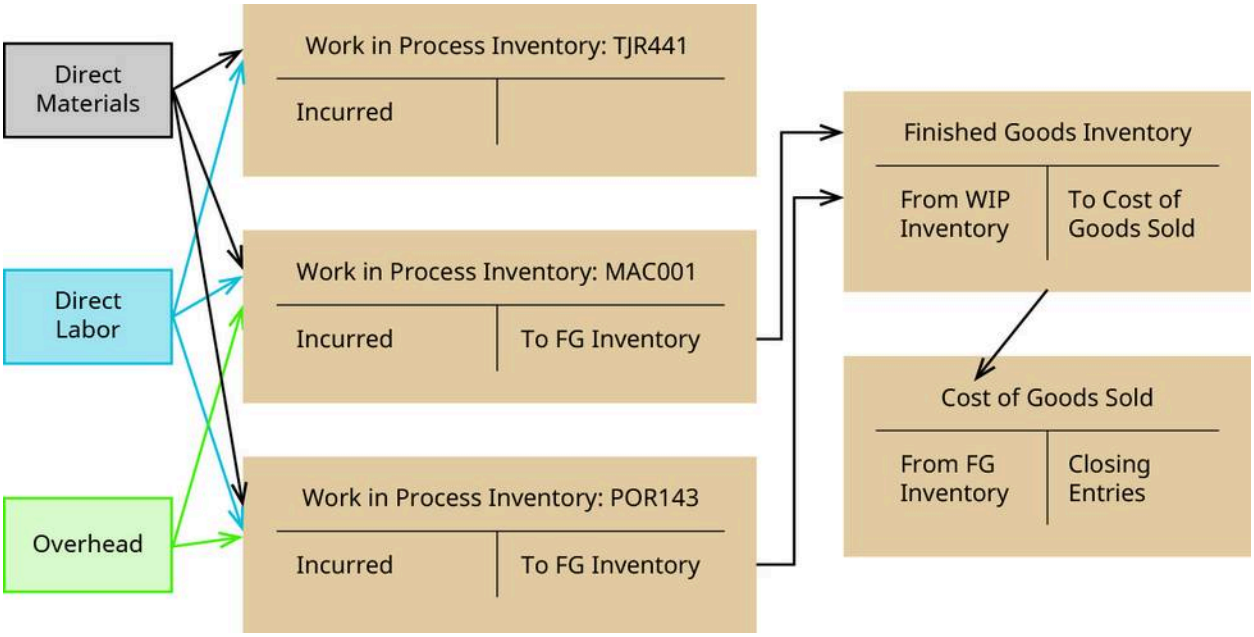


Figure 4.16 Flow of Manufacturing Costs under the Job Order Costing Method. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

YOUR TURN

Tracking the Flow with Selected T-Accounts

Use the transaction letters to show the flow in and out of the T-accounts. Note: some items may be used more than once. Also, not every possible T-account entry is required in this exercise. For example, for the purchase of raw materials, the credit entry for either cash or accounts payable is not required.

Raw Materials Inventory: Vinyl	Work in Process Inventory
Factory Wage Expense	Cost of Goods Sold
Manufacturing Overhead	Finished Goods Inventory

A. Purchase raw materials inventory
B. Factory wage expense incurred
C. Issue raw materials inventory to Job P33
D. Factory wage allocated to Job P33

E. Factory wage allocated to overhead
F. Job P33 completed
G. Job P33 sold

Solution

Raw Materials Inventory: Vinyl	
A	C

Work in Process Inventory	
C E	D

Factory Wage Expense	
B	E F

Cost of Goods Sold	
G	

Manufacturing Overhead	
F	

Finished Goods Inventory	
D	G

4.4 Compute a Predetermined Overhead Rate and Apply Overhead to Production

Job order cost systems maintain the actual direct materials and direct labor for each individual job. Since production consists of overhead—indirect materials, indirect labor, and other overhead—we need a methodology for applying that overhead. Unfortunately, the nature of indirect material, indirect labor, and other overhead expenses makes it impossible to determine the exact amount of overhead for each specific job. For example, how do you know the cost of electricity and heat for manufacturing one job? And, if you did, is it fair to say products manufactured in January are more expensive than the same product manufactured in March because of heat expense?

Fundamental Characteristics of the Overhead Determination Environment

Added to these issues is the nature of establishing an overhead rate, which is often completed months before being applied to specific jobs. Establishing the overhead allocation rate first requires management to identify which expenses they consider manufacturing overhead and then to estimate the manufacturing overhead for the next year. Manufacturing overhead costs include all manufacturing costs except for direct materials and direct labor. Therefore, in order to estimate manufacturing overhead, management must estimate the future purchase prices of dozens, or sometimes hundreds, of individual components, such as utilities, raw materials, contract labor, or diesel fuel. Estimating overhead costs is difficult because many costs fluctuate significantly from when the overhead allocation rate is established to when its actual application occurs during the production process. You can envision the potential problems in creating an overhead allocation rate within these circumstances.

Before demonstrating the calculation of a predetermined overhead allocation rate, let's review the basic

principles of revenue recognition and expense. In accounting, there are three ways to recognize expenses:

1. Direct relationship between the expense and the associated revenue. This method is used for many costs, and the expense is recognized when a direct relationship exists. For example, sales commission expenses can be directly traced to product sales, and a commission expense is recorded when a sale is made.
2. Systematic and rational allocation of expenses. This approach is used when costs exist and there is an expected benefit, even though the costs cannot be directly traced to the benefit. The assigning of expenses to a product or time period must be done in an objective and consistent manner. Examples of such expenses would include equipment rental for a factory or property insurance for the factory.
   Both of these expenses (direct relationship and systematic and rational) are also examples of the types of expenses that compose manufacturing overhead. An example of the current revenue recognition principle is a company paying $4,800 a year for property insurance. Since production rates can vary month to month, most producers would allocate $400 each month for property insurance, and this cost would be incorporated into the total overhead costs anticipated when estimating a manufacturing overhead allocation rate.
   The direct benefit is that the product will be sold and the revenue recognized. The overhead is associated but cannot be directly traced to an individual product, so the overhead expenses need to be assigned in a systematic and rational manner.
3. Immediate recognition. This method is used when expenses exist but there is no direct expected benefit. In this case, the expense is recognized immediately. For example, research and development costs are necessary expenses but cannot be traced to a specific product, so they are expensed as incurred.

The allocation of overhead to the cost of the product is also recognized in a systematic and rational manner. The expected overhead is estimated, and an allocation system is determined. The actual costs are accumulated in a manufacturing overhead account. The overhead is then applied to the cost of the product from the manufacturing overhead account. The overhead used in the allocation is an estimate due to the timing considerations already discussed.

The application rate that will be used in a coming period, such as the next year, is often estimated months before the actual overhead costs are experienced. Often, the actual overhead costs experienced in the coming period are higher or lower than those budgeted when the estimated overhead rate or rates were determined. At this point, do not be concerned about the accuracy of the future financial statements that will be created using these estimated overhead allocation rates. You will learn in Determine and Disposed of Underapplied or Overapplied Overhead how to adjust for the difference between the allocated amount and the actual amount.

Despite improvements in technology and information flow, using the actual overhead to calculate the application rate is usually not possible because the actual overhead information is available too late for management to make decisions. Also, as you will learn, the results of the actual overhead costs, if they were available, could be misleading. Therefore, most manufacturing companies use predetermined overhead rates for these reasons:

- Overhead costs are not uniform throughout the year. An example is electricity costs that vary by weather and time of day.
- Some overhead costs are fixed, and the cost per unit varies with production. For example, rent may be $1,000 per month. If 500 units were made during one month, and 2,000 units were made the next month, the cost per unit would vary from $2 per unit to $0.50 per unit.
- The total number of units produced varies and is often known sooner than the cost of overhead. For example, a company may know it will have a contract to produce 100 custom units long before it knows the utility costs for the next year.

As previously described, a predetermined overhead rate is established prior to the beginning of the fiscal year and typically is not changed during the year. The predetermined rate is calculated as shown and is used to apply overhead costs to work in process:

$$\frac{\text{Estimated (budgeted) Overhead Cost}}{\text{Expected (budgeted) Level of Activity}} = \text{Predetermined Overhead Rate}$$

CONCEPTS IN PRACTICE

Overhead in the Movie Industry

The movie industry uses job order costing, and studios need to allocate overhead to each movie. Their amount of allocated overhead is not publicly known because while publications share how much money a movie has produced in ticket sales, it is rare that the actual expenses are released to the public.

It has been speculated that *Star Wars: The Force Awakens* cost $201,000,000, with $30,000,000 considered overhead. Studios have estimated that the higher the movie expenses, the more studio overhead is required, and it has also been estimated that 10% of the total cost is assigned to studio overhead.

Determining Estimated Overhead Cost

The estimated or budgeted overhead is the amount of overhead determined during the budgeting process and consists of manufacturing costs but, as you have learned, excludes direct materials and direct labor. Examples of manufacturing overhead costs include indirect materials, indirect labor, manufacturing utilities, and manufacturing equipment depreciation. Another way to view it is overhead costs are those production costs that are not categorized as direct materials or direct labor.

Selecting an Estimated Activity Base

As you have learned, the overhead needs to be allocated to the manufactured product in a systematic and rational manner. This allocation process depends on the use of a cost driver, which drives the production activity's cost. Examples can include labor hours incurred, labor costs paid, amounts of materials used in production, units produced, or any other activity that has a cause-and-effect relationship with incurred costs.

Direct labor hours, direct labor dollars, or machine hours are often chosen as the allocation base because those costs are associated with each product, and as the activity increases, so does the manufacturing overhead. In other words, the products that involve more direct labor hours, direct labor dollars, or machine hours also increase utility expenses, supervisor time (and thus indirect labor), equipment usage and the related depreciation expense, and so forth.

Traditionally, direct labor hours were used as the activity base, but technology continually decreases the amount of direct labor used in production, and machine hours or units produced have become more common activity bases. Management analyzes the costs and selects the activity as the **estimated activity base** because it drives the overhead costs of the unit.

Computing a Predetermined Overhead Rate

Dinosaur Vinyl uses the expenses from the prior two years to estimate the overhead for the upcoming year to be $250,000, as shown in Figure 4.17.

	Annual Estimate
Indirect labor	$ 5,000
Indirect materials	20,000
Utilities	75,000
Depreciation	90,000
Insurance	35,000
Interest expense	25,000
	$250,000

Figure 4.17 Dinosaur Vinyl's Estimated Overhead. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Dinosaur Vinyl also used its payroll records to estimate that it will spend $100,000 on direct labor. Using the predetermined overhead rate calculation, the overhead rate is $2.50 per direct labor dollar:

$$\frac{\textbf{Estimated (budgeted) Overhead Cost } (\$250{,}000)}{\textbf{Expected (budgeted) Level of Activity } (\$100{,}000)} = \textbf{\$2.50 per Direct Labor Dollar}$$

Over the fiscal year, the actual costs are recorded as debits into the account called manufacturing overhead. When the overhead is applied to the jobs, the amount is first calculated using the application rate. If the total labor paid for the job is $66, the overhead applied to the job is $2.50 times that amount, or $165. The entry to record the overhead for Job MAC001 is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Work in Process Inventory	165	
	Manufacturing Overhead		165
	To apply overhead to Job MAC001		

That amount is added to the cost of the job, and the amount in the manufacturing overhead account is reduced by the same amount. At the end of the year, the amount of overhead estimated and applied should be close, although it is rare for the applied amount to exactly equal the actual overhead. For example, Figure 4.18 shows the monthly costs, the annual actual cost, and the estimated overhead for Dinosaur Vinyl for the year. While the total amounts are close to each other, they are not exact.

	Monthly Actual	Annual Amount	Annual Estimate
Indirect labor	$ 375	$ 4,500	$ 5,000
Indirect materials	1,500	18,000	20,000
Utilities	7,000	84,000	75,000
Depreciation	7,500	90,000	90,000
Insurance	2,917	35,000	35,000
Interest expense	2,083	25,000	25,000
	$21,000	$256,500	$250,000

Figure 4.18 Dinosaur Vinyl's Actual and Estimated Overhead. While the total amounts are close to each other, they are not an exact match. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Calculating Manufacturing Overhead Cost for an Individual Job

Figure 4.18 shows the monthly manufacturing actual overhead recorded by Dinosaur Vinyl. As explained previously, the overhead is allocated to the individual jobs at the predetermined overhead rate of $2.50 per direct labor dollar when the jobs are complete. When Job MAC001 is completed, overhead is $165, computed as $2.50 times the $66 of direct labor, with the total job cost of $931, which includes $700 for direct materials, $66 for direct labor, and $165 for manufacturing overhead.

LINK TO LEARNING

Companies need to make certain the sales price is higher than the prime costs and the overhead costs. This can be a difficult task in industries in which overhead costs change. In some industries, the company has no control over the costs it must pay, like tire disposal fees. To ensure that the company is profitable, an additional cost is added and the price is modified as necessary. In this example, the guarantee offered by Discount Tire (https://openstax.org/l/50tireguarantee) does not include the disposal fee in overhead and increases that fee as necessary.

4.5 Compute the Cost of a Job Using Job Order Costing

To summarize the job order cost system, the cost of each job includes direct materials, direct labor, and manufacturing overhead. While the product is in production, the direct materials and direct labor costs are included in the work in process inventory. The direct materials are requested by the production department, and the direct material cost is directly attached to each individual job, as the materials are released from raw materials inventory. The cost of direct labor is recorded by the employees and assigned to each individual job. When the allocation base is known, usually when the product is completed, the overhead is allocated to the product on the basis of the predetermined overhead rate.

LINK TO LEARNING

The construction industry typically uses job order costing and accounts for its costing (https://openstax.org/l/50construction) in a manner similar to the businesses profiled in this chapter.

Determining the Costs of an Individual Job Using Job Order Costing

When a job is completed, the costs of the job—the direct materials, direct labor, and manufacturing overhead—are totaled on the job cost sheet, and the total amount is transferred to finished goods at the same time the product is transferred, either physically or legally, such as in the case of a home built by a contractor. Finally, when the product is sold, the sale is recorded at the sale price, while the cost is transferred from finished goods inventory to the cost of goods sold expense account. Figure 4.19 shows the flow of costs from raw materials inventory to cost of goods sold.

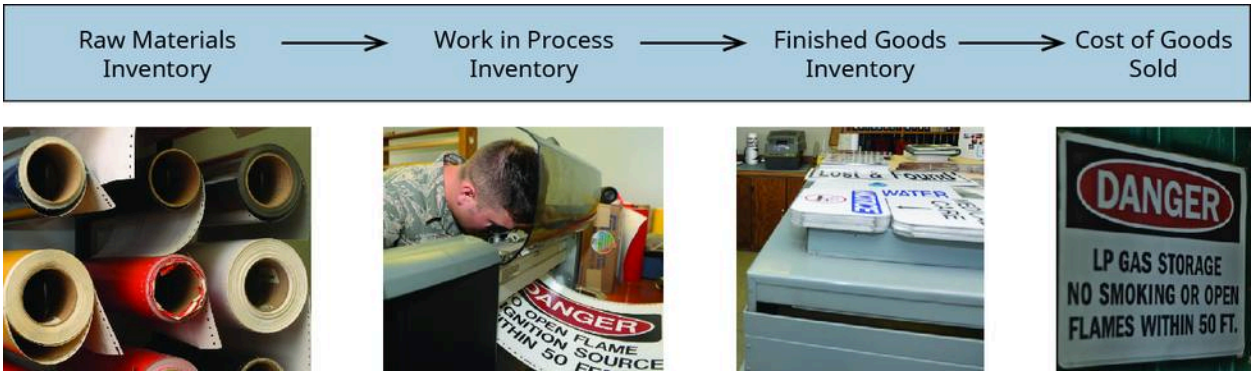


Figure 4.19 Flow of Costs during Production and Ultimate Sale or Transfer of Ownership. (credit "Raw Materials Inventory," "Work in Process Inventory," "Finished Goods Inventory": modifications of "160810-F-UY190-027.JPG," "160810-F-UY190-073.JPG," and "160810-F-UY190-105.JPG" by Jessica Weissman, Minot Air Force Base Public Affairs, Public Domain; credit Cost of Goods Sold": "Rustic Sign" by Grace Byrd/Flickr, CC BY 2.0)

At all points in the process, the work in process should include the cost of direct materials and direct labor.

When the job is completed and overhead assigned, the overhead allocation increases the cost of the work in process inventory. The cost of each individual job is maintained on a job cost sheet, and the total of all the work in process job cost sheets equals the work in process inventory and the statement of cost of goods manufactured, as you have learned.

A job cost sheet is a subsidiary ledger that identifies the individual costs for each job. Figure 4.20 shows the job cost sheet for Job MAC001.

JOB COST SHEET
Dinosaur Vinyl, Inc.

Job No.: 5416	**Customer No.:** 2501723
Customer: Macs & Cheese	**Date Started:** Dec. 22, 2018
Units Ordered: 1	**Date Completed:** Feb. 22, 2019

Direct Material	Units	Price	Amount
Vinyl	1	$300	$ 300
Black printing ink	2	50	100
Red printing ink	1	60	60
Gold printing ink	1	60	60
Grommets	12	10	120
Framing wood	40	1.5	60
Total Direct Materials			$ 700
Direct Labor	**Hours**	**Wage Rate**	**Amount**
Material Handler	1	$ 15	$ 15
Print Technician	1	15	15
Production Assistants	2	18	36
Total Direct Labor			$ 66
Manufacturing Overhead	**Base Units**	**Rate**	**Amount**
Direct Labor Cost	66	$ 2.5	$ 165
Total Manufacturing Overhead			$ 165
Total Job Cost			$ 931
Total Revenue			$2,000

Figure 4.20 Job Cost Sheet for Job MAC001. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Sample Cost Information for Dinosaur Vinyl

Dinosaur Vinyl worked on three jobs during the month: POR143, MAC001, and TRJ441, and a fourth Job SWM505 had been finished and moved to the finished goods inventory account during the previous month.

At the beginning of the month, the company had a beginning raw materials inventory balance of $2,500, and during the month, it purchased an additional $10,500, giving it a total of $13,000 in raw materials available for use in production.

The following example will examine four different production jobs. Each of the four will be at beginning stages at either the beginning of the current month or the end of the current month.

1. Job POR 143: This job was the only work in process inventory at the beginning of the current month, and it had $1,000 in direct material costs, and $0 of direct labor costs already allocated to the work in process inventory. During the current month, additional direct materials of $200 and direct labor of $150 were added to POR143. An overhead cost of $375 was applied to POR143 at the predetermined overhead rate of $2.50 per direct labor dollar. It was finished during the month and transferred to the finished goods inventory. The sale was not finalized during the month, so it continues to be part of the finished goods inventory.
2. Job MAC 001: This job was started and completed during the month. Since the job began in and was

completed in the same month, there was no beginning balance in the work in process inventory. During the month it incurred $700 in direct materials costs, $66 in direct labor, and $165 of overhead applied to the job before it was transferred to the finished goods inventory upon completion. The sale was finalized during the month at a sale price of $2,000, so the costs were transferred from finished goods inventory to cost of goods sold.

3. Job TRJ441: This job was started during the current month. Its costs consist of $500 in direct material cost, $150 in direct labor expenses, and $375 in applied overhead. The job remains in the work in process inventory awaiting assembly.
4. Job SWM505: At the beginning of the month, this job was completed and already in the finished goods inventory at a cost of $1,531. Since it was completed, it did not incur any additional costs in the current month. It was sold during the month for $3,500, and the costs were transferred from the finished goods inventory to cost of goods sold.

The cost of raw materials used is calculated as shown:

Beginning RM inventory	$ 2,500
Purchases	10,500
Materials available for use	13,000
Ending RM inventory	11,600
Materials used in production	$ 1,400

The individual job cost sheets show the $1,400 worth of materials used in production:

POR143	$ 200
MAC001	700
TJR441	500
	$1,400

The cost of goods manufactured is accounted for as shown:

	POR143	MAC001	TJR441	Total
Materials added	$200	$700	$ 500	$1,400
Labor added	150	66	150	366
Overhead applied	375	165	375	915
Total added during month	$725	$931	$1,025	$2,681
	POR143	**MAC001**	**TJR441**	**Total**
Beginning WIP inventory	$1,000	$ 0	$ 0	$1,000
Materials, labor, & OH added	725	931	1,025	2,681
Manufacturing costs incurred	1,725	931	1,025	3,681
Ending WIP inventory	0	0	1,025	1,025
Cost of Goods Manufactured	$1,725	$931	$ 0	$2,656

Notice the costs for Job TJR441 are included in the work in process inventory, whereas the costs for POR143 and MAC001 were transferred to the cost of goods manufactured. The costs of the jobs transferred are shown in the cost of goods sold and the finished goods inventory:

	SWM505	POR143	MAC001	TJR441	Total
Beginning FG inventory	$1,531	$ 0	$ 0	$0	$1,531
Cost of Goods Manufactured	0	1,725	931	0	2,656
Goods Available for Sale	1,531	1,725	931	0	4,187
Ending FG inventory	0	1,725	0	0	1,725
Cost of Goods Sold	$1,531	$ 0	$931	$0	$2,462

Mechanics of Job Order Costing for Dinosaur Vinyl

The amounts in raw materials, work in process, and finished goods inventories compose the total cost for each account, whereas the job cost sheets contain the costs for each individual job. A summary of the jobs for Dinosaur Vinyl is given in Figure 4.21.

	SWM505	POR143	MAC001	TJR441	Total
Beginning balance	$1,531	$1,000	$ 0	$ 0	$2,531
Direct materials		200	700	500	1,400
Direct labor		150	66	150	366
Overhead applied		375	165	375	915
Total cost	$1,531	$1,725	$931	$1,025	$5,212
Status	Sold	Finished	Sold	Unfinished	
Final account location	COGS	FG	COGS	WIP	

Beginning WIP

Total for WIP, FG, & COGS

Figure 4.21 Summary of Dinosaur Vinyl's Jobs during the Year. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

THINK IT THROUGH

Allocating Costs

A manufacturing company has incurred these costs:

Purchase raw materials inventory	$15,000
Issue raw materials inventory to Job A	3,000
Factory wage expense incurred	23,000
Factory wage allocated to Job A	2,000
Factory wage allocated to overhead	500
Manufacturing overhead incurred	7,500
Manufacturing overhead allocated to Job A	1,000

What is the cost allocated to Job A? For any costs not used, explain why they are not used.

4.6 Determine and Dispose of Underapplied or Overapplied Overhead

As you've learned, the actual overhead incurred during the year is rarely equal to the amount that was applied to the individual jobs. Thus, at year-end, the manufacturing overhead account often has a balance, indicating overhead was either overapplied or underapplied.

If, at the end of the term, there is a debit balance in manufacturing overhead, the overhead is considered **underapplied overhead**. A debit balance in manufacturing overhead shows either that not enough overhead was applied to the individual jobs or overhead was underapplied. If, at the end of the term, there is a credit balance in manufacturing overhead, more overhead was applied to jobs than was actually incurred. This shows the actual amount was **overapplied overhead**.

The actual overhead costs are recorded through a debit to manufacturing overhead. The same account is credited when overhead is applied to the individual jobs in production, as shown:

Manufacturing Overhead	
Actual Cost	Applied Costs

Since the overhead is first recorded in the manufacturing overhead account, then applied to the individual jobs, traced through finished goods inventory, and eventually transferred to cost of goods sold, the year-end balance is eliminated through an adjusting entry, offsetting the cost of goods sold. If manufacturing overhead has a debit balance, the overhead is underapplied, and the resulting amount in cost of goods sold is understated. The adjusting entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold Manufacturing Overhead	$$$	 $$$

If manufacturing overhead has a credit balance, the overhead is overapplied, and the resulting amount in cost of goods sold is overstated. The adjusting entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Manufacturing Overhead Cost of Goods Sold *Application of overhead to Cost of Goods Sold*	$$$	 $$$

Returning to our example, at the end of the year, Dinosaur Vinyl had actual overhead expenses of $256,500 and applied overhead expenses of $250,000, as shown:

Manufacturing Overhead	
Actual Costs $256,500	Applied Costs $250,000

Since manufacturing overhead has a debit balance, it is underapplied, as it has not been completely allocated. The adjusting journal entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold Manufacturing Overhead *Application of underapplied overhead to Cost of Goods Sold*	6,500	 6,500

If the overhead was overapplied, and the actual overhead was $248,000 and the applied overhead was $250,000, the entry would be:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Manufacturing Overhead Cost of Goods Sold *Application of overapplied overhead to Cost of Goods Sold*	2,000	 2,000

To adjust for overapplied or underapplied manufacturing overhead, some companies have a more complicated, three-part allocation to work in process, finished goods, and cost of goods sold. This method is typically used in the event of larger variances in their balances or in bigger companies. (You will learn more

about this in future cost or advanced managerial accounting courses.)

YOUR TURN

Kraken Boardsports



Figure 4.22 (credit: modification of images provided courtesy of Kraken Boardsports, CC BY 4.0)

Kraken Boardsports manufactures winches for snow and ski boarders to snow ski without a mountain or water ski without a lake (Figure 4.22). End-of-year data show these overhead expenses:

Indirect materials	$25,000
Indirect labor	31,750
Depreciation of factory equipment	50,000
Factory utility expenses	17,500
Factory supervisor salaries	85,000

Kraken Boardsports had 6,240 direct labor hours for the year and assigns overhead to the various jobs at the rate of $33.50 per direct labor hour.

How much overhead was overapplied or underapplied during the year? What would be the journal entry to adjust manufacturing overhead?

Solution

The total overhead incurred is the total of:

Indirect materials	$ 25,000
Indirect labor	31,750
Depreciation of factory equipment	50,000
Factory utility expenses	17,500
Supervisor salaries	85,000
Actual overhead incurred	$209,250

The total overhead applied is $209,040, which is calculated as:

$33.50/direct labor hours × 6,240 direct labor hours.

The balance in manufacturing overhead is a debit balance of $210:

	Manufacturing Overhead	
	Actual Costs	Applied Costs
	$209,250	$209,040
Balance	210	

The adjusting journal entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold	210	
	Manufacturing Overhead		210

LINK TO LEARNING

Job order costing and overhead allocation are not new methods of accounting and apply to governmental units as well. See it applied in this 1992 report on *Accounting for Shipyard Costs and Nuclear Waste Disposal Plans* (https://openstax.org/l/50shipyard) from the United States General Accounting Office.

4.7 Prepare Journal Entries for a Job Order Cost System

Although you have seen the job order costing system using both T-accounts and job cost sheets, it is necessary to understand how these transactions are recorded in the company's general ledger.

Journal Entries to Move Direct Materials, Direct Labor, and Overhead into Work in Process

Dinosaur Vinyl keeps track of its inventory and orders additional inventory to have on hand when the production department requests it. This inventory is not associated with any particular job, and the purchases stay in raw materials inventory until assigned to a specific job. For example, Dinosaur Vinyl purchased an additional $10,000 of vinyl and $500 of black ink to complete Macs & Cheese's billboard. If the purchase is made on account, the entry is as shown:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Raw Materials Inventory: Vinyl	10,000	
	Raw Materials Inventory: Black ink	500	
	Accounts Payable		10,500
	To record purchase of vinyl and ink inventory		

As shown in Figure 4.20, for the production process for job MAC001, the job supervisor submitted a materials requisition form for $300 in vinyl, $100 in black ink, $60 in red ink, and $60 in gold ink. For the finishing process for Job MAC001, $120 in grommets and $60 in finishing wood were requisitioned. The entry to reflect these actions is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Work in Process Inventory	700	
	Raw Materials Inventory: Vinyl		300
	Raw Materials Inventory: Black ink		100
	Raw Materials Inventory: Red ink		60
	Raw Materials Inventory: Gold ink		60
	Raw Materials Inventory: Grommets		120
	Raw Materials Inventory: Finishing wood		60
	To record requisition of vinyl and ink inventory		

The production department employees work on the sign and send it over to the finishing/assembly department when they have completed their portion of the job.

The direct cost of factory labor includes the direct wages paid to the employees and all other payroll costs associated with that labor. Typically, this includes wages and the payroll taxes and fringe benefits directly tied to those wages. The accounting system needs to keep track of the labor and the other related expenses assigned to a particular job. These records are typically kept in a time ticket submitted by employees daily.

On April 10, the labor time sheet totaling $30 is recorded for Job MAC001 through this entry:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Work in Process Inventory (MAC001)	30	
	Factory Wages Payable		30
	To record labor for Job MAC001		

The assembly personnel in the finishing/assembly department complete Job MAC001 in two hours. The labor is recorded as shown:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Work in Process Inventory	36	
	Factory Wages Payable		36
	To record labor for Job MAC001		

Indirect materials also have a materials requisition form, but the costs are recorded differently. They are first transferred into manufacturing overhead and then allocated to work in process. The entry to record the indirect material is to debit manufacturing overhead and credit raw materials inventory.

Indirect labor records are also maintained through time tickets, although such work is not directly traceable to a specific job. The difference between direct labor and indirect labor is that the indirect labor records the debit to manufacturing overhead while the credit is to factory wages payable.

Dinosaur Vinyl's time tickets indicate that $4,000 in indirect labor costs were incurred during the period. The entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Manufacturing Overhead	4,000	
	Factory wages payable		4,000
	To record indirect labor for WIP inventories		

Dinosaur Vinyl also records the actual overhead incurred. As shown in Figure 4.18, manufacturing overhead

costs of $21,000 were incurred. The entry to record these expenses increases the amount of overhead in the manufacturing overhead account. The entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Manufacturing Overhead	21,000	
	Supplies Inventory		1,500
	Utilities Payable		7,000
	Accumulated Depreciation		7,500
	Prepaid Insurance		2,917
	Interest Payable		2,083
	To record April's overhead expenses		

The amount of overhead applied to Job MAC001 is $165. The process of determining the manufacturing overhead calculation rate was explained and demonstrated in Accounting for Manufacturing Overhead. The journal entry to record the manufacturing overhead for Job MAC001 is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Work in Process Inventory	165	
	Manufacturing Overhead		165
	To apply overhead to Job MAC001		

Journal Entry to Move Work in Process Costs into Finished Goods

When each job and job order cost sheet have been completed, an entry is made to transfer the total cost from the work in process inventory to the finished goods inventory. The total cost of the product for Job MAC001 is $931 and the entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Finished Goods Inventory	931	
	Work in Process Inventory		931
	To recognize completion of Job MAC001		

Journal Entries to Move Finished Goods into Cost of Goods Sold

When the sale has occurred, the goods are transferred to the buyer. The product is transferred from the finished goods inventory to cost of goods sold. A corresponding entry is also made to record the sale. The sign for Job MAC001 had a sales price of $2,000 and a cost of $931. These are the entries to record the transfer of goods and sale to the buyer:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
	Cost of Goods Sold	931	
	Finished Goods Inventory		931
	To recognize sale of Job MAC001		
	Accounts Receivable	2,000	
	Sales		2,000
	To recognize sale of Job MAC001		

The resulting accounting is shown on the company's income statement:

DINOSAUR VINYL, INC. Income Statement	
Sales	$2,000
Cost of Goods Sold	931
Gross profit	$1,069

THINK IT THROUGH

Ongoing Overapplied Overhead

At the end of each year, manufacturing overhead is analyzed, and an adjusting entry is made to dispose of the under- or overapplied overhead. How would you advise a company that has had overapplied overhead for each of the last five years?

4.8 Explain How a Job Order Cost System Applies to a Nonmanufacturing Environment

Job order cost systems can be used beyond the manufacturing realm and are often used in the production of services. The same cost tracking and journaling techniques apply, as the outcome still consists of materials, labor, and overhead. However, the terminology changes in a nonmanufacturing environment. For example, a movie production studio and an accounting firm produce movies and financial statement audits, respectively, instead of manufacturing units.

Fundamentals of the Job Order Costing Method for Service Entities

Instead of being dependent on materials, service industries depend on labor. Since their work is labor intensive, it makes sense to use labor as an activity base with billable hours often as the best allocation base. For example, in an audit, there often will be several accountants, with differing levels of experience and expertise involved in the assignment. The accounting firms have more billable hours at the staff level and fewer billable hours at the partner level. And since the firm bills the partner's time at a significantly higher rate than the staff, it makes sense to apply overhead at the billable hours instead of the billable costs.

In service industries, there is no manufacturing overhead because they are not manufacturing a product, but instead are providing a service. Accordingly, overhead is called **operating overhead**.

Another terminology difference is the inventory accounts. The jobs are considered movies or assignments in process, and are transferred to a cost of service sold account instead of to a finished goods inventory.

CONCEPTS IN PRACTICE

Tracking Costs in Healthcare

Healthcare is one of the industries that keeps track of materials, such as medicine. In this industry, direct labor is shown to the patient as the cost of the provider, such as a physician, physician assistant, or nurse practitioner. Indirect labor includes all other personnel from front desk staff to the nurse who gathers vital signs or a technician who performs tests. Patients do not see the overhead cost on their bill, but it is built into the invoice as part of the practitioner or testing fees.

Service Entity Use of a Job Order Costing System

To understand how a service provider uses a job order cost system, let's consider the case of IFixIT. IFixIT Systems is a Sony-authorized repair provider that fixes audiovisual equipment brought in by customers. IFixIT

requires customers to pay $50 to diagnose the problem. IFixIt pays its employees $25 per hour and assigns overhead equal to its direct labor cost. The customers' bills do not show overhead and are instead itemized as parts plus labor, where the cost for parts is the original cost plus a markup, and the labor rate is $80 per hour.

A customer brought in his TV and paid the $50 diagnostic fee. IFixIT determined a new power cord was needed. To fix it, IFixIT purchases the part from its suppliers at $42 and pays $75 in direct labor for 3 hours at $25 per hour. Overhead is applied equal to the direct labor cost of $75. The customer is charged $310, consisting of $70 for the part and 3 hours of labor at a rate of $80 per hour. IFixIT records the journal entries shown:

JOURNAL

Date	Account	Debit	Credit
	Cash	50	
	Diagnosis revenue		50
	Diagnosis of Sony Bravia for Job 4740325		
	Repair in process	42	
	Accounts Payable		42
	Purchase of new power cord for Job 4740325		
	Repair in process	75	
	Salaries Payable		75
	Assigning labor to Job 4740325		
	Repair in process	75	
	Operating overhead		75
	Assigning overhead costs to Job 4740325		
	Cost of completed repair	192	
	Repair in Process		192
	Completion of Job 4740325		
	Cash	310	
	Repair revenue		310
	Completion of Job 4740325		

ETHICAL CONSIDERATIONS

Subcontractor Misrepresentation of Costs of Jobs Used to Overbill Clients

Construction is a typical industry where job order costing and related accounting misstatements can be used to commit fraud. A construction subcontractor might overstate the units of production accomplished, the units of labor, or the equipment actually used.[2] This occurs most commonly with subcontractor fraud, where the subcontractor does not perform the work but bills for it anyway.

Another complicating issue is that many subcontractors are disadvantaged business enterprises that are required by law to be included in governmental construction contracts. In Chicago, for example, McHugh Construction paid $12 million in fines to settle the claims that its disadvantaged business enterprise subcontractor did not perform work.[3] The subcontractor received a prison sentence, and a related party was put on probation. An accountant had to prepare the invoices that allowed this common type of scheme to operate.

2 Jim Schmid and Todd F. Taggart, "The Most Common Types of Construction Fraud," *Construction Business Owner*, November 2, 2011, http://www.constructionbusinessowner.com/insurance/risk-management/most-common-types-construction-fraud.
3 Kim Slowey, "Chicago Subcontractor Sentenced to 1-year Prison Term for DBE Fraud Scheme," *Construction Dive*, March 20, 2017, https://www.constructiondive.com/news/chicago-subcontractor-sentenced-to-1-year-prison-term-for-dbe-fraud-scheme/438441/.

Key Terms

conversion costs total of labor and overhead for a product; the costs that "convert" the direct material into the finished product

cost driver activity that is the reason for the increase or decrease of another cost; examples include labor hours incurred, labor costs paid, amounts of materials used in production, units produced, machine hours, or any other activity that has a cause-and-effect relationship with incurred costs

cost of goods manufactured manufacturing costs incurred less the ending work in process inventory

cost of goods sold expense account that houses all costs associated with getting a product ready for sale

direct labor labor directly related to the manufacturing of the product or the production of a service

direct materials materials used in the manufacturing process that can be traced directly to the product

estimated activity base total amount of the activity for the year

expense recognition principle (also, matching principle) matches expenses with associated revenues in the period in which the revenues were generated

indirect labor labor not directly involved in the active conversion of materials into finished products or the provision of services

indirect materials materials used in production but not efficiently traceable to a specific unit of production

job cost sheet document created for each job that includes all material, labor, and overhead costs for that job

job order costing information system that traces the individual costs directly to the final product and not to production departments

loss leader product sold at a price that is often less than the cost of producing it in order to entice customers to buy accessories that are necessary for its use

manufacturing costs total of all costs expended in the manufacturing process; generally consists of direct material, direct labor, and manufacturing overhead

manufacturing overhead costs incurred in the production process that are not economically feasible to measure as direct material or direct labor costs; examples include indirect material, indirect labor, utilities, and depreciation

materials requisition form form showing which specific raw materials and costs are transferred from raw materials inventory to work in process inventory

operating overhead overhead account used for service industries

overapplied overhead situation when the overhead applied to the individual jobs is greater than the actual overhead; when overhead is overapplied, the manufacturing overhead has a credit balance

period costs typically related to a particular time period instead of attached to the production of an asset; treated as an expense in the period incurred (examples include many sales and administrative expenses)

prime costs direct material expenses and direct labor costs

time ticket document used to record the particular job worked on by each employee

underapplied overhead situation when the overhead applied to the individual jobs is less than the actual overhead; when overhead is underapplied, the manufacturing overhead has a debit balance

Summary

4.1 Distinguish between Job Order Costing and Process Costing

- Job order costing (JOC) is the optimal costing method for producing custom goods or when it is easy to identify the cost directly with the product.
- A JOC system assigns costs to each individual job as the costs are incurred, so that at all points in the manufacturing process, the costs assigned to that particular job are known.

4.2 Describe and Identify the Three Major Components of Product Costs under Job Order Costing

- Direct materials are requested on a materials requisition form and recorded on the job cost sheet when transferred from raw materials inventory to the work in process inventory.
- Time tickets are used to accumulate the labor associated with particular jobs and assigned to those jobs on the job cost sheet.
- Manufacturing overhead costs are accumulated in the manufacturing overhead account and assigned to the individual jobs using the predetermined overhead rate.

4.3 Use the Job Order Costing Method to Trace the Flow of Product Costs through the Inventory Accounts

- Materials used in production include the beginning raw materials inventory and purchases, less the ending inventory. This amount is the amount added to the work in process inventory.
- The cost of goods manufactured includes the beginning work in process inventory, the materials used in production, the direct labor assigned to each job, and the manufacturing overhead costs assigned, less the costs remaining in the work in process inventory. This amount is transferred to the finished goods inventory.
- The cost of goods sold include the beginning finished goods inventory and the cost of goods manufactured during the period, less the ending inventory.
- When the job is completed, the costs are transferred from the work in process inventory to the finished goods inventory.
- When the jobs are sold, the costs are transferred from the finished goods inventory to the cost of goods sold.

4.4 Compute a Predetermined Overhead Rate and Apply Overhead to Production

- Expenses are recognized when they have a direct relationship with the associated revenue, when there is a systematic and rational method to allocate them, or immediately when there is no expected benefit.
- The estimated activity base is typically direct labor dollars or direct labor hours, and is based on an allocation base that increases or decreases as overhead increases or decreases.
- The predetermined overhead rate is the estimated overhead divided by the activity base.

4.5 Compute the Cost of a Job Using Job Order Costing

- Costs from the materials requisition sheet and time tickets are recorded on the job cost sheet.
- Overhead is allocated from the manufacturing overhead account to the individual jobs and recorded on the job cost sheet.
- Each job has its own job cost sheet, showing the materials, labor, and overhead for each job.

4.6 Determine and Dispose of Underapplied or Overapplied Overhead

- Overhead is allocated to individual jobs based on the estimated overhead costs for the year and may be overapplied or underapplied for the year.
- Overhead is underapplied when not all of the costs accumulated in the manufacturing overhead account are applied during the year.
- Overhead is overapplied when more overhead is applied to the jobs than was actually incurred.
- The amount of overhead overapplied or underapplied is adjusted into the cost of goods sold account.

4.7 Prepare Journal Entries for a Job Order Cost System

- Job cost sheets record the material, labor, and overhead costs for each job, whereas journal entries actually transfer the costs into the work in process inventory, the finished goods inventory, and cost of goods sold.

4.8 Explain How a Job Order Cost System Applies to a Nonmanufacturing Environment

- Job order costing can be used in nonmanufacturing companies and with the same techniques, even though there are not any inventory accounts.

Multiple Choice

1. LO 4.1 Which of the following product situations is better suited to job order costing than to process costing?
   A. Each product batch is exactly the same as the prior batch.
   B. The costs are easily traced to a specific product.
   C. Costs are accumulated by department.
   D. The value of work in process is based on assigning standard costs.

2. LO 4.1 A job order costing system is most likely used by which of the following?
   A. a pet food manufacturer
   B. a paper manufacturing company
   C. an accounting firm specializing in tax returns
   D. a stereo manufacturing company

3. LO 4.1 Which of the following is a prime cost?
   A. indirect materials
   B. direct labor
   C. administrative expenses
   D. factory depreciation expenses

4. LO 4.1 Which of the following is a conversion cost?
   A. raw materials
   B. direct materials
   C. administrative expenses
   D. factory depreciation expenses

5. LO 4.1 During production, to what are the costs in job order costing applied?
   A. manufacturing overhead
   B. cost of goods sold
   C. each individual product
   D. each individual department

6. LO 4.2 Which document lists the inventory that will be removed from the raw materials inventory?
   A. job cost sheet
   B. purchase order
   C. materials requisition form
   D. receiving document

7. LO 4.2 Which document shows the cost of direct materials, direct labor, and overhead applied for each specific job?
   A. job cost sheet
   B. purchase order
   C. materials requisition form
   D. receiving document

8. LO 4.2 Which document lists the total direct materials used in a specific job?
A. job cost sheet
B. purchase order
C. materials requisition form
D. receiving document

9. LO 4.2 Which document lists the total direct labor used in a specific job?
A. job cost sheet
B. purchase order
C. employee time ticket
D. receiving document

10. LO 4.4 Assigning indirect costs to specific jobs is completed by which of the following?
A. applying the costs to manufacturing overhead
B. using the predetermined overhead rate
C. using the manufacturing costs incurred
D. applying the indirect labor to the work in process inventory

11. LO 4.4 In a job order cost system, which account shows the overhead used by the company?
A. work in process inventory
B. finished goods inventory
C. cost of goods sold
D. manufacturing overhead

12. LO 4.7 In a job order cost system, raw materials purchased are debited to which account?
A. raw materials inventory
B. work in process inventory
C. finished goods inventory
D. cost of goods sold

13. LO 4.7 In a job order cost system, overhead applied is debited to which account?
A. work in process inventory
B. finished goods inventory
C. manufacturing overhead
D. cost of goods sold

14. LO 4.7 In a job order cost system, factory wage expense is debited to which account?
A. raw materials inventory
B. work in process inventory
C. finished goods inventory
D. cost of goods sold

15. LO 4.7 In a job order cost system, utility expense incurred is debited to which account?
A. work in process inventory
B. finished goods inventory
C. manufacturing overhead
D. cost of goods sold

16. LO 4.7 In a job order cost system, indirect labor incurred is debited to which account?
A. work in process inventory

B. finished goods inventory
C. manufacturing overhead
D. cost of goods sold

17. LO 4.8 The activity base for service industries is most likely to be _____.
A. machine hours
B. administrative salaries
C. direct labor cost
D. direct labor hours

Questions

1. LO 4.1 A printing company manufactures notebooks of various sizes. The company manufactures 3,000 notebooks each day. Should the company use process costing or job order costing?

2. LO 4.1 Burnham Industries incurs these costs for the month:

Direct materials	$2,000
Direct labor	3,000
Factory depreciation expense	3,500
Factory utilities expense	750
CEO's salary	4,000

What is the prime cost?

3. LO 4.1 Choco's Chocolates incurs these costs for the month:

Direct materials	$15,000
Direct labor	25,000
Factory depreciation expense	45,000
Factory utilities expense	2,000
Payroll staff's salary	15,000

What is the conversion cost?

4. LO 4.1 How do job order costing and process costing differ with respect to recording direct materials and direct labor?

5. LO 4.1 Why are product costs assigned to the product and period costs immediately expensed?

6. LO 4.3 Is the cost of goods manufactured the same as the cost of goods sold?

7. LO 4.3 From beginning to end, place these items in the order of the flow of goods.
A. cost of goods sold
B. raw materials inventory
C. finished goods inventory
D. work in process inventory

8. LO 4.4 How is the predetermined overhead rate determined?

9. LO 4.4 How is the predetermined overhead rate applied?

10. LO 4.5 Why are the overhead costs first accumulated in the manufacturing overhead account instead of in the work in process inventory account?

11. LO 4.6 Why is the manufacturing overhead account debited as expenses are recognized and then credited when overhead is applied?

12. LO 4.7 Match the concept on the left to its correct description.

A. job order costing	i. computes the overhead applied to each job
B. materials requisition sheet	ii. source document indicating the number of hours an employee worked on specific jobs
C. overapplied overhead	iii. source document indicating the raw materials assigned to a specific production job
D. predetermined overhead rate	iv. the cost accounting system used by pet food manufacturers
E. process costing	v. the cost accounting system used by law firms
F. time ticket	vi. the result when the actual overhead is less than the amount assigned to each specific job
G. underapplied overhead	vii. the result when the actual overhead is more than the amount assigned to each specific job

13. LO 4.8 When compared to manufacturing companies, service industries do not generally use _____ as a component of product cost.

Exercise Set A

EA 1. LO 4.1 Little Things manufactures toys. For each item listed, identify whether it is a product cost, a period cost, or not an expense.

A. internet provider services
B. material expense
C. raw materials inventory
D. production equipment rental
E. showroom rental
F. factory employee salary
G. Human Resource Director salary

EA 2. LO 4.2 Table 4.3 shows a list of expenses involved in the production of custom, professional lacrosse sticks.

A. For each item listed, state whether the cost should be applied to manufacturing or sales and administration.
B. If the cost is a manufacturing cost, state whether it is direct materials, direct labor, or manufacturing overhead.
C. If the cost is a manufacturing overhead cost, state whether it is indirect materials, indirect labor, or another type of manufacturing overhead.

Expenses Involved in Lacrosse Stick Production

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Carbon, fiberglass			
Administrative building rent			
Accountant salary			
Factory building depreciation			
Strings for the pocket			
Advertising			
Production supervisor salary			
Paint for sticks			
Research and development costs			
Wages of person who strings the sticks			
Cutting machine depreciation			

Table 4.3

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Human resources salaries			
Factory maintenance			

Table 4.3

EA 3. LO 4.2 Burnham Industries incurs these costs for the month:

Direct materials	$2,000
Direct labor	3,000
Factory depreciation expense	3,500
Factory utilities expense	750
CEO's salary	4,000

A. What is the prime cost?
B. What is the conversion cost?

EA 4. LO 4.3 Marzoni's records show raw materials inventory had a beginning balance of $200 and an ending balance of $300. If the cost of materials used during the month was $900, what were the purchases made during the month?

EA 5. LO 4.3 Sterling's records show the work in process inventory had a beginning balance of $4,000 and an ending balance of $3,000. How much direct labor was incurred if the records also show:

Materials used	$1,500
Overhead applied	500
Cost of goods manufactured	7,500

EA 6. LO 4.3 Logo Gear purchased $2,250 worth of merchandise during the month, and its monthly income statement shows cost of goods sold of $2,000. What was the beginning inventory if the ending inventory was $1,000?

EA 7. LO 4.4 A company estimates its manufacturing overhead will be $750,000 for the next year. What is the predetermined overhead rate given the following independent allocation bases?

A. Budgeted direct labor hours: 60,000
B. Budgeted direct labor expense: $1,500,000
C. Estimated machine hours: 100,000

EA 8. LO 4.4 Job order cost sheets show the following costs assigned to each job:

	Job 13	Job 14	Job 15
Direct materials	$7,560	$1,525	$3,290
Direct labor	3,760	3,824	3,796

The company assigns overhead at $1.25 for each direct labor dollar spent. What is the total cost for each of the jobs?

EA 9. LO 4.4 A new company started production. Job 10 was completed, and Job 20 remains in production. Here is the information from job cost sheets from their first and only jobs so far:

Job 10	Hours	Total Cost
Direct materials		$ 765
Direct labor	75	1,575
Manufacturing overhead		60
Total cost		$2,400

Job 20	Hours	Total Cost
Direct materials		$ 145
Direct labor	113	2,373
Manufacturing overhead		90
Total cost		$2,608

Using the information provided,

A. What is the balance in work in process?
B. What is the balance in the finished goods inventory?
C. If manufacturing overhead is applied on the basis of direct labor hours, what is the predetermined overhead rate?

EA10. LO 4.5 K company production was working on Job 1 and Job 2 during the month. Of the $780 in direct materials, $375 in materials was requested for Job 1. Direct labor cost, including payroll taxes, are $23 per hour, and employees worked 18 hours on Job 1 and 29 hours on Job 2. Overhead is applied at the rate of $20 per direct labor hours. Prepare job order cost sheets for each job.

EA11. LO 4.7 A company has the following transactions during the week.

- Purchase of $1,000 raw materials inventory
- Assignment of $500 of raw materials inventory to Job 5
- Payroll for 20 hours with $1,000 assigned to Job 5
- Factory utility bills of $750
- Overhead applied at the rate of $10 per hour

What is the cost assigned to Job 5 at the end of the week?

EA12. LO 4.7 During the month, Job AB2 used specialized machinery for 450 hours and incurred $500 in utilities on account, $300 in factory depreciation expense, and $100 in property tax on the factory. Prepare journal entries for the following:

A. Record the expenses incurred.
B. Record the allocation of overhead at the predetermined rate of $1.50 per machine hour.

EA13. LO 4.7 Job 113 was completed at a cost of $5,000, and Job 85 was completed at a cost of $3,000 and sold on account for $4,500. Prepare journal entries for the following:

A. Completion of Job 113.
B. Completion and sale of Job 85.

EA14. LO 4.7 A company's individual job sheets show these costs:

	Job 131	Job 132	Job 133
Direct materials	$4,585	$8,723	$1,575
Direct labor	2,385	2,498	2,874

Overhead is applied at 1.25 times the direct labor cost. Use the data on the cost sheets to perform these tasks:

A. Apply overhead to each of the jobs.
B. Prepare an entry to record the assignment of direct materials to work in process.
C. Prepare an entry to record the assignment of direct labor to work in process.
D. Prepare an entry to record the assignment of manufacturing overhead to work in process.

EA15. LO 4.7 A summary of material requisition slips and time tickets, along with the overhead allocation, show these costs:

Job No.	Material Requisition Slips	Factory Labor Time Tickets	Overhead Applied
131	$ 505	$ 200	$ 70
132	251	260	91
133	393	180	63
134	340	300	105
Not job specific	76	145	0
	$1,565	$1,085	$329

A. Prepare an entry to record the assignment of direct material to work in process.
B. Prepare an entry to record the assignment of direct labor to work in process.
C. Prepare an entry to record the assignment of manufacturing overhead to work in process.



Exercise Set B

EB 1. LO 4.1 Abuah Goods manufactures clothing. For each item listed, identify whether it is a product cost, a period cost, or not an expense.

A. pins to keep materials together while garment is being manufactured
B. real estate taxes on store
C. advertising expense
D. product inspector wages
E. shirts for sale
F. Chief Financial Officer salary
G. cost of goods sold

EB 2. LO 4.2 Choco's Chocolates incurs the following costs for the month:

Direct materials	$15,000
Direct labor	25,000
Factory depreciation expense	45,000
Factory utilities expense	2,000
Payroll staff's salary	15,000

A. What is the prime cost?
B. What is the conversion cost?

EB 3. LO 4.2 The table shows a list of expenses involved in the production of custom snowboard bindings.

A. For each item listed, state if the cost is manufacturing or sales and administration.
B. If the cost is a manufacturing cost, state if it is direct materials, direct labor, or manufacturing overhead.
C. If the cost is a manufacturing overhead cost, state if it is indirect materials, indirect labor, or another type of manufacturing overhead.

Snowboard Binding Production Costs

Snowboard Bindings Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Aluminum			
Factory building rent			
Fiberglass framework for each pair of bindings			
Accountant salary			
Administration building depreciation			
Straps			
Advertising			
Production supervisor salary			
Glue			
Research and development costs			
Inspector wages			
Metal shaping machine depreciation			

Table 4.4

Snowboard Bindings Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Human resources salaries			
Factory repair			

Table 4.4

EB 4. LO 4.3 Masonry's records show the raw materials inventory had purchases of $1,000 and an ending raw materials inventory balance of $200. If the cost of materials used during the month was $900, what was the beginning inventory?

EB 5. LO 4.3 Steinway's records show their work in process inventory had a beginning balance of $3,000 and an ending balance of $3,500. How much overhead was applied if the records also show the following:

Materials used	$2,500
Direct labor	5,000
Cost of goods manufactured	7,700

EB 6. LO 4.3 Langston's purchased $3,100 of merchandise during the month, and its monthly income statement shows a cost of goods sold of $3,000. What was the beginning inventory if the ending inventory was $1,250?

EB 7. LO 4.4 A company estimates its manufacturing overhead will be $840,000 for the next year. What is the predetermined overhead rate given each of the following independent allocation bases?

A. Budgeted direct labor hours: 90,615
B. Budgeted direct labor expense: $750,000
C. Estimated machine hours: 150,000

EB 8. LO 4.4 Job order cost sheets show the following costs assigned to each job:

	Job 131	Job 132	Job 133	Total
Direct materials	$3,485	$39,853	$2,301	$45,639
Direct labor	2,353	34,245	2,037	38,635

The company assigns overhead at twice the direct labor cost. What is the total cost for each job?

EB 9. LO 4.4 A new company started production. Job 1 was completed, and Job 2 remains in production. Here is the information from the job cost sheets from their first and only jobs so far:

Job 1	Hours	Total Cost
Direct materials		$ 375
Direct labor	231	5,313
Manufacturing overhead		4,620
Total cost		$10,308

Job 2	Hours	Total Cost
Direct materials		$ 405
Direct labor	85	1,955
Manufacturing overhead		1,700
Direct materials		$4,060

Using the information provided,

A. What is the balance in work in process?
B. What is the balance in finished goods inventory?
C. If manufacturing overhead is applied on the basis of direct labor hours, what is the predetermined overhead rate?

EB10. LO 4.5 Inez has the following information relating to Job AA5. Direct material cost was $200,000, direct labor was $36,550, and overhead applied on the basis of direct labor hours was $73,100. What was the predetermined overhead rate using the labor rate of $17 per hour?

EB11. LO 4.6 A company has the following information relating to its production costs:

Machine hours	25,000
Direct labor cost	$550,000
Indirect labor	45,000
Plant maintenance	259,300
Plant supervision	90,000
Plant depreciation	150,000
Plant utilities	48,000
Indirect materials	5,000

Compute the actual and applied overhead using the company's predetermined overhead rate of $23.92 per machine hour. Was the overhead overapplied or underapplied, and by how much?

EB12. LO 4.7 A company has the following transactions during the week.

- Purchase of $3,000 raw materials inventory
- Assignment of $700 of raw materials inventory to Job 7
- Payroll for 10 hours and $3,000 is assigned to Job 7
- Factory depreciation of $1,750
- Overhead applied at the rate of $200 per hour

What is the cost assigned to Job 7 at the end of the week?

EB13. LO 4.7 During the month, Job Arch2 used specialized machinery for 350 hours and incurred $700 in utilities on account, $400 in factory depreciation expense, and $200 in property tax on the factory. Prepare journal entries for the following:

A. Record the expenses incurred.
B. Record the allocation of overhead at the predetermined rate of $1.50 per machine hour.

EB14. LO 4.7 Job 113 was completed at a cost of $7,500, and Job 85 was completed at a cost of $2,300 and sold on account for $4,500. Prepare journal entries for the following:

A. Completion of Job 113.
B. Completion and sale of Job 85.

EB15. LO 4.7 A company's individual job sheets show these costs:

	Job 298	Job 299	Job 300
Direct material	$2,228	$23,945	$ 4,231
Direct labor	2,391	23,492	3,413
Overhead	?	?	?
Total	$8,803	$88,548	$13,617

Overhead is applied at 1.75 times the direct labor cost. Use the data on the cost sheets to perform these tasks:

A. Apply overhead to each of the jobs.
B. Prepare an entry to record the assignment of direct material to work in process.
C. Prepare an entry to record the assignment of direct labor to work in process.
D. Prepare an entry to record the assignment of manufacturing overhead to work in process.

EB16. LO 4.7 A summary of materials requisition slips and time tickets, along with the overhead allocation, show these costs:

Job Number	Material Requisition Slips	Factory Labor Time Tickets	Overhead Applied
AA001	$ 3,423	$ 5,004	$1,750
AA002	4,342	4,530	1,568
AA003	3,431	5,345	1,813
AA004	3,421	2,423	840
Not specific to one job	570	3,353	
	$15,187	$20,655	$5,971

A. Prepare an entry to record the assignment of direct material to work in process.
B. Prepare an entry to record the assignment of direct labor to work in process.
C. Prepare an entry to record the assignment of manufacturing overhead to work in process.

Problem Set A

PA 1. LO 4.1 For each item listed, state whether a job order costing system or process costing system would be best.

A. cereal
B. team uniforms
C. houses
D. beach chairs
E. plastic
F. restaurant-specific pizza boxes
G. sneakers customized with number and colors

PA 2. LO 4.4 York Company is a machine shop that estimated overhead will be $50,000, consisting of 5,000 hours of direct labor. The cost to make Job 0325 is $70 in aluminum and two hours of labor at $20 per hour. During the month, York incurs $50 in indirect material cost, $150 in administrative labor, $300 in utilities, and $250 in depreciation expense.

A. What is the predetermined overhead rate if direct labor hours are considered the cost driver?
B. What is the cost of Job 0325?
C. What is the overhead incurred during the month?

PA 3. LO 4.4 Pocono Cement Forms expects $900,000 in overhead during the next year. It does not know whether it should apply overhead on the basis of its anticipated direct labor hours of 60,000 or its expected machine hours of 30,000. Determine the product cost under each predetermined allocation rate if the last job incurred $1,550 in direct material cost, 90 direct labor hours, and 75 machine hours. Wages are paid at $16 per hour.

PA 4. LO 4.5 Job cost sheets show the following information:

Job	January	February	March	Completed	Sold
AA2	$2,500	$1,200		February	Not sold
AA4	4,838			January	February
AA5		3,250		February	March
AA3		3,409	$2,319	April	Not sold
Total	7,338	7,859	2,319		

What are the balances in the work in process inventory, finished goods inventory, and cost of goods sold for January, February, and March?

PA 5. LO 4.5 Complete the information in the cost computations shown here:

Raw Materials	
Beginning inventory	$ 342
Purchases	1,533
Materials available for use	?
Ending inventory	321
Materials used in production	?

Work in Process Inventory	
Beginning inventory	$ 932
Materials used in production	?
Direct labor	1,535
Overhead applied	?
Manufacturing costs incurred	22,441
Ending inventory	935
Cost of Goods Manufactured	?

Finished Goods Inventory	
Beginning inventory	?
Cost of Goods Manufactured	?
Goods Available for Sale	$25,002
Ending inventory	?
Cost of Goods Sold	21,788

PA 6. LO 4.5 During the year, a company purchased raw materials of $77,321, and incurred direct labor costs of $125,900. Overhead is applied at the rate of 75% of the direct labor cost. These are the inventory balances:

	Beginning	Ending
Raw materials inventory	$ 17,433	$ 16,428
Work in process inventory	241,439	234,423
Finished goods inventory	312,842	342,384

Compute the cost of materials used in production, the cost of goods manufactured, and the cost of goods sold.

PA 7. LO 4.5 Freeman Furnishings has summarized its data as shown:

Depreciation of factory building	$100,000
Factory real estate taxes	15,000
Factory utility expenses	85,000
Indirect materials	32,000
Indirect labor	25,000
Direct labor cost	85,000
Direct labor hours incurred	23,500
Estimated direct labor hours	24,000
Raw materials purchased	$350,000
Raw materials, beginning inventory	30,000
Raw materials, ending inventory	28,000
Work in process, beginning inventory	51,000
Work in process, ending inventory	67,000
Estimated overhead	270,000

Compute the cost of goods manufactured, assuming that the overhead is allocated based on direct labor hours.

PA 8. LO 4.6 Coop's Stoops estimated its annual overhead to be $85,000 and based its predetermined overhead rate on 24,286 direct labor hours. At the end of the year, actual overhead was $90,000 and the total direct labor hours were 24,100. What is the entry to dispose of the overapplied or underapplied overhead?

PA 9. LO 4.6 Mountain Peaks applies overhead on the basis of machine hours and reports the following information:

	Budget	Actual
Overhead	$350,000	$352,000
Machine hours	50,000	49,000
Direct materials		$210,000
Direct labor		$350,000

A. What is the predetermined overhead rate?
B. How much overhead was applied during the year?
C. Was overhead over- or underapplied, and by what amount?
D. What is the journal entry to dispose of the over- or underapplied overhead?

PA10. LO 4.6 The actual overhead for a company is $74,539. Overhead was based on 6,000 direct labor hours and was $2,539 underapplied for the year.
A. What is the overhead application rate per direct labor hour?
B. What is the journal entry to dispose of the underapplied overhead?

PA11. LO 4.6 When setting its predetermined overhead application rate, Tasty Box Meals estimated its overhead would be $100,000 and would require 25,000 machine hours in the next year. At the end of the year, it found that actual overhead was $102,000 and required 26,000 machine hours.
A. Determine the predetermined overhead rate.
B. What is the overhead applied during the year?
C. Prepare the journal entry to eliminate the underapplied or overapplied overhead.

PA12. LO 4.7 The following data summarize the operations during the year. Prepare a journal entry for each transaction.
A. Purchase of raw materials on account: $3,000
B. Raw materials used by Job 1: $500

C. Raw materials used as indirect materials: $100
D. Direct labor for Job 1: $300
E. Indirect labor incurred: $50
F. Factory utilities incurred on account: $700
G. Adjusting entry for factory depreciation: $250
H. Manufacturing overhead applied as percent of direct labor: 200%
I. Job 1 is transferred to finished goods
J. Job 1 is sold: $3,000
K. Manufacturing overhead is overapplied: $100

PA13. LO 4.7 The following events occurred during March for Ajax Company. Prepare a journal entry for each transaction.

A. Materials were purchased on account for $35,429.
B. Materials were requisitioned to begin work on Job C15 in the amount of $25,259.
C. Direct labor expense for Job C15 was $24,129.
D. Actual overhead was incurred on account of $32,852.
E. Factory overhead was charged to Job C15 at the rate of 200% of direct labor.
F. Job C15 was transferred to finished goods at $97,646.
G. Job C15 was sold on account for $401,000.

PA14. LO 4.8 A movie production studio incurred the following costs related to its current movie:

A. Purchased office supplies on account: $33,000
B. Issued direct supplies: $22,512
C. Issued indirect supplies: $7,535
D. Time tickets showing direct labor: $32,503,230
E. Time tickets showing indirect labor: $574,326
F. Utilities expense on account: $957,323
G. Overhead applied: 10% of direct labor cost

Create journal entries for the listed transactions.

Problem Set B

PB 1. LO 4.1 For each item listed, state whether a job order costing system or process costing system would be best.

A. television repair
B. cell phone charge cords
C. glassware with company logo
D. dog food
E. golf balls
F. hotel signs to welcome guests
G. highlighters and pens

PB 2. LO 4.4 Rulers Company is a neon sign company that estimated overhead will be $60,000, consisting of 1,500 machine hours. The cost to make Job 416 is $95 in neon, 15 hours of labor at $13 per hour, and five machine hours. During the month, it incurs $95 in indirect material cost, $130 in administrative labor, $320 in utilities, and $350 in depreciation expense.

A. What is the predetermined overhead rate if machine hours are considered the cost driver?
B. What is the cost of Job 416?
C. What is the overhead incurred during the month?

PB 3. LO 4.4 Event Forms expects $120,000 in overhead during the next year. It doesn't know whether it should apply overhead on the basis of its anticipated direct labor hours of 6,000 or its expected machine hours of 5,000. What would be the product cost under each predetermined allocation rate if the last job incurred $3,500 in direct material cost, 55 direct labor hours, and 55 machine hours? Wages are paid at $17 per hour.

PB 4. LO 4.5 Summary information from a company's job cost sheets shows the following information:

Job	April	May	June	Completed	Sold
BB3	$3,500	$1,500		May	Not sold
BB4	9,231			April	May
BB5		2,540		May	June
BB6		3,230	$1,434	July	Not sold

What are the balances in the work in process inventory, finished goods inventory, and cost of goods sold for April, May, and June?

PB 5. LO 4.5 Complete the information in the cost computations shown here:

Beginning Inventory	$74,323
Purchases	?
Materials available for use	?
Ending inventory	?
Materials used in production	78,413

Work in Process Inventory	
Beginning inventory	$253,210
Materials used in production	?
Direct labor	125,900
Overhead applied	94,425
Manufacturing costs incurred	?
Ending inventory	242,932
Cost of goods manufactured	?

Finished Goods Inventory	
Finished goods inventory	?
Beginning inventory	$333,149
Cost of goods manufactured	309,016
Goods available for sale	?
Ending inventory	354,235
Cost of goods sold	287,930

PB 6. LO 4.5 During the year, a company purchased raw materials of $77,321 and incurred direct labor costs of $125,900. Overhead is applied at the rate of 75% of the direct labor cost. These are the inventory balances:

	Beginning	Ending
Raw materials inventory	$ 15,394	$ 17,432
Work in process inventory	57,304	53,721
Finished goods inventory	120,432	132,432

Compute the cost of materials used in production, the cost of goods manufactured, and the cost of goods sold.

PB 7. LO 4.5 Freeman Furnishings has summarized its data as shown. Direct labor hours will be used as the activity base to allocate overhead:

Raw materials purchased	$320,000
Raw materials, beginning inventory	15,000
Raw materials, ending inventory	14,000
Work in process, beginning inventory	35,000
Work in process, ending inventory	37,000
Estimated overhead	300,000
Depreciation of factory building	50,000
Factory real estate taxes	7,382
Factory utility expenses	45,000
Indirect materials	20,000
Indirect labor	11,000
Direct labor cost	100,000
Direct labor hours incurred	24,000
Estimated direct labor hours	25,000

Compute the cost of goods manufactured.

PB 8. LO 4.6 Queen Bee's Honey, Inc., estimated its annual overhead to be $110,000 and based its predetermined overhead rate on 27,500 direct labor hours. At the end of the year, actual overhead was $106,000 and the total direct labor hours were 29,000. What is the entry to dispose of the overapplied or underapplied overhead?

PB 9. LO 4.6 Mountain Tops applies overhead on the basis of direct labor hours and reports the following information:

	Budget	Actual
Overhead	$450,000	$452,000
Direct labor hours	75,000	77,000
Direct materials		$195,000
Direct labor		$333,865

A. What is the predetermined overhead rate?
B. How much overhead was applied during the year?
C. Was overhead overapplied or underapplied, and by what amount?
D. What is the journal entry to dispose of the overapplied or underapplied overhead?

PB10. LO 4.6 The actual overhead for a company is $73,175. Overhead was based on 4,500 machine hours and was $3,325 overapplied for the year.
A. What is the overhead application rate per direct labor hour?
B. What is the journal entry to dispose of the underapplied overhead?

PB11. LO 4.6 When setting its predetermined overhead application rate, Tasty Turtle estimated its overhead would be $75,000 and manufacturing would require 25,000 machine hours in the next year. At the end of the year, it found that actual overhead was $74,000 and manufacturing required 24,000 machine hours.
A. Determine the predetermined overhead rate.
B. What is the overhead applied during the year?
C. Prepare the journal entry to eliminate the under- or overapplied overhead.

PB12. LO 4.7 The following data summarize the operations during the year. Prepare a journal entry for each transaction.
A. Purchase of raw materials on account: $1,500
B. Raw materials used by Job 1: $400

C. Raw materials used as indirect materials: $50
D. Direct labor for Job 1: $200
E. Indirect labor incurred for Job 1: $30
F. Factory utilities incurred on account: $500
G. Adjusting entry for factory depreciation: $200
H. Manufacturing overhead applied as percent of direct labor: 100%
I. Job 1 is transferred to finished goods
J. Job 1 is sold: $1,000
K. Manufacturing overhead is underapplied: $100

PB13. LO 4.7 The following events occurred during March for Ajax Company. Prepare a journal entry for each transaction.

A. Materials were purchased on account for $5,429.
B. Materials were requisitioned to begin work on Job C15 in the amount of $2,500.
C. Direct labor expense for Job C15 was $4,250.
D. Actual overhead was incurred on account for $5,385.
E. Factory overhead was charged to Job C15 at the rate of 200% direct labor.
F. Job C15 was transferred to finished goods at $15,250.
G. Job C15 was sold on account for $28,000.

PB14. LO 4.8 A leather repair shop incurred the following expenses while repairing luggage for a major airline.

A. Time cards showing direct labor: $750
B. Time cards showing indirect labor: $100
C. Purchased repair supplies on account: $1,500
D. Issued indirect supplies: $350
E. Utilities expense on account: $24,000
F. Overhead applied: 100% of direct labor costs

Journalize the listed transactions.



Thought Provokers

TP 1. LO 4.1 Can a company use both job order costing and process costing? Why or why not?

TP 2. LO 4.4 If a job order cost system tracks the direct materials and direct labor, why doesn't it track the actual overhead used for a specific job?

TP 3. LO 4.5 What are the similarities in calculating the cost of materials used in production, the cost of goods manufactured, and the cost of goods sold?

TP 4. LO 4.6 If a company bases its predetermined overhead rate on 100,000 machine hours, and it actually has 100,000 machine hours, would there be an underapplied or overapplied overhead?

TP 5. LO 4.7 How do the job cost sheets act as a subsidiary ledger for the work in process inventory if journal entries are not made to the job cost sheets?

TP 6. LO 4.8 How is a job order cost system used in a service industry?



5

Process Costing

Figure 5.1 David and William's Family Cookie Venture. (credit: modification of "Big Giant cookies" by Sayuri Miss/Flickr, CC BY 2.0)

Chapter Outline

- LO **5.1** Compare and Contrast Job Order Costing and Process Costing
- LO **5.2** Explain and Identify Conversion Costs
- LO **5.3** Explain and Compute Equivalent Units and Total Cost of Production in an Initial Processing Stage
- LO **5.4** Explain and Compute Equivalent Units and Total Cost of Production in a Subsequent Processing Stage
- LO **5.5** Prepare Journal Entries for a Process Costing System

Why It Matters

David and William's family has used a secret family recipe for generations to make amazing chocolate chip cookies. While in college, they helped their grandmother, who used only locally sourced products, make and sell the cookies to a local restaurant. They helped her become more efficient, discovered how to retain the quality taste while making larger batches, and developed a plate-sized version that could be decorated similar to a birthday cake. After creating an equally successful peanut butter cookie recipe, David and William decided to expand the business and sell to high-end grocers as well as to a second restaurant. They found it was optimal in terms of cost, efficiency, and quality to produce 100 cookies per batch for each regular-sized cookie and 5 cookies per batch for the large cookies. They surveyed restaurants and grocery stores and determined that each flavor should be offered in four different package sizes. They also analyzed the marketability at various sale prices. David and William now know they need to use their information to identify the costs associated with making the cookies. They need to know the cost to produce one unit of their product in order to price their cookies correctly, determine the optimal product mix, manage efficiency and process improvement, and make other management decisions.

5.1 Compare and Contrast Job Order Costing and Process Costing

As you've learned, job order costing is the optimal accounting method when costs and production specifications are not identical for each product or customer but the direct material and direct labor costs can

easily be traced to the final product. Job order costing is often a more complex system and is appropriate when the level of detail is necessary, as discussed in Job Order Costing. Examples of products manufactured using the job order costing method include tax returns or audits conducted by a public accounting firm, custom furniture, or, in a comprehensive example, semitrucks. At the Peterbilt factory in Denton, Texas, the company can build over 100,000 unique versions of their semitrucks without making the same truck twice.

Process costing is the optimal costing system when a standardized process is used to manufacture identical products and the direct material, direct labor, and manufacturing overhead cannot be easily or economically traced to a specific unit. Process costing is used most often when manufacturing a product in batches. Each department or production process or batch process tracks its direct material and direct labor costs as well as the number of units in production. The actual cost to produce each unit through a process costing system varies, but the average result is an adequate determination of the cost for each manufactured unit. Examples of items produced and accounted for using a form of the process costing method could be soft drinks, petroleum products, or even furniture such as chairs, assuming that the company makes batches of the same chair, instead of customizing final products for individual customers.

For example, small companies, such as David and William's, and large companies, such as Nabisco, use similar cost-determination processes. In order to understand how much each product costs—for example, Oreo cookies—Nabisco uses process costing to track the direct materials, direct labor, and manufacturing overhead used in the manufacturing of its products. Oreo production has six distinct steps or departments: (1) make the cookie dough, (2) press the cookie dough into a molding machine, (3) bake the cookies, (4) make the filling and apply it to the cookies, (5) put the cookies together into a sandwich, and (6) and place the cookies into plastic trays and packages. Each department keeps track of its direct materials used and direct labor incurred, and manufacturing overhead applied to facilitate determining the cost of a batch of Oreo cookies.



As previously mentioned, process costing is used when similar items are produced in large quantities. As such, many individuals immediately associate process costing with assembly line production. Process costing works best when products cannot be distinguished from each other and, in addition to obvious production line products like ice cream or paint, also works for more complex manufacturing of similar products like small engines. Conversely, products in a job order cost system are manufactured in small quantities and include custom jobs such as custom manufacturing products. They can also be legal or accounting tasks, movie production, or major projects such as construction activities.

The difference between process costing and job order costing relates to how the costs are assigned to the products. In either costing system, the ability to obtain and analyze cost data is needed. This results in the costing system selected being the one that best matches the manufacturing process.

A job order cost system is often more expensive to maintain than a basic process costing system, since there is a cost associated with assigning the individual material and labor to the product. Thus, a job order cost system is used for custom jobs when it is easy to determine the cost of materials and labor used for each job. A process cost system is often less expensive to maintain and works best when items are identical and it is difficult to trace the exact cost of materials and labor to the final product. For example, assume that your company uses three production processes to make jigsaw puzzles. The first process glues the picture on the cardboard backing, the second process cuts the puzzle into pieces, and the final process loads the pieces into the boxes and seals them. Tracing the complete costs for the batch of similar puzzles would likely entail three steps, with three separate costing system components. In this environment, it would be difficult and not economically feasible to trace the exact materials and the exact labor to each individual puzzle; rather, it would be more efficient to trace the costs per batch of puzzles.

The costing system used typically depends on whether the company can most efficiently and economically trace the costs to the job (favoring job order costing system) or to the production department or batch

(favoring a process costing system).

While the costing systems are different from each other, management uses the information provided to make similar managerial decisions, such as setting the sales price. For example, in a job order cost system, each job is unique, which allows management to establish individual prices for individual projects. Management also needs to establish a sales price for a product produced with a process costing system, but this system is not designed to stop the production process and individually cost each batch of a product, so management must set a price that will work for many batches of the product.

In addition to setting the sales price, managers need to know the cost of their products in order to determine the value of inventory, plan production, determine labor needs, and make long- and short-term plans. They also need to know the costs to determine when a new product should be added or an old product removed from production.

In this chapter, you will learn when and why process costing is used. You'll also learn the concepts of conversion costs and equivalent units of production and how to use these for calculating the unit and total cost of items produced using a process costing system.

Basic Managerial Accounting Terms Used in Job Order Costing and Process Costing

Regardless of the costing system used, manufacturing costs consist of direct material, direct labor, and manufacturing overhead. Figure 5.2 shows a partial organizational chart for Rock City Percussion, a drumstick manufacturer. In this example, two groups—administrative and manufacturing—report directly to the chief financial officer (CFO). Each group has a vice president responsible for several departments. The organizational chart also shows the departments that report to the production department, illustrating the production arrangement. The material storage unit stores the types of wood used (hickory, maple, and birch), the tips (nylon and felt), and packaging materials.

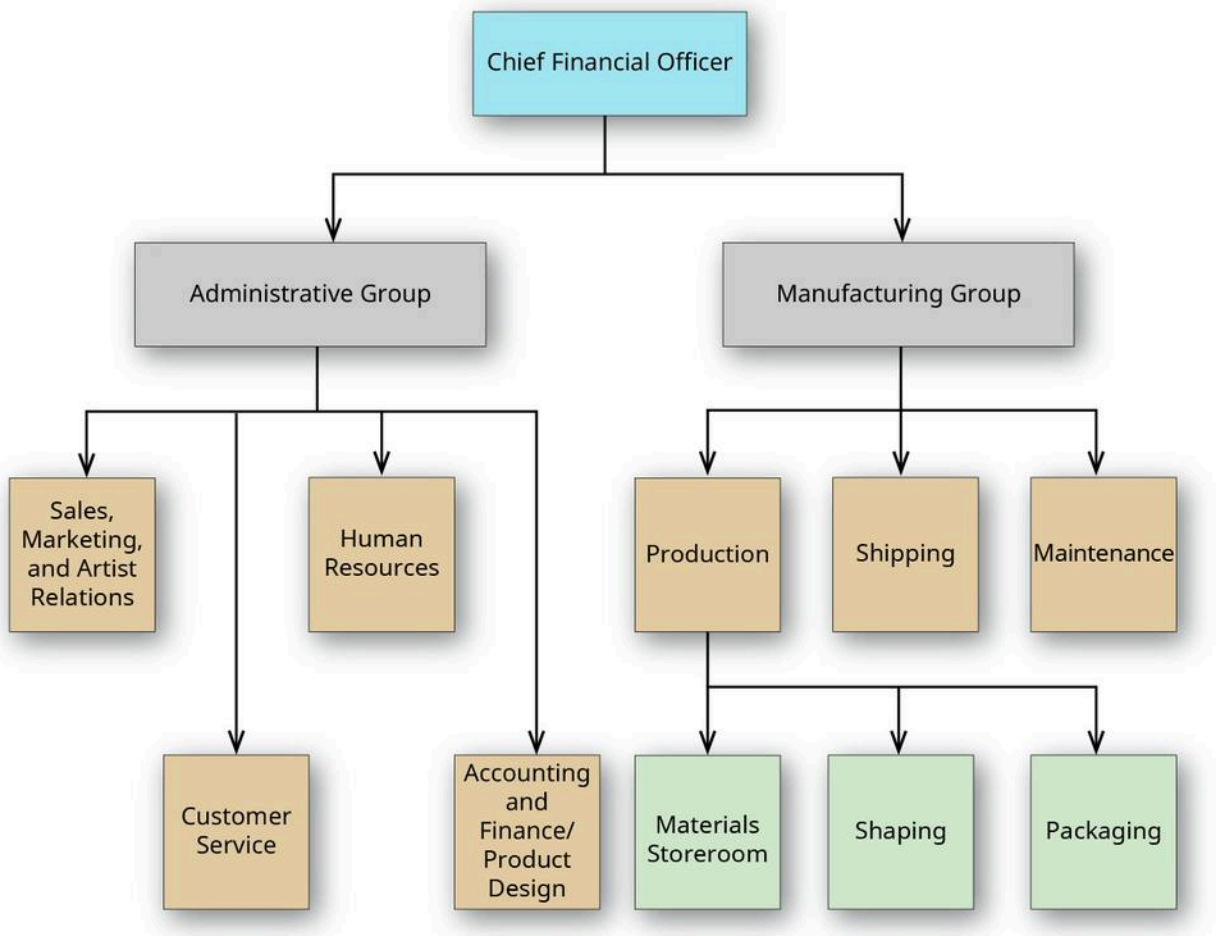


Figure 5.2 The Organizational Chart for Rock City Percussion. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Understanding the company's organization is an important first step in any costing system. Next is understanding the production process. The most basic drumstick is made of hickory and has a wooden tip. When the popular size 5A stick is manufactured, the hickory stored in the materials storeroom is delivered to the shaping department where the wood is cut into pieces, shaped into dowels, and shaped into the size 5A shape while under a stream of water. The sticks are dried, and then sent to the packaging department, where the sticks are embossed with the Rock City Percussion logo, inspected, paired, packaged, and shipped to retail outlets such as Guitar Center. The manufacturing process is described in Figure 5.3.

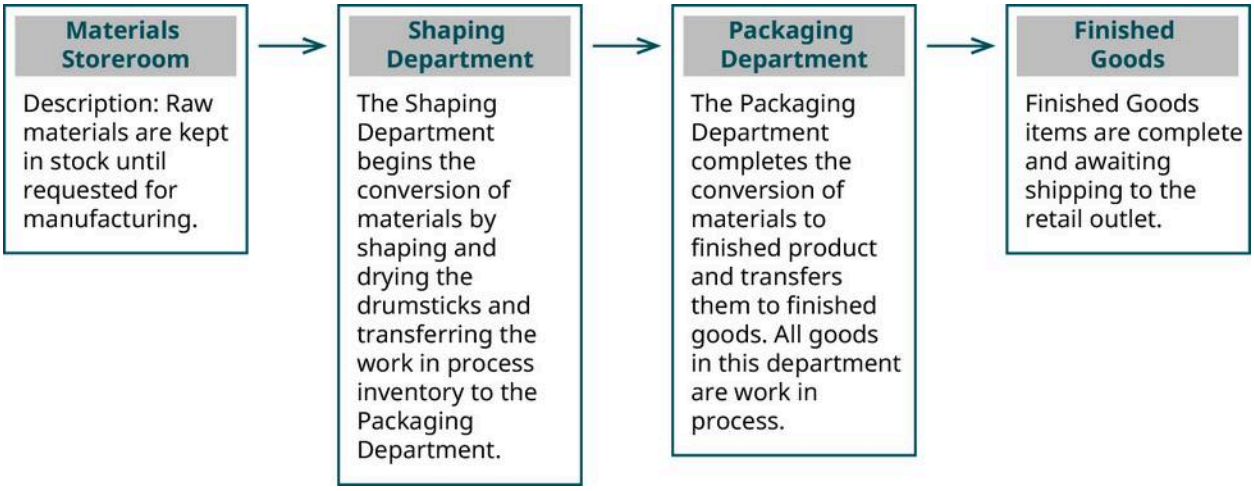


Figure 5.3 Rock City Percussion Manufacturing Process. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The different units within Rock City Percussion illustrate the two main cost categories of a manufacturing company: manufacturing costs and administrative costs.

LINK TO LEARNING

Understanding the full manufacturing process for a product helps with tracking costs. This video on how drumsticks are made (https://openstax.org/l/50drumsticks) shows the production process for drumsticks at one company, starting with the raw wood and ending with packaging.

Manufacturing Costs

Manufacturing costs or **product costs** include all expenses required to manufacture the product: direct materials, direct labor, and manufacturing overhead. Since process costing assigns the costs to each department, the inventory at the end of the period includes the finished goods inventory, and the work in process inventory for each manufacturing department. For example, using the departments shown in Figure 5.3, raw materials inventory is the cost paid for the materials that remain in the storeroom until requested.

While still in production, the work in process units are moved from one department to the next until they are completed, so the work in process inventory includes all of the units in the shaping and packaging departments. When the units are completed, they are transferred to finished goods inventory and become costs of goods sold when the product is sold.

When assigning costs to departments, it is important to separate the product costs from the **period costs**, which are those that are typically related with a particular time period, instead of attached to the production of an asset. Management often needs additional information to make decisions and needs the product costs further categorized as prime costs or conversion costs (Figure 5.4). **Prime costs** are costs that include the primary (or direct) product costs: direct material and direct labor. **Conversion costs** are the costs necessary to convert direct materials into a finished product: direct labor and manufacturing overhead, which includes other costs that are not classified as direct materials or direct labor, such as plant insurance, utilities, or property taxes. Also, note that direct labor is considered to be a component of both prime costs and conversion costs.

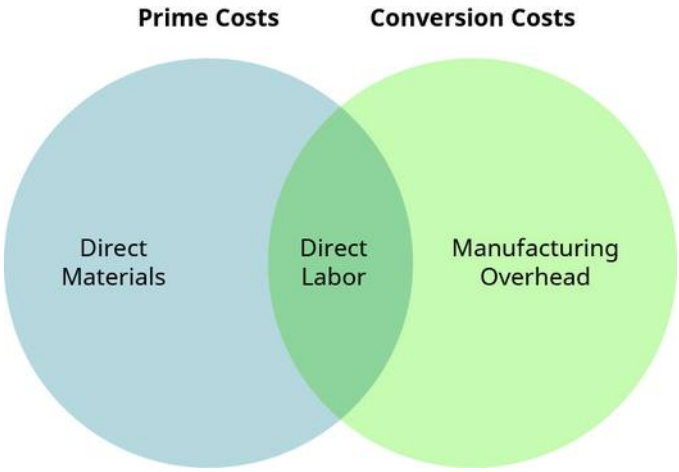


Figure 5.4 Prime Costs and Conversion Costs. Product costs can be categorized as prime costs (direct product costs) or conversion costs (costs incurred while converting the materials into a finished product). Direct labor is accounted for in both categories. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Job order costing tracks prime costs to assign direct material and direct labor to individual products (jobs). Process costing also tracks prime costs to assign direct material and direct labor to each production department (batch). Manufacturing overhead is another cost of production, and it is applied to products (job order) or departments (process) based on an appropriate activity base.

ETHICAL CONSIDERATIONS

The Unethical Bakery Accountant[1,2]

According to the Federal Bureau of Investigation (FBI), "Sandy Jenkins was a shy, daydreaming accountant at the Collin Street Bakery, the world's most famous fruitcake company. He was tired of feeling invisible, so he started stealing—and got a little carried away." Being unethical netted the accountant ten years in federal prison, and his wife Kay was sentenced to five years' probation and 100 hours of community service, and she was required to write a formal apology to the bakery. According to the FBI, "Jenkins spent over $11 million on a Black American Express card alone—roughly $98,000 per month over the course of the scheme—for a couple that had a legitimate income, through the Bakery, of approximately $50,000 per year."

How did this happen? *Texas Monthly* reports that Sandy found a way to write unapproved checks in the accounting system. He implemented his accounting system and created checks that were "signed" by the owner of the company, Bob McNutt. McNutt was perplexed as to why his bakery was not more profitable year after year. The accountant was stealing the money while making the stolen checks appear to be paying for material costs or operating costs. According to *Texas Monthly*, "Once Sandy was sure that nobody had noticed the first fraudulent check, he tried it again. And again and again. Each time, Sandy would repeat the scheme, pairing his fraudulent check with one that appeared legitimate. Someone would have to closely examine the checks to see any discrepancies, and that seemed unlikely." The multimillion dollar fraud was exposed when another accountant looked closely at the checks and noticed discrepancies.

Selling and Administrative Expenses

Selling and administrative (S&A) expenses are period costs, which means that they are recorded in the period in which they were incurred. Selling and administrative expenses typically are not directly assigned to

1 Katy Vine. "Just Desserts." *Texas Monthly*. October 2010. https://features.texasmonthly.com/editorial/just-desserts/
2 Federal Bureau of Investigation (FBI). "Former Collin Street Bakery Executive and Wife Sentenced." September 16, 2015. https://www.fbi.gov/contact-us/field-offices/dallas/news/press-releases/former-collin-street-bakery-executive-and-wife-sentenced

the items produced or services provided and include costs of departments not directly associated with manufacturing but necessary to operate the business. The selling costs component of S&A expenses is related to the promotion and sale of the company's products, while administrative expenses are related to the administration of the company. Some examples of S&A expenses include marketing costs; administration building rent; the chief executive officer's salary expense; and the accounting, payroll, and data processing department expenses.

These general rules for S&A expenses, however, have their exceptions. For example, some items that are classified as overhead, such as plant insurance, are period costs but are classified as overhead and are attached to the items produced as product costs.

The **expense recognition principle** is the primary reason to separate the costs of production from the other expenses of the company. This principle requires costs to be recorded in the period in which they are incurred. The costs are expensed when matched to the revenue with which they are associated; this is commonly referred to as *having the expenses follow the revenues*.

Period costs are expensed during the period in which they are incurred; this allows a company to apply the administrative and other expenses shown on the income statement to the same period in which the company earns income. Under generally accepted accounting principles (GAAP), separating the production costs and assigning them to the department results in the costs of the product staying with the work in process inventory for each department. This follows the expense recognition principle because the cost of the product is expensed when revenue from the sale is recognized.

Equivalent Units

In a process cost system, costs are maintained by each department, and the method for determining the cost per individual unit is different than in a job order costing system. Rock City Percussion uses a process cost system because the drumsticks are produced in batches, and it is not economically feasible to trace the direct labor or direct material, like hickory, to a specific drumstick. Therefore, the costs are maintained by each department, rather than by job, as they are in job order costing.

How does an organization determine the cost of each unit in a process costing environment? The costs in each department are allocated to the number of units produced in a given period. This requires determination of the number of units produced, but this is not always an easy process. At the end of the accounting period, there typically are always units still in production, and these units are only partially complete. Think of it this way: At midnight on the last day of the month, all accounting numbers need to be determined in order to process the financial statements for that month, but the production process does not stop at the end of each accounting period. However, the number of units produced must be calculated at the end of the accounting period to determine the number of **equivalent units**, or the number of units that would have been produced if the units were produced sequentially and in their entirety in a particular time period. The number of equivalent units is different from the number of actual units and represents the number of full or whole units that could have been produced given the amount of effort applied. To illustrate, consider this analogy. You have five large pizzas that each contained eight slices. Your friends served themselves, and when they were finished eating, there were several partial pizzas left. In equivalent units, determine how many whole pizzas are left if the remaining slices are divided as shown in Figure 5.5.

- Pie 1 had one slice
- Pie 2 had two slices
- Pie 3 had two slices
- Pie 4 had three slices
- Pie 5 had eight slices

Together, there are sixteen slices left. Since there are eight slices per pizza, the leftover pizza would be considered two full equivalent units of pizzas. The equivalent unit is determined separately for direct materials

and for conversion costs as part of the computation of the per-unit cost for both material and conversion costs.

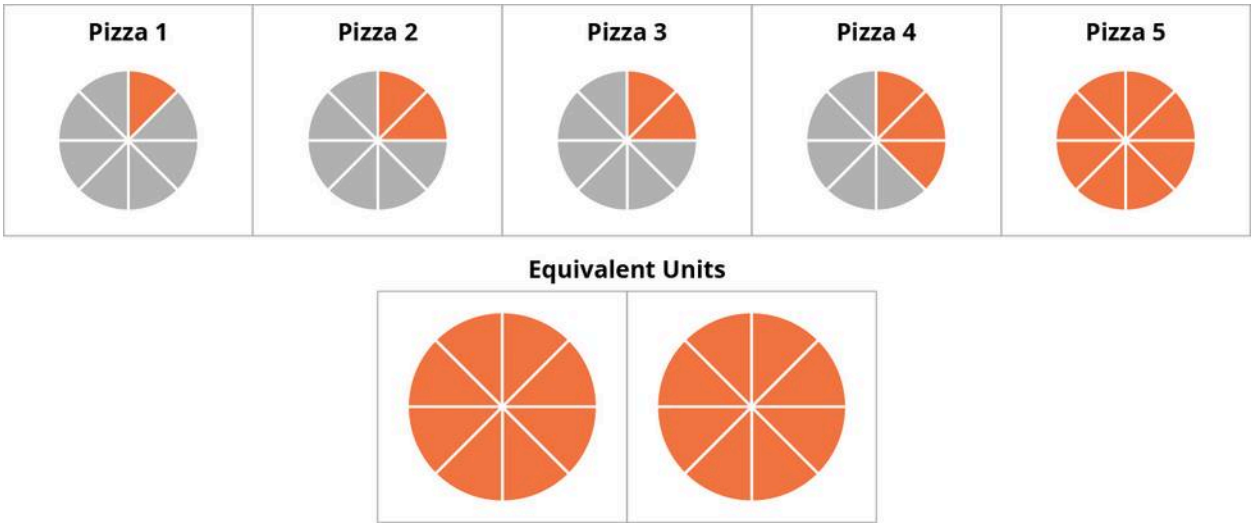


Figure 5.5 Equivalent Units. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Major Characteristics of Process Costing

Process costing is the optimal system for a company to use when the production process results in many similar units. It is used when production is continuous or occurs in large batches and it is difficult to trace a particular input cost to a specific individual product.

For example, before David and William found ways to make five large cookies per batch, their family always made one large cookie per batch. In order to make five cookies at a time, they had to gather the ingredients and baking materials, including five bowls and five cookie sheets. The exact amount of ingredients for one large cookie was mixed in each separate bowl and then placed on the cookie sheet. When this method was used, it was easy to establish that exactly one egg, two cups of flour, three-quarter cup of chocolate chips, three-quarter cup of sugar, one-quarter teaspoon salt, and so forth, were in each cookie. This made it easy to determine the exact cost of each cookie. But if David and William used one bowl instead of five bowls, measured the ingredients into it and then divided the dough into five large cookies, they could not know for certain that each cookie has exactly two cups of flour. One cookie may have 1 7/8 cups and another may have 1 15/16 cups, and one cookie may have a few more chocolate chips than another. It is also impossible to trace the chocolate chips from each bag to each cookie because the chips were mixed together. These variations do not affect the taste and are not important in this type of accounting. Process costing is optimal when the products are relatively homogenous or indistinguishable from one another, such as bottles of vegetable oil or boxes of cereal.

Often, process costing makes sense if the individual costs or values of each unit are not significant. For example, it would not be cost effective for a restaurant to make each cup of iced tea separately or to track the direct material and direct labor used to make each eight-ounce glass of iced tea served to a customer. In this scenario, job order costing is a less efficient accounting method because it costs more to track the costs per eight ounces of iced tea than the cost of a batch of tea. Overall, when it is difficult or not economically feasible to track the costs of a product individually, process costing is typically the best cost system to use.

Process costing can also accommodate increasingly complex business scenarios. While making drumsticks may sound simple, an immense amount of technology is involved. Rock City Percussion makes 8,000 hickory sticks per day, four days each week. The sticks made of maple and birch are manufactured on the fifth day of the week. It is difficult to tell the first drumstick made on Monday from the 32,000th one made on Thursday, so a computer matches the sticks in pairs based on the tone produced.

Process costing measures and assigns the costs to the associated department. The basic 5A hickory stick consists only of hickory as direct material. The rest of the manufacturing process involves direct labor and manufacturing overhead, so the focus is on properly assigning those costs. Thus, process costing works well for simple production processes such as cereal, rubber, and steel, and for more complicated production processes such as the manufacturing of electronics and watches, if there is a degree of similarity in the production process.

In a process cost system, each department accumulates its costs to compute the value of work in process inventory, so there will be a work in process inventory for each manufacturing or production department as well as an inventory cost for finished goods inventory. Manufacturing departments are often organized by the various stages of the production process. For example, blending, baking, and packaging could each be categorized as manufacturing or production departments for the cookie producer, while cutting, assembly, and finishing could be manufacturing or production departments with accompanying costs for a furniture manufacturer. Each department, or process, will have its own work in process inventory account, but there will only be one finished goods inventory account.

There are two methods used to compute the values in the work in process and finished goods inventories. The first method is the weighted-average method, which includes all costs (costs incurred during the current period and costs incurred during the prior period and carried over to the current period). This method is often favored, because in the process cost production method there often is little product left at the end of the period and most has been transferred out. The second method is the first-in, first-out (FIFO) method, which calculates the unit costs based on the assumption that the first units sold come from the prior period's work in process that was carried over into the current period and completed. After these units are sold, the newer completed units can then be sold. The theory is similar to the FIFO inventory valuation process that you learned about in Inventory (http://openstax.org/books/principles-financial-accounting/pages/10-why-it-matters). (Since the FIFO process costing method is more complicated than the weighted-average method, the FIFO method is typically covered in more advanced accounting courses.)

With processing, it is difficult to establish how much of each material, and exactly how much time is in each unit of finished product. This will require the use of the equivalent unit computation, and management selects the method (weighted average or FIFO) that best fits their information system.

Process costing can also be used by service organizations that provide homogeneous services and often do not have inventory to value, such as a hotel reservation system. Although they have no inventory, the hotel might want to know its costs per reservation for a period. They could allocate the total costs incurred by the reservation system based on the number of inquiries they served. For example, assume that in a year they incurred costs of $200,000 and served 50,000 potential guests. They could determine an average cost by dividing costs by number of inquiries, or $200,000/50,000 = $4.00 per potential guest.

In the case of a not-for-profit company, the same process could be used to determine the average costs incurred by a department that performs interviews. The department's costs would be allocated based on the number of cases processed. For example, assume a not-for-profit pet adoption organization has an annual budget of $180,000 and typically matches 900 shelter animals with new owners each year. The average cost would be $200 per match.

Similarities between Process Costing and Job Order Costing

Both process costing and job order costing maintain the costs of direct material, direct labor, and manufacturing overhead. The process of production does not change because of the costing method. The costing method is chosen based on the production process.

In job order cost production, the costs can be directly traced to the job, and the job cost sheet contains the total expenses for that job. Process costing is optimal when the costs cannot be traced directly to the job. For example, it would be impossible for David and William to trace the exact amount of eggs in each chocolate

chip cookie. It is also impossible to trace the exact amount of hickory in a drumstick. Even two sticks made sequentially may have different weights because the wood varies in density. These types of manufacturing are optimal for the process cost system.

The similarities between job order cost systems and process cost systems are the product costs of materials, labor, and overhead, which are used determine the cost per unit, and the inventory values. The differences between the two systems are shown in Table 5.1.

Differences between Job Order Costing and Process Costing

Job Order Costing	Process Costing
Product costs are traced to the product and recorded on each job's individual job cost sheet.	Product costs are traced to departments or processes.
Each department tracks its expenses and adds them to the job cost sheet. As jobs move from one department to another, the job cost sheet moves to the next department as well.	Each department tracks its expenses, the number of units started or transferred in, and the number of units transferred to the next department.
Unit costs are computed using the job cost sheet.	Unit costs are computed using the departmental costs and the equivalent units produced.
Finished goods inventory includes the products completed but not sold, and all incomplete jobs are work in process inventory.	Finished goods inventory is the number of units completed at the per unit cost. Work in process inventory is the cost per unit and the equivalent units remaining to be completed.

Table 5.1

CONCEPTS IN PRACTICE

Choosing Between Process Costing and Job Order Costing

Process costing and job order costing are both acceptable methods for tracking costs and production levels. Some companies use a single method, while some companies use both, which creates a hybrid costing system. The system a company uses depends on the nature of the product the company manufactures.

Companies that mass produce a product allocate the costs to each department and use process costing. For example, General Mills uses process costing for its cereal, pasta, baking products, and pet foods. Job order systems are custom orders because the cost of the direct material and direct labor are traced directly to the job being produced. For example, Boeing uses job order costing to manufacture planes.

When a company mass produces parts but allows customization on the final product, both systems are used; this is common in auto manufacturing. Each part of the vehicle is mass produced, and its cost is calculated with process costing. However, specific cars have custom options, so each individual car costs the sum of the specific parts used.

THINK IT THROUGH

Direct or Indirect Material

Around Again is a wooden frame manufacturer. Wood and fastener metals are typically added at the beginning of the process and are easily tracked as direct material. Sometimes, after inspection, the product needs to be reworked and additional pieces are added. Because the frames have already been through each department, the additional work is typically minor and often entails simply adding an additional fastener to keep the back of the frame intact. Other times, all the frame needs is additional glue for a corner piece.

How does a company differentiate between direct and indirect material? Many direct material costs, as the wood in the frame, are easy to identify as direct costs because the material is identifiable in the final product. But not all readily identifiable material is a direct material cost.

Technology makes it easy to track costs as small as one fastener or ounce of glue. However, if each fastener had to be requisitioned and each ounce of glue recorded, the product would take longer to make and the direct labor cost would be higher. So, while it is possible to track the cost of each individual product, the additional information may not be worth the additional expense. Managerial accountants work with management to decide which products should be accounted for as direct material and tracked individually, versus which should be considered indirect material and allocated to the departments through overhead application.

Should Around Again consider the fasteners or glue added after inspection as direct material or indirect material?

5.2 Explain and Identify Conversion Costs

In a processing environment, there are two concepts important to determining the cost of products produced. These are the concepts of equivalent units and conversion costs. As you have learned, equivalent units are the number of units that would have been produced if one unit was completed before starting a second unit. For example, four units that are one-fourth finished would equal one equivalent unit. Conversion costs are the labor and overhead expenses that "convert" raw materials into a completed unit. Each department tracks its conversion costs in order to determine the quantity and cost per unit (see Explain and Compute Equivalent Units and Total Cost of Production in an Initial Processing Stage details the steps to determine cost per unit; we discuss this concept in more detail later). Management often uses the cost information generated to set the sales price; to set standard usage data and price for material, labor, and overhead; and to allow management to evaluate the efficiency of production and plan for the future.

Definition of Conversion Costs

Conversion costs are the total of direct labor and factory overhead costs. They are combined because it is the labor and overhead together that convert the raw material into the finished product. Remember that factory, manufacturing, or organizational overhead (you might see all three terms in practice) is composed of three sources: indirect materials, indirect labor, and all other overhead costs that are not indirect materials or indirect labor. Materials are often added in stages at discrete points of production, such as at the beginning, middle, or end of a process, but conversion is usually applied equally throughout the process. For example, in the opening example, David and William do not add direct material (ingredients) evenly throughout the cookie-making process. They are all added at the beginning of the production process, so they begin with the direct materials but add labor and overhead throughout the rest of the process.

Conversion costs can be explained through the process of making Just Born's Peeps. Just Born makes 5.5 million Peeps per day using three ingredients and the following process:[3]

1. Use machines to add and mix the sugar, corn syrup, and gelatin into a mixture called a slurry. Send slurry through a whipper to give the marshmallow its fluffy texture.
2. Color the sugar.
3. Deposit marshmallows on sugar-coated belts in the Peep shape. Send Peeps on belts through a wind tunnel that stirs up the sugar to coat the entire shape.
4. Add eyes, and inspect.
5. Move the Peeps via belt into their appropriate tray, and wrap with cellophane.

In the Peep-making process, the direct materials of sugar, corn syrup, gelatin, color, and packaging materials are added at the beginning of steps 1, 2, and 5. While the fully automated production does not need direct labor, it does need indirect labor in each step to ensure the machines are operating properly and to perform inspections (step 4).

Mechanics of Applying Conversion Costs

Let's return to our drumstick example to learn how to work with conversion costs. Rock City Percussion has two departments critical to manufacturing drumsticks: the shaping and packaging departments.

The shaping department uses only wood as its direct material and water as its indirect material. In the shaping department, the material is added first. Then, machines cut the wood underwater into dowels, separate them, and move them to machines that shape the dowels into drumsticks. These machines need electricity to operate and personnel to monitor and adjust the processes and to maintain the equipment. When the shaping is finished, a conveyer belt transfers the sticks to the finishing department.

Since the drumsticks are made by performing one process on one batch at a time, instead of producing one stick at a time from start to finish, it is difficult to determine the exact materials, labor, and overhead for a single pair of drumsticks. It is easier to track the materials and conversion costs for one batch and have those costs follow the batch to the next process.

Therefore, once the batch of sticks gets to the second process—the packaging department—it already has costs attached to it. In other words, the packaging department receives both the drumsticks and their related costs from the shaping department. For the basic size 5A stick, the packaging department adds material at the beginning of the process. The 5A uses only packaging sleeves as its direct material, while other types may also include nylon, felt, and/or the ingredients for the proprietary handgrip. Direct labor and manufacturing overhead are used to test, weigh, and sound-match the drumsticks into pairs.

Thus, at the end of the accounting period, there are two work in process inventories: one in the shaping department and one in the packaging department.

Direct materials are added at the beginning of shaping and packaging departments, so the work in process inventory for those departments is 100% complete with regard to materials, but it is not complete with regard to conversion costs. If they were 100% complete with regard to conversion costs, then they would have been transferred to the next department.

LINK TO LEARNING

Management needs to understand its costs in order to set prices, budget for the upcoming year, and evaluate performance. Sometimes individuals become managers due to their knowledge of the production process but not necessarily the costs. Managers can view this information on the importance of identifying

3 Just Born. "Marshmallow Peeps Factory Tour." n.d. http://www.justborn.com/resource/corporate/popups/virtualTour.cfm

prime and conversion costs (https://openstax.org/l/50PrimeConvCost) from Investopedia, a resource for managers.

5.3 Explain and Compute Equivalent Units and Total Cost of Production in an Initial Processing Stage

As described previously, process costing can have more than one work in process account. Determining the value of the work in process inventory accounts is challenging because each product is at varying stages of completion and the computation needs to be done for each department. Trying to determine the value of those partial stages of completion requires application of the equivalent unit computation. The equivalent unit computation determines the number of units if each is manufactured in its entirety before manufacturing the next unit. For example, forty units that are 25% complete would be ten (40 × 25%) units that are totally complete.

Direct material is added in stages, such as the beginning, middle, or end of the process, while conversion costs are expensed evenly over the process. Often there is a different percentage of completion for materials than there is for labor. For example, if material is added at the beginning of the process, the forty units that are 100% complete with respect to material and 25% complete with respect to conversion costs would be the same as forty units of material and ten units (40 × 25%) completed with conversion costs.

For example, during the month of July, Rock City Percussion purchased raw material inventory of $25,000 for the shaping department. Although each department tracks the direct material it uses in its own department, all material is held in the material storeroom. The inventory will be requisitioned for each department as needed.

During the month, Rock City Percussion's shaping department requested $10,179 in direct material and started into production 8,700 hickory drumsticks of size 5A. There was no beginning inventory in the shaping department, and 7,500 drumsticks were completed in that department and transferred to the finishing department. Wood is the only direct material in the shaping department, and it is added at the beginning of the process, so the work in process (WIP) is considered to be 100% complete with respect to direct materials. At the end of the month, the drumsticks still in the shaping department were estimated to be 35% complete with respect to conversion costs. All materials are added at the beginning of the shaping process. While beginning the size 5A drumsticks, the shaping department incurred these costs in July:

Direct materials	$10,179
Direct labor	15,176
Applied overhead	7,000
Total cost	$32,355

These costs are then used to calculate the equivalent units and total production costs in a four-step process.

Step One: Determining the Units to Which Costs Will Be Assigned

In addition to the equivalent units, it is necessary to track the units completed as well as the units remaining in ending inventory. A similar process is used to account for the costs completed and transferred. Reconciling the number of units and the costs is part of the process costing system. The reconciliation involves the total of beginning inventory and units started into production. This total is called "units to account for," while the total of beginning inventory costs and costs added to production is called "costs to be accounted for." Knowing the total units or costs to account for is helpful since it also equals the units or costs transferred out plus the amount remaining in ending inventory.

When the new batch of hickory sticks was started on July 1, Rock City Percussion did not have any beginning

inventory and started 8,700 units, so the total number of units to account for in the reconciliation is 8,700:

Units to Account For	Units
Beginning work in process	0
Units started into production	8,700
Total units to account for	8,700

The shaping department completed 7,500 units and transferred them to the testing and sorting department. No units were lost to **spoilage**, which consists of any units that are not fit for sale due to breakage or other imperfections. Since the maximum number of units that could possibly be completed is 8,700, the number of units in the shaping department's ending inventory must be 1,200. The total of the 7,500 units completed and transferred out and the 1,200 units in ending inventory equal the 8,700 possible units in the shaping department.

Units Accounted For	Units
Completed and transferred out	7,500
Ending work in process	1,200
Total units to account for	8,700

Step Two: Computing the Equivalent Units of Production

All of the materials have been added to the shaping department, but all of the conversion elements have not; the numbers of equivalent units for material costs and for conversion costs remaining in ending inventory are different. All of the units transferred to the next department must be 100% complete with regard to that department's cost or they would not be transferred. So the number of units transferred is the same for material units and for conversion units. The process cost system must calculate the equivalent units of production for units completed (with respect to materials and conversion) and for ending work in process with respect to materials and conversion.

For the shaping department, the materials are 100% complete with regard to materials costs and 35% complete with regard to conversion costs. The 7,500 units completed and transferred out to the finishing department must be 100% complete with regard to materials and conversion, so they make up 7,500 (7,500 × 100%) units. The 1,200 ending work in process units are 100% complete with regard to material and have 1,200 (1,200 × 100%) equivalent units for material. The 1,200 ending work in process units are only 35% complete with regard to conversion costs and represent 420 (1,200 × 35%) equivalent units.

Work in Process Completion %		100%	35%
Units accounted for	Total Units	Material Units	Conversion Units
Completed and transferred out	7,500	7,500	7,500
Ending work in process	1,200	1,200	420
Total equivalent units for shaping	8,700	8,700	7,920

Step Three: Determining the Cost per Equivalent Unit

Once the equivalent units for materials and conversion are known, the cost per equivalent unit is computed in a similar manner as the units accounted for. The costs for material and conversion need to reconcile with the total beginning inventory and the costs incurred for the department during that month.

Costs to Account For	Materials	Conversion	Total
Beginning work in process	$ 0	$ 0	$ 0
Incurred during the period	$10,179	$22,176	$32,355
Total costs to account for	$10,179	$22,176	$32,355
Equivalent units	8,700	7,920	
Cost per equivalent unit	$ 1.17	$ 2.80	$ 3.97

The total materials costs for the period (including any beginning inventory costs) is computed and divided by the equivalent units for materials. The same process is then completed for the total conversion costs. The total of the cost per unit for material ($1.17) and for conversion costs ($2.80) is the total cost of each unit transferred to the finishing department ($3.97).

Step Four: Allocating the Costs to the Units Transferred Out and Partially Completed in the Shaping Department

Now you can determine the cost of the units transferred out and the cost of the units still in process in the shaping department. To calculate the goods transferred out, simply take the units transferred out times the sum of the two equivalent unit costs (materials and conversion) because all items transferred to the next department are complete with respect to materials and conversion, so each unit brings all its costs. But the ending WIP value is determined by taking the product of the work in process material units and the cost per equivalent unit for materials plus the product of the work in process conversion units and the cost per equivalent unit for conversion.

Transferred out costs	(7,500 units × $1.17) + (7,500 units × $2.80) = $29,775
Ending work in process: materials	(1,200 × $1.17) = $1,404
Ending work in process: conversion	(420 × $2.80) = $1,176
Ending work in process: total	$1,404 + $1,176 = $2,580

This information is accumulated in a **production cost report**. This report shows the costs used in the preparation of a product, including the cost per unit for materials and conversion costs, and the amount of work in process and finished goods inventory. A complete production cost report for the shaping department is illustrated in Figure 5.6.

Units to account for	Units		
Beginning work in process	—		
Units started into production	8,700		
Total units to account for	8,700		
Work in process completion %	100%	35%	
Units accounted for	**Materials Units**	**Conversion Units**	**Total**
Completed and transferred out	7,500	7,500	7,500
Ending work in process	1,200	420	1,200
Total units to account for	8,700	7,920	8,700
Costs to account for	**Materials**	**Conversion**	**Total**
Beginning work in process	$ 0	$ 0	$ 0
Incurred during the period	$10,179	$22,176	$32,355
Total costs to account for	$10,179	$22,176	$32,355
Equivalent units	8,700	7,920	
Cost per equivalent unit	$ 1.17	$ 2.80	$ 3.97
	Materials	**Conversion**	**Total**
Value of ending work in process	$ 1,404	$ 1,176	$ 2,580
Completed and transferred	8,775	21,000	29,775
Total costs	$10,179	$22,176	$32,355

Figure 5.6 Production Cost Report for the Shaping Department. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

YOUR TURN

Calculating Inventory Transferred and Work in Process Costs

Kyler Industries started a new batch of paint on October 1. The new batch consists of 8,700 cans of paint, of which 7,500 was completed and transferred to finished goods. During October, the manufacturing process recorded the following expenses: direct materials of $10,353; direct labor of $17,970; and applied overhead of $9,000. The inventory still in process is 100% complete with respect to materials and 30% complete with respect to conversion. What is the cost of inventory transferred out and work in process? Assume that there is no beginning work in process inventory.

Solution

Units to account for	Units		
Beginning work in process	0		
Units started into production	8,700		
Total units to be account for	8,700		
	Materials	**Conversion**	**Total**
Work in process completion %	100%	30%	
Units accounted for			
Completed and transferred out	7,500	7,500	7,500
Ending work in process	1,200	360	1,200
Total units to account for	8,700	7,860	8,700
Costs to account for	**Materials**	**Conversion**	**Total**
Beginning work in process	0	0	0
Incurred during the period	$10,353	$26,970	$37,323
Total costs to account for	$10,353	$26,970	$37,323
Equivalent units	8,700	7,860	
Cost per equivalent unit	$ 1.19	$ 3.43	$ 4.62

5.4 Explain and Compute Equivalent Units and Total Cost of Production in a Subsequent Processing Stage

In many production departments, units are typically transferred from the initial stage to the next stage in the process. When the units are transferred, the accumulated cost per unit is transferred along with them. Since the unit being produced includes work from all of the prior departments, the transferred-in cost is the cost of the work performed in all earlier departments.

When the hickory size 5A drumsticks have completed the shaping process, they are transferred to the packaging department along with the inventory costs of $29,775. The inventory costs of $29,775 were $8,775 for materials and $21,000 for conversion costs and were calculated in Figure 5.6. During the month of July, Rock City Percussion purchased raw material inventory of $2,000 for the packaging department. As with the shaping department, the packaging department tracks its costs and requisitions the raw material from the material storeroom. The packaging department has computed direct material costs of $2,000, direct labor costs of $13,000, and applied overhead of $9,100, for a total of $22,100 in conversion costs. Equivalent units are computed for this department, and a new cost per unit is computed.

	Materials	**Conversion**	**Total**
Beginning work in process	$ 1,600	$ 6,580	$ 8,180
Transferred in	8,775	21,000	29,775
Added this month	2,000	22,100	24,100
Total	$12,375	$49,680	$62,055

As with calculating the equivalent units and total cost of production in the initial processing stage, there are four steps for calculating these costs in a subsequent processing stage.

Step One: Determining the Stage 2 Units to Which Costs Will Be Assigned

In the initial manufacturing department, there is beginning inventory, and units are started in production. In subsequent stages, instead of starting new units, units are transferred in from the prior department, but the accounting process is the same. Returning to the example, Rock City Percussion had a beginning inventory of 750 units in the packaging department. When the 7,500 sticks are transferred into the packaging department

from the shaping department, the total number of units to account for in the reconciliation is 8,250, which is the total of the beginning WIP and the units transferred in:

Units to Account For	Units
Beginning work in process	750
Units transferred in	7,500
Total units to account for	8,250

The reconciliation of units to account for are the same for each department. The units that were completed and transferred out plus the ending inventory equal the total units to account for. The packaging department for Rock City Percussion completed 6,500 units and transferred them into finished goods inventory. Since the maximum number of units to possibly be completed is 8,250 and no units were lost to spoilage, the number of units in the packaging department's ending inventory must be 1,750. The total of the 6,500 units completed and transferred out and the 1,750 units in ending inventory equal the 8,250 possible units in the packaging department.

Units Accounted For	Units
Completed and transferred out	6,500
Ending work in process	1,750
Total units to account for	8,250

Step Two: Computing the Stage 2 Equivalent Units of Production

The only direct material added in the packaging department for the 5A sticks is packaging. The packaging materials are added at the beginning of the process, so all the materials have been added before the units are transferred out, but all of the conversion elements have not. As a result, the number of equivalent units for material costs and for conversion costs remaining in ending inventory is different for the testing and sorting department. As you've learned, all of the units transferred to the next department must be 100% complete with regard to that department's cost, or they would not be transferred. The process cost system must calculate the equivalent units of production for units completed (with respect to materials and conversion) and for ending WIP with respect to materials and conversion.

For the packaging department, the materials are 100% complete with regard to materials costs and 40% complete with regard to conversion costs. The 6,500 units completed and transferred out to the finishing department must be 100% complete with regard to materials and conversion, so they make up 6,500 (6,500 × 100%) units. The 1,750 ending WIP units are 100% complete with regard to material and have 1,750 (1,750 × 100%) equivalent units for material. The 1,750 ending WIP units are only 40% complete with regard to conversion costs and represent 700 (1,750 × 40%) equivalent units.

Work in process completion %		100%	40%
Units accounted for	Total Units	Material Units	Conversion Units
Completed and transferred out	6,500	6,500	6,500
Ending work in process	1,750	1,750	700
Total equivalent units for packaging	8,250	8,250	7,200

Step Three: Determining the Stage 2 Cost per Equivalent Unit

Once the equivalent units for materials and conversion are known for the packaging department, the cost per equivalent unit is computed in a manner similar to the calculation for the units accounted for. The costs for material and conversion need to reconcile with the department's beginning inventory and the costs incurred for the department during that month.

Costs to account for	Materials	Conversion	Total
Beginning work in process	$ 1,600	$ 6,580	$ 8,180
Incurred during the period	$10,775	$43,100	$53,875
Total costs to account for	$12,375	$49,680	$62,055
Equivalent units	8,250	7,200	15,450
Cost per equivalent unit	$ 1.50	$ 6.90	$ 8.40

The total materials costs for the period (including any beginning inventory costs) are computed and divided by the equivalent units for materials. The same process is then completed for the total conversion costs. The total of the cost per unit for materials ($1.50) and for conversion costs ($6.90) is the total cost of each unit transferred to the testing and sorting department.

Step Four: Allocating the Costs to the Units in the Finishing Department

Now you can determine the cost of the units transferred out and the cost of the units still in process in the finishing department. For the goods transferred out, simply take the units transferred out times the sum of the two equivalent unit costs (materials and conversion) because all items transferred to the next department are complete with respect to materials and conversion, so each unit brings all its costs. But the ending WIP value is determined by taking the product of the work in process materials units and the cost per equivalent unit for materials plus the product of the work in process conversion units and the cost per equivalent unit for conversion.

Transferred out costs	(6,500 units × $1.50) + (6,500 units × $6.90) = $54,600
Ending work in process: materials	(1,750 × $1.50) = $2,625
Ending work in process: conversion	(700 × $6.90) = $4,830
Ending work in process: total	$2,625 + $4,830 = $7,455

LINK TO LEARNING

Knowing the cost to produce a unit is critical to management's decisions. Sometimes that knowledge leads to management's decision to stop production, but sometimes that decision isn't as simple as it seems. The cost to produce a penny is more than one cent, and yet, the United States still makes pennies. See this article from *Forbes* that explains the difference among *cost*, *worth*, and *value* (https://openstax.org/l/50CostValWorth) to learn more.

5.5 Prepare Journal Entries for a Process Costing System

Calculating the costs associated with the various processes within a process costing system is only a part of the accounting process. Journal entries are used to record and report the financial information relating to the transactions. The example that follows illustrates how the journal entries reflect the process costing system by recording the flow of goods and costs through the process costing environment.

Purchased Materials for Multiple Departments

Each department within Rock City Percussion has a separate work in process inventory account. Raw materials totaling $33,500 were ordered prior to being requisitioned by each department: $25,000 for the shaping department and $8,500 for the packaging department. The July 1 journal entry to record the purchases on account is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Raw Materials Inventory Accounts Payable *To record purchase of raw materials*	33,500	33,500

Direct Materials Requisitioned by the Shaping and Packaging Departments and Indirect Material Used

During July, the shaping department requisitioned $10,179 in direct material. Similar to job order costing, indirect material costs are accumulated in the manufacturing overhead account. The overhead costs are applied to each department based on a predetermined overhead rate. In the example, assume that there was an indirect material cost for water of $400 in July that will be recorded as manufacturing overhead. The journal entry to record the requisition and usage of direct materials and overhead is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Shaping Department Manufacturing Overhead Materials Inventory *To record direct and indirect material cost for July in the Shaping Department*	10,179 400	10,579

During July, the packaging department requisitioned $2,000 in direct material and overhead costs for indirect material totaled $300 for the month of July. The journal entry to record the requisition and usage of materials is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Packaging Department Manufacturing Overhead Materials Inventory *To record direct and indirect material cost for July in the Packaging Department*	2,000 300	2,300

Direct Labor Paid by All Production Departments

During July, the shaping department incurred $15,000 in direct labor costs and $600 in indirect labor. The journal entry to record the labor costs is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Shaping Department Manufacturing Overhead Wages Payable *To record direct and indirect labor for July in the Shaping Department*	15,000 600	15,600

During July, the packaging department incurred $13,000 of direct labor costs and indirect labor of $1,000. The journal entry to record the labor costs is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Packaging Department	13,000	
	Manufacturing Overhead	1,000	
	Wages Payable		14,000
	To record direct and indirect labor for July in the Packaging Department		

Applied Manufacturing Overhead to All Production Departments

Manufacturing overhead includes indirect material, indirect labor, and other types of manufacturing overhead. It is difficult, if not impossible, to trace manufacturing overhead to a specific product, and yet, the total cost per unit needs to include overhead in order to make management decisions.

Overhead costs are accumulated in a manufacturing overhead account and applied to each department on the basis of a predetermined overhead rate. Properly allocating overhead to each department depends on finding an activity that provides a fair basis for the allocation. It needs to be an activity common to each department and influential in driving the cost of manufacturing overhead. In traditional costing systems, the most common activities used are machine hours, direct labor in dollars, or direct labor in hours. If the number of machine hours can be related to the manufacturing overhead, the overhead can be applied to each department based on the machine hours. The formula for overhead allocation is:

$$\text{Overhead Allocation} = \frac{\text{Estimated Overhead Costs (\$)}}{\text{Expected Annual Activity (machine hours)}}$$

Rock City Percussion determined that machine hours is the appropriate base to use when allocating overhead. The estimated annual overhead cost is $340,000 per year. It was also estimated that the total machine hours will be 34,000 hours, so the allocation rate is computed as:

$$\frac{\text{Estimated Overhead Costs (\$340,000)}}{\text{Expected Annual Activity (34,000)}} = \text{\$10 per machine hour}$$

The shaping department used 700 machine hours, and with an overhead application rate of $10 per direct labor hour, the journal entry to record the overhead allocation is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Shaping Department	7,000	
	Manufacturing Overhead		7,000
	To record overhead applied to the Shaping Department		

The finishing department used 910 machine hours, and with an overhead application rate of $10 per direct labor hour, the journal entry to record the overhead allocation is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Finishing Department	9,100	
	Manufacturing Overhead		9,100
	To record overhead applied to the Finishing Department		

Transferred Costs of Finished Goods from the Shaping Department to the Packaging Department

When the units are transferred from the shaping department to the packaging department, they are transferred at $3.97 per unit, as calculated previously. The amount transferred from the shaping department is the same amount listed on the production cost report in Figure 5.6. The journal entry is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Work in Process Inventory: Finishing Department	29,775	
	Work in Process Inventory: Shaping Department		29,775
	To record the weighted-average method of the cost of goods transferred from the Shaping Department to the Packaging Department		

Transferred Goods from the Packaging Department to Finished Goods

The computation of inventory for the packaging department is shown in Figure 5.7.

Costs to Account For		**Units to Account For**	
Beginning work in process	$ 8,180	Beginning WIP	750
Direct material	2,000	Units started into production	7,500
Direct labor	13,000	Total units to account for:	8,250
Applied overhead	9,100		
Transferred in	29,775		
Total costs to account for	$62,055		

Step 1: Units to Assign Costs

Completed and transferred out	6,500
Ending work in process	1,750
Total units to account for	8,250

Step 2: Computing Equivalent Units of Production

Work in process completion %		40%	
	Material Units	**Conversion Units**	**Total**
Completed and transferred out	6,500	6,500	6,500
Ending work in process	1,750	700	1,750
Total units to account for	8,250	7,200	8,250

Step 3: Determining Cost per Equivalent Unit

Costs to account for	**Materials**	**Conversion**	**Total**
Beginning work in process	$ 1,600	$ 6,580	$ 8,180
Incurred during the period	$10,775	$43,100	$53,875
Total costs to account for	$12,375	$49,680	$62,055
Equivalent units	8,250	7,200	
Cost per equivalent unit for department	$ 1.50	$ 6.90	$ 8.40

Step 4: Allocating the Costs to Units Transferred to Finished Goods and Partially Completed Units

Transferred-out costs	(6,500 units × $1.50) + (6,500 units × $8.40) =	$54,600
End work in process: materials	(1,750 × $1.50) = $2,625	
End work in process: conversion	(700 × $6.90) = $4,830	
End work in process: total	$2,625 + $4,830 =	7,455
Total Costs Accounted For		$62,055

Figure 5.7 Inventory Computation for Packaging Department. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

The value of the inventory transferred to finished goods in the production cost report is the same as in the journal entry:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Finished Goods Inventory	54,600	
	Work in Process Inventory: Packaging Department		54,600
	To transfer completed goods into Finished Goods		

Recording the Cost of Goods Sold Out of the Finished Goods Inventory

Each unit is a package of two drumsticks that cost $8.40 to make and sells for $24.99. There are two transactions when recording a sale. One entry is to transfer the inventory from finished goods inventory to cost of goods sold and is at the cost of the product. The second transaction is to record the sale at the sales price. The compound entry to record both transactions for the sale of 500 units on account is:

JOURNAL			
Date	**Account**	**Debit**	**Credit**
July 1	Cost of Goods Sold	4,200	
	Finished Goods Inventory		4,200
	To record the manufacturing costs of items sold		
July 1	Accounts Receivable	12,495	
	Sales		12,495
	To record the sale of 500 units		

LINK TO LEARNING

The importance of properly recording the production process is illustrated in this report on work in process inventory (https://openstax.org/l/50WorkInProcess) from InventoryOps.com.

Key Terms

conversion cost total of labor and overhead for a product; the costs that "convert" the direct material into the finished product

equivalent units number of units that would have been produced if the units were produced sequentially and in their entirety in a particular time period

expense recognition principle (also, matching principle) matches expenses with associated revenues in the period in which the revenues were generated

manufacturing costs (also, product costs) total of all costs expended in the manufacturing process; generally consists of direct material, direct labor, and manufacturing overhead

period costs typically related to a particular time period instead of attached to the production of an asset; treated as an expense in the period incurred (examples include many sales and administrative expenses)

prime costs direct material expenses and direct labor costs

process costing costing system used when a standardized process is used to manufacture identical products and the direct material, direct labor, and manufacturing overhead cannot be traced to a specific unit

product costs all expenses required to manufacture the product: direct materials, direct labor, and manufacturing overhead

production cost report shows the costs used in the preparation of a product, including the cost per unit for materials and conversion costs and the amount of work in process and finished goods inventory

selling and administrative (S&A) expenses period costs not directly assigned to the items produced or services provided; include costs of departments not directly associated with manufacturing but necessary to operate the business

spoilage any units that are not fit for sale due to breakage or other imperfections

Summary

5.1 Compare and Contrast Job Order Costing and Process Costing

- The three categories of costs incurred in producing an item are direct material, direct labor, and manufacturing overhead.
- Process costing is the system of accumulating costs within each department for large-volume, mass-produced units.
- Process costing often groups direct labor and manufacturing overhead as conversion costs.
- Costs under GAAP are categorized as period costs when they are not related to production and instead cover a time period.
- Selling and administrative costs are period costs related to the sales of products and management of the company and are not directly tied to a specific product.
- Process costing determines the cost per unit through the use of equivalent units, or the number of units that would have been produced if production was sequential instead of in batches.

5.2 Explain and Identify Conversion Costs

- Conversion costs are the costs of direct labor and manufacturing overhead used to convert raw materials into a finished product.
- Materials are added during various stages of the manufacturing process, such as the beginning or end, while conversion of the product from raw material into finished goods is considered to occur uniformly through the process. Thus, it is possible for a product to have all of its materials and not be complete.
- Equivalent units for direct materials can be different than the equivalent units for conversion costs because materials are added in steps through the manufacturing process, while conversion costs are incurred evenly throughout the process.

5.3 Explain and Compute Equivalent Units and Total Cost of Production in an Initial Processing Stage

- Process costing has a work in process inventory account for each department.
- Equivalent units of production for materials may differ from the equivalent units for conversion costs.
- The total units to account for is the number of units in the beginning work in process inventory plus the number of units started into production; this total also represents the sum of the number of units completed and the number of units in the ending work in process inventory.
- The cost per equivalent unit for materials is the total of the material costs for the beginning work in process inventory and the total of material costs incurred during the period.
- The cost per equivalent unit for conversion costs is the total of the conversion costs for the beginning work in process inventory and the total of conversion costs incurred during the period.
- The cost of units transferred to the next department is the number of units transferred times the total of the cost per equivalent unit of material plus the cost per equivalent unit for conversion costs.

5.4 Explain and Compute Equivalent Units and Total Cost of Production in a Subsequent Processing Stage

- The total units to account for is the number of units in the beginning work in process inventory plus the number of units transferred from the prior department; this total also represents the number of units completed plus the number of units in the ending work in process inventory.
- The cost per equivalent units for materials is the total of the material costs for the beginning work in process inventory plus the cost of material transferred in to the department plus the total of material costs incurred during the period.
- The cost per equivalent unit for conversion costs is the total of the conversion costs for the beginning work in process inventory plus the conversion costs transferred in plus the total of conversion costs incurred during the period.

5.5 Prepare Journal Entries for a Process Costing System

- Traditional journal entries show the purchase of material and the incurring of overhead costs.
- Each department records the transfer of material from the storeroom into production, its direct labor costs, the application of overhead, and the transfer of goods to the next department or finished goods.
- The value of the inventory transferred to the next department or to finished goods equals the amount listed as transferred on the production cost report.

Multiple Choice

1. LO 5.1 Which of the following production characteristics is better suited for process costing and *not* job order costing?

A. Each product batch is distinguishable from the prior batch.
B. The costs are easily traced to a specific product.
C. Costs are accumulated by department.
D. The value of work in process is the direct material used, the direct labor incurred, and the overhead applied to the job in process.

2. LO 5.1 A process costing system is most likely used by which of the following?

A. airplane manufacturing
B. a paper manufacturing company
C. an accounting firm specializing in tax returns
D. a hospital

3. LO 5.1 Which of the following is a prime cost?

A. direct labor
B. work in process inventory
C. administrative labor
D. factory maintenance expenses

4. LO 5.1 Which of the following is a conversion cost?
A. raw materials
B. direct labor
C. sales commissions
D. direct material used

5. LO 5.1 During production, how are the costs in process costing accumulated?
A. to cost of goods sold
B. to each individual product
C. to manufacturing overhead
D. to each individual department

6. LO 5.2 Which is *not* needed to compute equivalent units of production?
A. the percentage of completion for inventory still in process
B. the number of units transferred out
C. the number of units started and completed
D. the material cost per unit

7. LO 5.2 What is the cost of direct labor if the conversion costs are $330,000 and manufacturing overhead is $275,000?
A. $55,000
B. $275,000
C. $330,000
D. $605,000

8. LO 5.2 What is the conversion cost to manufacture insulated travel cups if the costs are: direct materials, $17,000; direct labor, $33,000; and manufacturing overhead, $70,000?
A. $16,000
B. $50,000
C. $103,000
D. $120,000

9. LO 5.2 Which of the following lists contains only conversion costs for an inflatable raft manufacturing corporation?
A. vinyl for raft, machine operator, electricity, insurance
B. machine operator, electricity, depreciation, plastic for air valves
C. machine operator, electricity, depreciation, insurance
D. vinyl for raft, electricity, insurance, plastic for air valves

10. LO 5.3 Direct material costs $3 per unit, direct labor costs $5 per unit, and overhead is applied at the rate of 100% of the direct labor cost. What is the value of the inventory transferred to the next department if beginning inventory was 2,000 units; 9,000 units were started; and 1,000 units were in ending inventory?
A. $1,000
B. $13,000
C. $130,000

D. $20,000

11. LO 5.3 Beginning inventory and direct material cost added during the month total $55,000. What is the value of the ending work in process inventory if beginning inventory was 2,000 units; 9,000 units were started; and 1,000 units were in ending inventory?

A. $1,000
B. $5,000
C. $50,000
D. $55,000

12. LO 5.3 The initial processing department had a beginning inventory of 750 units and an ending inventory of 1,350 units, and it started 9,500 units into production. How many were transferred out to the next department?

A. 750
B. 1,350
C. 8,900
D. 10,250

13. LO 5.3 There were 1,000 units in ending inventory after transferring 16,000 units to finished goods inventory. If the beginning inventory was 2,000 units, how many units were started in process?

A. 1,000
B. 2,000
C. 15,000
D. 17,000

14. LO 5.4 The costs to be accounted for consist of which of the following?

A. costs of the units added during the period
B. costs of the units in ending inventory
C. costs of the units started and transferred during the period
D. costs of the units in the beginning inventory and costs added during the period

15. LO 5.4 Which of the following is the step in which materials, labor, and overhead are detailed?

A. determining the units to which costs are assigned
B. determining the equivalent units of production
C. determining the cost per equivalent units
D. allocating the costs to the units transferred out and the units partially completed

16. LO 5.5 The journal entry to record the $500 of work in process ending inventory that consists of $300 of direct materials, $50 of manufacturing overhead, and $150 of direct labor is which of the following?

A.

Work in Process Inventory	500	
Accounts Payable		500

(When added to the process)

B.

Work in Process Inventory	500	
Materials Inventory		300
Wages Payable		150
Manufacturing Overhead		50

(As incurred or applied)

C.

Cost of Goods Sold	500	
Work in Process Inventory		500

(When transferred)

D.

Accounts Payable	500	
Work in Process Inventory		500

(When added to the process)

17. LO 5.5 Assigning indirect costs to departments is completed by _____.
A. applying the predetermined overhead rate
B. debiting the manufacturing costs incurred
C. applying the costs to manufacturing overhead
D. applying the costs to work in process inventory

18. LO 5.5 In a process costing system, which account shows the overhead assigned to the department?
A. cost of goods sold
B. finished goods inventory
C. raw material inventory
D. work in process inventory

19. LO 5.5 In a process cost system, factory depreciation expense incurred is debited to _____.
A. finished goods inventory
B. work in process inventory
C. manufacturing overhead
D. cost of goods sold

Questions

1. LO 5.1 Explain how process costing differs from job order costing.

2. LO 5.1 Would a pharmaceutical manufacturer use process or job order costing? Why?

3. LO 5.1 Which costs are assigned using the weighted-average method?

4. LO 5.1 What is the primary purpose of process costing?

5. LO 5.2 What is the difference between prime costs and conversion costs?

6. LO 5.2 Explain conversion costs using an example.

7. LO 5.2 Why are there conversion costs in both job order costing and process costing?

8. LO 5.2 What are equivalent units of production, and how are they used in process costing?

9. LO 5.2 How can there be a different number of equivalent units for materials as compared to conversion costs?

10. LO 5.2 Why is the number of equivalent units for materials only sometimes equal to the equivalent units for conversion?

11. LO 5.3 What are the four steps involved in determining the cost of inventory transferred from one department to the next and the cost of work in process inventory?

12. LO 5.3 What is the weighted-average method for computing the equivalent units of production?

13. LO 5.4 How does process costing treat the costs transferred in from another department?

14. LO 5.4 Why does each department have its own work in process inventory?

15. LO 5.5 Match each term with its description.

A. conversion costs	i. total of direct material costs and direct labor costs
B. cost of goods sold	ii. manufacturing costs of the items sold
C. cost of production report	iii. number of units produced if each unit was produced sequentially
D. cost per equivalent unit	iv. total of direct labor costs, indirect labor costs, indirect material costs, and manufacturing overhead
E. equivalent units of production	v. where costs in a process cost system are reported before being applied to the product
F. manufacturing department	vi. detailed listing of the total costs of the product including the value of work in process
G. prime costs	vii. cost of materials or conversion for a specific department during production
H. transferred out costs	viii. product of the total cost per unit and the number of units completed and transferred during the time period

16. LO 5.5 How is manufacturing overhead handled in a process cost system?

17. LO 5.5 How are predetermined overhead rates used in process costing?

Exercise Set A

EA 1. LO 5.1 How many units were started into production in a period if there were zero units of beginning work in process inventory, 1,100 units in ending work in process inventory, and 21,500 completed and transferred out units?

EA 2. LO 5.1 A company started a new product, and in the first month started 100,000 units. The ending work in process inventory was 20,000 units that were 100% complete with materials and 75% complete with conversion costs. There were 100,000 units to account for, and the equivalent units for materials was $6 per unit while the equivalent units for conversion was $8 per unit. What is the value of the inventory transferred out, using the weighted-average inventory method?

EA 3. LO 5.3 Given the following information, determine the equivalent units of ending work in process for materials and conversion under the weighted-average method:

- beginning inventory of 2,500 units is 100% complete with regard to materials and 60% complete with regard to conversion
- 18,000 units were started during the period
- 17,500 units were completed and transferred

- ending inventory is 100% complete with materials and 65% complete with conversion

EA 4. LO 5.3 There were 1,700 units in beginning inventory that were 40% complete with regard to conversion. During the month, 8,550 units were started and 9,000 were transferred to finished goods. The ending work in process was 60% complete with regard to conversion costs, and materials are added at the beginning of the process. What is the total amount of equivalent units for materials and conversion at the end of the month using the weighted-average method?

EA 5. LO 5.3 A company has 1,500 units in ending work in process that are 30% complete after transferring out 10,000 units. All materials are added at the beginning of the process. If the cost per unit is $4 for materials and $7 for conversion, what is the cost of units transferred out and in ending work in process inventory using the weighted-average method?

EA 6. LO 5.3 There were 2,400 units in ending work in process inventory that were 100% complete with regard to material and 25% complete with regard to conversion costs. Ending work in process inventory had a cost of $9,000 and a per-unit material cost of $2. What was the conversion cost per unit using the weighted-average method?

EA 7. LO 5.3 How many units must be in ending inventory if beginning inventory was 15,000 units, 55,000 units were started, and 57,000 units were completed and transferred out?

EA 8. LO 5.3 How many units must have been completed and transferred if beginning inventory was 75,000 units, ending inventory was 72,000 units, and 290,000 units were started?

EA 9. LO 5.3 Using the weighted-average method, compute the equivalent units of production if the beginning inventory consisted of 20,000 units; 55,000 units were started in production; and 57,000 units were completed and transferred to finished goods inventory. For this process, materials are added at the beginning of the process, and the units are 35% complete with respect to conversion.

EA10. LO 5.3 Using the weighted-average method, compute the equivalent units of production for a new company that started 85,000 units into production and transferred 67,000 to the second department. Assume that beginning inventory was 0. Conversion is considered to occur evenly throughout the process, while materials are added at the beginning of the process. The ending inventory for Equivalent Units: Conversion is 9,000 units.

EA11. LO 5.3 Mazomanie Farm completed 20,000 units during the quarter and has 2,500 units still in process. The units are 100% complete with regard to materials and 55% complete with regard to conversion costs. What are the equivalent units for materials and conversion?

EA12. LO 5.3 What are the total costs to account for if a company's beginning inventory had $231,432 in materials, $186,450 in conversion costs, and added direct material costs ($4,231,392), direct labor ($2,313,392), and manufacturing overhead ($1,156,696)?

EA13. LO 5.3 A company started the month with 8,329 units in work in process inventory. It started 23,142 units and had an ending inventory of 9,321. The units were 100% complete to materials and 67% complete with conversion. How many units were transferred out during the period?

EA14. LO 5.4 A production department within a company received materials of $10,000 and conversion costs of $10,000 from the prior department. It added material of $27,200 and conversion costs of $53,000. The equivalent units are 20,000 for material and 18,000 for conversion. What is the unit cost for materials and conversion?

EA15. LO 5.4 Production data show 35,920 units were transferred out of a stage of production and 6,150 units remained in ending WIP inventory that was 100% complete to material and 35% complete to conversion. The unit material cost is $5 for material and $8 for conversion. What is the amount of inventory transferred out and remaining in ending work in process inventory?

EA16. LO 5.5 Overhead is assigned to the manufacturing department at the rate of $10 per machine hour. There were 3,500 machine hours during October in the shaping department and 2,500 in the packaging department. Prepare the journal entry to apply overhead to the manufacturing departments.

EA17. LO 5.5 Prepare the journal entry to record the factory wages of $28,000 incurred for a single production department assuming payment will be made in the next pay period.

EA18. LO 5.5 Prepare the journal entry to record the transfer of 3,000 units from the packaging department to finished goods if the material cost per unit is $4 and the conversion cost per unit is $5.50.

EA19. LO 5.5 Prepare the journal entry to record the sale of 2,000 units that cost $8 per unit and sold for $15 per unit.

Exercise Set B

EB 1. LO 5.3 Given the following information, determine the equivalent units of ending work in process for materials and conversion using the weighted-average method:

- Beginning inventory of 750 units is 100% complete with regard to materials and 30% complete with regard to conversion.
- 9,500 units were started during the period.
- 8,900 units were completed and transferred.
- Ending inventory is 100% complete with regard to materials and 68% complete with regard to conversion.

EB 2. LO 5.3 There were 2,000 units in beginning inventory that were 70% complete with regard to conversion. During the month, 15,000 units were started, and 16,000 were transferred to finished goods. The ending work in process was 55% complete with regard to conversion costs, and materials are added at the beginning of the process. What is the total amount of equivalent units for materials and conversion at the end of the month using the weighted-average method?

EB 3. LO 5.3 A company has 100 units in ending work in process that are 40% complete after transferring out 750 units. If the cost per unit is $5 for materials and $2.50 for conversion, what is the cost of units transferred out and in ending work in process inventory using the weighted-average method?

EB 4. LO 5.3 There were 1,500 units in ending work in process inventory that were 100% complete with regard to material and 60% complete with regard to conversion costs. Ending work in process inventory had a cost of $7,200 and a per-unit material cost of $3. What was the conversion cost per unit using the weighted-average method?

EB 5. LO 5.3 Using the weighted-average method, compute the equivalent units of production if the beginning inventory consisted of 20,000 units, 55,000 units were started in production, and 57,000 units were completed and transferred to finished goods inventory. For this process, materials are 70% complete and the units are 30% complete with respect to conversion.

EB 6. LO 5.3 What are the total costs to account for if a company's beginning inventory had $23,432 in materials and $18,450 in conversion costs, and added direct material costs ($41,392), direct labor ($23,192), and manufacturing overhead ($62,500)?

EB 7. LO 5.3 A company started the month with 4,519 units in work in process inventory. It started 15,295 units and had an ending inventory of 4,936. The units were 100% complete to materials and 30% complete with conversion. How many units were transferred out during the period?

EB 8. LO 5.4 A production department within a company received materials of $7,000 and conversion costs of $5,000 from the prior department. It added material of $78,400 and conversion costs of $47,000. The equivalent units are 5,000 for material and 4,000 for conversion. What is the unit cost for materials and

conversion?

EB 9. LO 5.4 Production data show 15,200 units were transferred out of a stage of production and 3,500 units remained in ending WIP inventory that was 100% complete to material and 60% complete to conversion. The unit material cost is $9 for material and $4 for conversion. What is the amount of inventory transferred out and remaining in ending work in process inventory?

EB10. LO 5.5 Overhead is assigned to the manufacturing department at the rate of $5 per machine hour. There were 3,000 machine hours used in the molding department. Prepare the journal entry to apply overhead to the manufacturing department.

EB11. LO 5.5 Prepare the journal entry to record the factory wages of $25,000 incurred in the processing department and $15,000 incurred in the production department assuming payment will be made in the next pay period.

EB12. LO 5.5 Prepare the journal entry to record the transfer of 3,500 units from the separation department to the mash department if the material cost per unit is $2 and the conversion cost per unit is $5.

EB13. LO 5.5 Prepare the journal entry to record the sale of 700 units that cost $5 per unit and sold for $15 per unit.

Problem Set A

PA 1. LO 5.1 The following product costs are available for Haworth Company on the production of chairs: direct materials, $15,500; direct labor, $22,000; manufacturing overhead, $16,500; selling expenses, $6,900; and administrative expenses, $15,200.

A. What are the prime costs?
B. What are the conversion costs?
C. What is the total product cost?
D. What is the total period cost?
E. If 7,750 equivalent units are produced, what is the equivalent material cost per unit?
F. If 22,000 equivalent units are produced, what is the equivalent conversion cost per unit?

PA 2. LO 5.1 The following product costs are available for Arrez Company on the production of DVD cases: direct materials, $1,450; direct labor, $15.50; manufacturing overhead, applied at 150% of direct labor cost; selling expenses, $1,550; and administrative expenses, $950. The direct labor hours worked for the month are 90 hours.

A. What are the prime costs?
B. What are the conversion costs?
C. What is the total product cost?
D. What is the total period cost?
E. If 1,450 equivalent units are produced, what is the equivalent material cost per unit?
F. What is the equivalent conversion cost per unit?

PA 3. LO 5.3 Pant Risers manufactures bands for self-dressing assistive devices for mobility-impaired individuals. Manufacturing is a one-step process where the bands are cut and sewn. This is the information related to this year's production:

Units to Account For	Units	Materials	Conversion
Beginning WIP inventory	500	500	250
Started	20,500		
To account for	21,000		

Ending inventory was 100% complete as to materials and 70% complete as to conversion, and the total

materials cost is $57,540 and the total conversion cost is $36,036. Using the weighted-average method, what are the unit costs if the company transferred out 17,000 units? What is the value of the inventory transferred out and the value of the ending WIP inventory?

PA 4. LO 5.3 During March, the following costs were charged to the manufacturing department: $14,886 for materials; $14,656 for labor; and $13,820 for manufacturing overhead. The records show that 30,680 units were completed and transferred, while 2,400 remained in ending inventory. There were 33,080 equivalent units of material and 31,640 of conversion costs. Using the weighted-average method, what is the cost of inventory transferred and the balance in work in process inventory?

PA 5. LO 5.3 Materials are added at the beginning of a production process, and ending work in process inventory is 30% complete with respect to conversion costs. Use the information provided to complete a production cost report using the weighted-average method.

Costs to Account For	
Beginning inventory: materials	$ 10,000
Beginning inventory: conversion	19,000
Direct materials	50,000
Direct labor	75,000
Applied overhead	37,248
Total costs to account for	$191,248

Units to Account For	
Beginning work in process	5,000
Units started into production	20,000
Transferred out	19,000

PA 6. LO 5.3 Narwhal Swimwear has a beginning work in process inventory of 13,500 units and transferred in 130,000 units before ending the month with 14,000 units that were 100% complete with regard to materials and 30% complete with regard to conversion costs. The cost per unit of material is $5.80 and the cost per unit for conversion is $8.20 per unit. Using the weighted-average method, what is the amount of material and conversion costs assigned to the department for the month?

PA 7. LO 5.3 The following data show the units in beginning work in process inventory, the number of units started, the number of units transferred, and the percent completion of the ending work in process for conversion. Given that materials are added at the beginning of the process, what are the equivalent units for material and conversion costs for each quarter using the weighted-average method? Assume that the quarters are independent.

Quarter	Beginning Work in Process	Started	Transferred Out	Conversion (%)
1	—	134,250	133,250	25
2	523	132,593	132,541	24
3	338	133,245	133,443	20
4	489	134,843	134,232	23

PA 8. LO 5.4 The finishing department started the month with 700 units in WIP inventory. It received 2,200 units from the molding department and transferred out 2,150 units. How many units were in process at the end of the month?

PA 9. LO 5.4 The packaging department began the month with 500 units that were 100% complete with regard to material and conversion. It received 9,500 units from the processing department and ended the month with 750 units that were 100% complete with regard to materials and 30% complete with regard to conversion. With a $5 per unit cost for conversion and a $5 per unit cost for materials, what is the cost of the units transferred out and remaining in ending inventory?

PA10. LO 5.4 Production information shows these costs and units for the smoothing department in August.

	Cost		Units
Work in process			
Beginning balance: materials	$ 1,550	Beginning units	650
Beginning balance: conversion	2,500	Transferred in	1,780
Materials	7,441	Transferred out	1,810
Labor	14,520		
Overhead	7,930		

All materials are added at the beginning of the period. The ending work in process is 30% complete as to conversion. What is the value of the inventory transferred to finished goods and the value of the WIP inventory at the end of the month?

PA11. LO 5.4 Given the following information, prepare a production report with materials added at the beginning and ending work in process inventory being 25% complete with regard to conversion costs.

Costs to Account For	
Beginning inventory: materials	$ 5,000
Beginning inventory: conversion	1,000
Direct materials	24,170
Direct labor	44,500
Applied overhead	12,400
Transferred-in materials	8,000
Transferred-in conversion	10,200
Total costs to account for	$105,270

Units to Account For	
Beginning work in process	3,500
Units started into production	85,000
Transferred out	84,000

PA12. LO 5.4 Complete this production cost report:

	Units		
Beginning inventory	20,000		
Started during the month	75,000		
Total units to account for	?		
Completed and transferred out	70,000		
Ending work in process	?		
Total units accounted for	?		
Work in process completion %	100%	25%	
	Material Units	**Conversion Units**	**Total**
Completed and transferred out	70,000	70,000	70,000
Ending work in process	?	?	?
Total units accounted for	?	?	?
Costs to account for	**Materials**	**Conversion**	**Total**
Beginning work in process	$ 3,500	$16,000	$19,500
Transferred in	0	0	0
Incurred during the period	$25,000	$45,000	$70,000
Total costs to account for	?	?	?
Equivalent units	?	?	
Cost per equivalent unit for department	?	?	
Transferred-out costs		?	
End work in process: materials	?		
End work in process: conversion	?		
End work in process: total		?	
Total costs accounted for		?	

PA13. LO 5.5 Selected information from Skylar Studios shows the following:

Raw materials purchased	$ 5,000
Direct materials	$ 1,500
Direct labor	$11,000
Factory depreciation	$ 2,000
Total manufacturing overhead incurred	$15,000
Machine hours per month	28,000
Unit cost for material	$ 2
Unit cost of conversion	$ 8
Number units transferred	15,000

Prepare journal entries to record the following:

A. raw material purchased
B. direct labor incurred
C. depreciation expense (hint: this is part of manufacturing overhead)
D. raw materials used
E. overhead applied on the basis of $0.50 per machine hour
F. the transfer from department 1 to department 2

PA14. LO 5.5 Loanstar had 100 units in beginning inventory before starting 950 units and completing 800 units. The beginning work in process inventory consisted of $2,000 in materials and $5,000 in conversion costs before $8,500 of materials and $11,200 of conversion costs were added during the month. The ending WIP inventory was 100% complete with regard to materials and 40% complete with regard to conversion costs. Prepare the journal entry to record the transfer of inventory from the manufacturing department to the finished goods department.

Problem Set B

PB 1. LO 5.3 The following product costs are available for Stellis Company on the production of erasers: direct materials, $22,000; direct labor, $35,000; manufacturing overhead, $17,500; selling expenses, $17,600; and administrative expenses; $13,400.

A. What are the prime costs?
B. What are the conversion costs?
C. What is the total product cost?
D. What is the total period cost?
E. If 13,750 equivalent units are produced, what is the equivalent material cost per unit?
F. If 17,500 equivalent units are produced, what is the equivalent conversion cost per unit?

PB 2. LO 5.3 The following product costs are available for Kellee Company on the production of eyeglass frames: direct materials, $32,125; direct labor, $23.50; manufacturing overhead, applied at 225% of direct labor cost; selling expenses, $22,225; and administrative expenses, $31,125. The direct labor hours worked for the month are 3,200 hours.

A. What are the prime costs?
B. What are the conversion costs?
C. What is the total product cost?
D. What is the total period cost?
E. If 6,425 equivalent units are produced, what is the equivalent material cost per unit?
F. What is the equivalent conversion cost per unit?

PB 3. LO 5.3 Vexar manufactures nails. Manufacturing is a one-step process where the nails are forged. This is the information related to this year's production:

Units to Account For	Units	Materials	Conversion
Beginning work in process inventory	1,000	1,000	500
Started	41,000	?	?
To account for	42,000	?	?

Ending inventory was 100% complete as to materials and 70% complete as to conversion, and the total materials cost is $115,080 and the total conversion cost is $72,072. Using the weighted-average method, what are the unit costs if the company transferred out 34,000 units? Using the weighted-average method, prepare the company's process cost summary for the month.

PB 4. LO 5.3 During March, the following costs were charged to the manufacturing department: $22,500 for materials; $45,625 for labor; and $50,000 for manufacturing overhead. The records show that 40,000 units were completed and transferred, while 10,000 remained in ending inventory. There were 45,000 equivalent units of material and 42,500 units of conversion costs. Using the weighted-average method, prepare the company's process cost summary for the month.

PB 5. LO 5.3 Ardt-Barger has a beginning work in process inventory of 5,500 units and transferred in 25,000 units before ending the month with 3,000 units that were 100% complete with regard to materials and 80% complete with regard to conversion costs. The cost per unit of material is $5.45, and the cost per unit for conversion is $6.20 per unit. Using the weighted-average method, prepare the company's process cost summary for the month.

PB 6. LO 5.3 The following data show the units in beginning work in process inventory, the number of units started, the number of units transferred, and the percent completion of the ending work in process for conversion. Given that materials are added at the beginning of the process, what are the equivalent units for material and conversion costs for each quarter using the weighted-average method? Assume that the quarters are independent.

Quarter	Beginning Work in Process	Started	Transferred Out	Conversion (%)
1	—	255,000	235,000	50
2	2,500	275,000	250,000	40
3	3,600	290,000	275,000	45
4	6,500	300,000	280,000	60

PB 7. LO 5.3 The following data show the units in beginning work in process inventory, the number of units started, the number of units transferred, and the percent completion of the ending work in process for conversion. Given that materials are added 50% at the beginning of the process and 50% at the end of the process, what are the equivalent units for material and conversion costs for each quarter using the weighted-average method? Assume that the quarters are independent.

Quarter	Beginning Work in Process	Started	Transferred Out	Conversion (%)
1	—	134,250	133,250	25
2	523	132,593	132,541	24
3	338	133,245	133,443	20
4	489	134,843	134,232	23

PB 8. LO 5.3 The following data show the units in beginning work in process inventory, the number of units started, the number of units transferred, and the percent completion of the ending work in process for conversion. Given that materials are added 50% at the beginning of the process and 50% at the end of the process, what are the equivalent units for material and conversion costs for each quarter using the weighted-average method? Assume that the quarters are independent.

Quarter	Beginning Work in Process	Started	Transferred Out	Conversion (%)
1	—	255,000	235,000	50
2	2,500	275,000	250,000	40
3	3,600	290,000	275,000	45
4	6,500	300,000	280,000	60

PB 9. LO 5.4 The finishing department started the month with 600 units in WIP inventory. It received 1,500 units from the molding department and ended the month with 550 units still in process. How many units were transferred out?

PB10. LO 5.4 The packaging department began the month with 750 units that were 100% complete with regard to material and 25% complete with regard to conversion. It received 9,500 units from the processing department and ended the month with 500 units that were 100% complete with regard to materials and 75% complete with regard to conversion. With a $7 per unit material cost and a $4 per unit cost for conversion, what is the cost of the units transferred out and remaining in ending inventory?

PB11. LO 5.4 Production information shows these costs and units for the smoothing department in August.

	Cost		Units
Work in process			
Beginning balance: materials	$ 5,213	Beginning units	1,100
Beginning balance: conversion	8,321	Transferred in	24,500
Materials	10,403	Transferred out	1,300
Labor	16,000		
Overhead	$13,475		

What is the value of the inventory transferred out to finished goods and the value of the WIP inventory at the end of the month, assuming conversion costs are 30% complete?

PB12. LO 5.4 Given the following information, prepare a production report with materials added at the beginning and ending work in process inventory being 80% complete with regard to conversion costs.

Costs to Account For	
Beginning inventory: materials	$ 25,000
Beginning inventory: conversion	30,500
Direct materials	2,000
Direct labor	45,000
Applied overhead	11,840
Transferred-in materials	12,015
Transferred-in conversion	33,500
Total costs to account for	$159,855

Units to Account For	
Beginning work in process	4,000
Units started into production	9,500
Transferred out	9,600

PB13. LO 5.5 Selected information from Hernandez Corporation shows the following:

Raw materials purchased	$12,000
Direct materials	$ 3,500
Direct labor	$14,000
Factory depreciation	$ 4,200
Total manufacturing overhead incurred	$18,500
Machine hours per month	34,000
Unit cost for material	$ 2.75
Unit cost of conversion	$ 5.95
Number of units transferred	16,500

Prepare journal entries to record the following:

A. raw material purchased
B. direct labor incurred
C. depreciation expense (hint: this is part of manufacturing overhead)
D. raw materials used
E. overhead applied on the basis of $0.50 per machine hour
F. the transfer from department 1 to department 2

PB14. LO 5.5 Rexar had 1,000 units in beginning inventory before starting 9,500 units and completing 8,000 units. The beginning work in process inventory consisted of $5,000 in materials and $8,500 in conversion costs before $16,000 of materials and $18,500 of conversion costs were added during the month. The ending WIP inventory was 100% complete with regard to materials and 40% complete with regard to conversion costs. Prepare the journal entry to record the transfer of inventory from the manufacturing department to the finished goods department.



Thought Provokers

TP 1. LO 5.1 How would process costing exist in a service industry?

TP 2. LO 5.2 Why are labor and manufacturing overhead grouped together as conversion costs?

TP 3. LO 5.4 How is process costing for a single manufacturing department different from a manufacturing company with multiple departments?

TP 4. LO 5.5 What is different between the journal entries for process costing and that of job order costing?

Management accounting supervisor → Controller → CFO → CEO → Board of Directors

13. Specialization areas for management accountants includes budget analyst, financial analyst, accounting manager, controller, chief financial officer.
15. Professional business organizations that have a code of ethics include the American Institute of Public Accountants, the Association of Certified Fraud Examiners, the Financial Executives Institute, the American Marketing Association, and National Society of Professional Engineers.
17. Several accounting scandals involving publicly traded companies (Enron, WorldCom, and Arthur Andersen) led to the act. It was aimed particularly at public accounting organizations that performed audits of publicly traded corporations.

Chapter 2

Multiple Choice

1. C
3. A
5. A
7. D
9. D
11. B
13. D
15. D
17. C

Questions

1. Answers will vary but should include merchandising, service, and manufacturing businesses.
3. Answers will vary but should include a discussion of operating costs such as salaries and wages, advertising, rent, and office expenses.
5. Answers will vary but must include direct materials, direct labor, and manufacturing overhead.
7. Answers will vary but should include that fixed costs remain fixed in total across the relevant range, bounded by a minimum and maximum activity level.
9. Answers will vary but should include that prime costs are the direct material and direct labor costs, and conversion costs are direct labor and general factory overhead combined.
11. Answers will vary.

Chapter 3

Multiple Choice

1. A
3. C
5. B
7. C
9. D
11. B
13. C
15. C
17. C
19. D
21. C
23. D

Questions

1. Answers will vary. Responses should include that per-unit contribution margin is the amount by which a product's selling price exceeds it total variable cost per unit.
3. Answers will vary. Responses should include that contribution income statements express total contribution margin for a given level of activity and can be useful in making decisions about product pricing and optimal levels of activity.
5. Answers will vary. Responses should include the fact that the contribution margin ratio represents the percentage of every sales dollar available to cover fixed expenses. Businesses can use this ratio when projecting profit at various levels of sales revenue.
7. Answers will vary. Responses should include a description of how the CVP analysis information can be brought into a projected income statement that takes into account additional revenues and expenses of the business to create a "big picture" of what happens as a result of a change in cost, volume, and profit
9. Answers will vary. Responses should include the definition of sales mix as the relative proportions in which a company's products are sold as well as a description of how products within the sales mix have unique sales prices, variable costs, and contribution margins.
11. Answers will vary. Responses should include an explanation of how margin of safety allows the business to operate at a level where the risk of falling to or below the break-even point is low. There should also be some mention of the usefulness of the margin safety as an "alarm" for companies, such that when sales fall to the margin of safety level, action may be warranted.

Chapter 4

Multiple Choice

1. B
3. B
5. C
7. A
9. A
11. D
13. A
15. C
17. D

Questions

1. The company should use process costing. Since there are many similar items, process costing is a better fit than job order costing.
3. The conversion cost is $72,000: the sum of direct labor, factory depreciation expense, and utility expense.
5. The expense recognition principle requires that expenses follow the revenue. Product costs are assigned to the product because they are associated with the revenue from the sale of the product. The cost is transferred from inventory to cost of goods sold when the item is sold. This matches the revenue from the sale with the cost of the item being sold. Period costs are expensed when incurred because they are not related to a specific product but are instead related to the time period in which revenue is earned.
7. B, D, C, A.
9. Management uses the activity considered to be the cost driver and multiplies that rate by the activity for each specific job. The result is the amount of overhead applied to that specific job.
11. Expenses normally have a debit balance, and the manufacturing overhead account is debited when expenses are incurred to recognize the incurrence. When the expenses are allocated to the asset, the work in process inventory, the expense account manufacturing overhead is credited. This is in accordance with the expense recognition principle. The timing of the expense follows the revenue, and when the costs

are allocated to inventory, they become a part of the product's cost and are recognized when the asset is sold.

13. direct materials

Chapter 5

Multiple Choice

1. C
3. A
5. D
7. A
9. C
11. B
13. C
15. C
17. C
19. C

Questions

1. Answers will vary but should include the following:

Area	Process	Job Order
Types of jobs	Identical	Custom order
Quantity within each job	Large volume	Small volume
Cost accumulation	In each department	In each job

3. The weighted-average method assigns the beginning inventory and the costs added during the period. The weighted-average method does not differentiate between the beginning inventory and the units started in production. This is different from the FIFO method that accounts for the beginning inventory differently and separately from current period costs.
5. Prime costs and conversion costs both include labor. Prime costs are the direct costs, other than equipment, used in manufacturing and therefore are direct material and direct labor. Conversion costs are the costs involved in converting the direct material into the product and therefore are direct labor and manufacturing overhead.
7. Job order costing and process costing are the accounting systems used to record the costs expended to produce a product. Conversion costs are the direct labor and manufacturing overhead involved in the production process and exist regardless of the accounting system used.
9. While conversion typically occurs evenly throughout the process, materials are not typically added evenly, so the ending work in process can be different. For example, when materials are added at the beginning of the process, materials can be 100% complete and conversion can be 50% complete. Different completion percentages result in different equivalent units.
11. Step 1: Determine the units to which costs are assigned. Step 2: Compute the equivalent units of production. Step 3: Determine the cost per equivalent unit. Step 4: Allocate the costs to the units transferred out and the units partially completed.
13. The costs transferred in are treated in the same way as direct material that is added to production at the beginning of the process.
15. A. iv; B. ii; C. vi; D. vii; E. iii; F. v; G. i; H. viii.
17. Prior to the new year, a company computes the estimates of the annual overhead per department divided by the estimated driver for that department. A driver is the measure that increases the cost of overhead and is commonly direct labor hours, direct labor cost, or machine hours. The result is the predetermined